Virtual University of Pakistan

Federal Government University

PAST PAPERS /MCQs/ MID TERM /FINAL TERM/SOLVED QUESTIONS

MID TERM & FINAL TERM

QUIZ NO 1 | QUIZ NO 2 | GRAND QUIZ | MID TERM | FINAL TERM

MSC-ZOOLOGY

BIF401-Bioinformatics-I

PAST PAPERS

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covering Lesson 1- 5, 15- 25 and 80-86.	QUIZ NO 1 - 23-5-2021

1. Laboratory work using computers and computer-generated models generally offline is referred as:_____ **dry-lab**
2. In GenBank. specific gene can be searched by using? _____**All of the above**
3. Analyzing or comparing entire genome of species known as,_____ **genomics**
4. Bioinformatics is the interdisciplinary relative new development of _____**physics. chemistry. mathematics, computer science and biology**
5. RNA has Ribose sugar and DNA has _____ **De-oxyribose sugar**
6. Which of the following is untrue about the Unweighted Pair Group Method Using Arithmetic Average? _____**The distances between this new composite taxon and all remaining taxa are calculated to create a reduced matrix**
7. Bioinformatics can deliver: _____**All of the above**
8. GenBank and SWISSPORT are example of _____**Primary Database**
9. The study of full complement of proteins expressed by a genome is called _____**Proteomics**
10. How many reading frames exist for a double-stranded DNA sequence?_____ **Six**
11. Which one of the following methods can be used for statistical evaluation of data? _____ _____ **Both A&B**
12. How many proteins does the human body produce? _____**Hypothesis of wo gene = one protein So, there should at least *10,000**
13. How many proteins does the human body produce?_____ **Hypothesis of "One gene = two protein" So, there should at least ~40,000**
14. Translation is the process in which _____**Protein is formed on RNA template**
15. mRNA directs the building of proteins through a sequence of_ _____**Codons**
16. The procedure of aligning two sequences by searching of patterns that is in the same order in the sequences? _____**Sequence Alignment**
17. GenBank database is maintained by _____ _____**National Centre of Biotechnology Information (NCBI)**
18. Before starting to match nucleotides or amino acids, we need to find the _____ _____**location of conserved residues**
19. Which of the following does not describe local alignment_____ _____**local alignment have terminal gap**
20. **Tandem MS helps us to measure masses of _____.___ Fragments**
21. **____** are the unit of protein _____**Amino acids**
22. Choose the incorrect statement out of the following ._____ **tyrosine is a modified amino acid**
23. The identification of drugs through genomic study is known as _____ **Pharmagenatics**
24. Amino acids are chained together by _____ **peptide bond**
25. The alignment method suitable for finding out conserved patterns in DNA or protein sequences is _____ **Local alignment**
26. Which of the following is incorrect regarding Mass Spectrometry Protein Identification? _____ The proteolysis doesn't generate a pattern according to molecular weight
27. Amino acids are mostly synthesized from_____ **Alpha-ketoglutaric acids**
28. Which of the following is the character based phylogenetic tree construction method? _____ _____**maximum parsimony**
29. Pairwise Sequence Alignment is used to identify regions of similarity that may indicate _____**functional, structural and/or evolutionary relationships between two biological sequences**
30. In dot plot, when the two sequences have substantial regions of similarity many dots line up to form contiguous _____ lines._____ **Diagonal**
31. Knowing____ should be enough to find the required entry quickly_____ _____**accession number ' protein name or name of gene**
32. Bioinformatics is a becoming a popular science due to _____ **all of the above**
33. Which of the following is the correct order of organization of genetic material from largest to smallest? _____ **Genome, Chromosome, Gene, Nucleotide**

34. A collection of seeds, plants, or animals, maintained as a repository of genetic material, typically to preserve genetic diversity_____ **Gene Bank**
35. Number of chiral centers in isoleucine is_____ **2**
36. BLAST uses a _____ to find matching words, whereas FASTA identifies identical matching words using the _____._____ **substitution matrix, hashing procedure**
37. For which one of the following activities bioinformatics is used in modern biological sciences_____ **both a and b**
38. Glycoprotein antibodies produced by ___ plasma cell. _____ **WBC**
39. _____ is a biological information search engine maintained by NCBI._____ **ESEMBLE**
40. Which of the following does not describe dynamic programming?____ **The method can be useful in aligning protein sequences to protein sequences only**
41. In the Neighbor Joining step, The UPGMA method uses unweighted distances and assumes that all taxa have constant evolutionary rates._____ **TRUE**
42. Sequencing of genomic DNA is included in_____ **Structural genomics**
43. Human genome contains about_____ **3 billion base pairs**
44. Translation takes place before transcription._____ **FALSE**

Quiz no 1 December 2020

1. Scoring Matrices contains, _____**Both +VE and -VE values**
2. Progressive alignment methods use the dynamic programming method to build a MSA starting with the most related sequences and then progressively adding less related sequences or groups of sequences to the initial alignment?_____**TRUE**
3. The correct order of progressive alignment steps for multiple sequence alignment is ___ _____**Pairwise alignment, Guide tree, Branching order in tree**
4. Name any two frequently used scoring matrices _____**BLOSUM and PAM**
5. The ….tool compares nucleotide sequence against DNA database._____**blastn**
6. The…. E-Value. the more significant the hit. _____**Lower**
7. Your lab partner is using BLAST, and his best E-value is 3. This means that _____**There would be 3 matches that good in a database of this size by chance alone**
8. Which of the following is not a benefit or a factual of FASTA over BLAST? _____ _____**It gives more sensitive results**
9. In dynamic programming we can do multiple sequence alignment up to._____**21 sequences**
10. BLASTX uses protein sequences as queries to search against a protein sequence database. _____-_____**FALSE**
11. compares protein sequence against protein databases._____**blastp**
12. Which of the following is not a variant of BLAST?_____**TBLASTNX**
13. CLUSTALW is a more recent version of CLUSTAL with the W standing for _____**Weighting**
14. Which one of the following is not a Sequence Alignment Tool"?_____**rasmol**
15. Which of the following is untrue about the modification of PAM matrices?_ _____ _____**This took place in 1976**
16. The BLAST program was developed in_____**1990**
17. Which is the default scaring matrix used in BLAST?_____**BLOSUM 62**
18. In multiple sequence alignment (MSA), we have _____**Global alignment**
19. The scoring of gaps in a MSA (Multiple Sequence Alignment) has to be performed in a different manner from scoring gaps in 4 pair-wise alignment? _____**FALSE**
20. Which of the following doesn't describe PAM matrices? _____ _____**it stands for Percent Altered Mutation**

1. FASTA achieves alignment by using _____**short lengths of exact matches.**
2. Dot plots cannot capture _____**insertions, deletions and gap indications**
3. Dot plots employ dot matrix with two sequences plotted on _____**top and left of the matrix**
4. In multiple sequencing, we have _____**both**
5. CLUSTALW can run in two modes _____**slow/accurate, fast/appropriate.**

6. The correct order of progressive alignment for multiple sequencing alignment is _____**Pair wise alignment, Guide tree, branching order in tree.**

7. Blast can find sequencing from protein or nucleotide database _____**it can find known and unknown sequencing.**

8. is used to known about unknown sequences of nucleotide. __ **Next generation sequencing**

9. In multiple sequence algorithm _____**doing so many progressive alignment can be slow.**

10. : Structural information of a molecule obtained by _____**X ray crystallography & Nuclear magnetic resonance.**

11. Before starting to mach nucleotide or amino acid are need to find the _____**Location of conserved residues.**

12. Bioinformatics is an interdisciplinary relative new development of_____ _____**biology, mathematics, computer science, chemistry and physics.**

13. Fasta online software Can also perform genome and proteome similarity search ._____**true**

14. From Fasta we can obtain the information _____ _____**on the parent organism, function and evolution history.**

15. name any two frequently used scoring matrix _____**Blosum and pam.**

16. Amino acids are chained together by _____**peptide bonds**

17. To find out the unknown sequence of nucleotide and protein we use _____**blast.**

18. GenBank and Uniprot, have ___**A large number of sequences are available in publically .**

19. Protein databank is a _____**data base**

20. In dynamic programming we can do multiple sequence alignment upto _____**21 sequence** .

21. How many types of secondary structure can be formed possible of RNA? _____**5.**

22. RNA structures have the ….quantity of free energy. _____- **lowest (or close)**

23. In a dot matrix, two sequences to be compared are written in the _____ of the matrix.__ _____ **Horizontal and vertical axis**

24. RNA can play role due to its structure . _____**all of above.**

25. Rna contains …..sugar? _____ **ribose**

26. Thymine is replaced by **…..**in rna. _____ **uracil**

27. RNA molecules readily degrade due to their **….**shelf life . _____ **short**

28. Each Nucleotide in rna has …..to the backbone . _____ **covalently bonded**

29. If a problem can be broken into subproblems which are reused several times, the problem possesses _____ property._____ **Overlapping subproblems**

30. When dynamic programming is applied to a problem, it takes far less time as compared to other methods that don't take advantage of overlapping subproblems._____ **TRUE**

31. Nucleotide in RNA are joined by_____ **5'3' phosphodiester bond**

32. …is the free energy of an RNA molecule available for reaction . _____**Gibbs Free Energy"**

33. The structure of rna sampled can be measured by._____**Atomic Force Microscopy**

34. The 3'-end of ….has a poly A tail _____ **mRNA**

35. The role of **….**Carry genetic information from DNA to Ribosomes where proteins are being assembled . _____ **mRNA**

36. Messenger rna is only ….of rna content in the cell. _____**5-10%**

37. Gaps are produced by _____**insertion or deletion**

38. Example of energy base _____**Zuker algorithm.**

39. .. forms of DNA rending sequence. _____**two**

40. Translation involves coding of protein by …at ribosome. _____**RNA**

41. some medicine have side effect on certain patient _____ _____**Frontiers in personalized medicine**

42. By comparing sequence we can get _____**similarity, specific difference, relationship, evolutionary insight.**

43. Non clustering method include _____**Maximum parsimony.**

44. Destabilizing energy of a nucleotide has _____**(+) value.**

45. Comparison of protein sequence can be done with…_____**Uniprot**

46. Function is a property of the RNA structure,which is formed from the____ **RNA sequences.**
47. Structural information of a molecule obtained by _____ **resonance.**
48. Protein have… amino acid. _____**20**
49. If sequence are 9 then according formula it will be _____**92**
50. Fast is _____**available online.**
51. Fasta was developed in_____ **1988.**
52. Bioinformatics is the inter disciplinary relative new_____ **developing field.**
53. Genbank can be reached by…online website or … bank. __**www.ncbi.nim.nih.gov/gene**
54. Before starting to mach nucleotide or amino acid are need to find the_____ _____ **Location of conserved residues.**
55. some medicine have side effect on certain patient _____ _____**Frontiers in personalized medicine.**
56. Identity is the number of nucleotide or amino acid which match exactly between two biological sequence.
57. Alignment are represented by diagonals in dot matrix.
58. Needleman wunsch man algorithm we use for…… _____**Global alignment.**
59. Exons in protein tend to be more conserved then the_____ **intron.**
60. In MSA the distance between 2 sequences in equal to_____ **NO of identical residue/number of aligned residue.**
61. All the changes which accrued in our cell …….. _____**metabolism**
62. NJ algorithm is used to produc ……. _____**Optimal RNA 2´ sturucture.**
63. Types of RNA…present in organism. _____ **5**
64. The online search is for DNA and RNA is…….. _____**genebank**
65. In multiple sequence algorithm _____**doing so many progressive alignment can be slow.**
66. DNA/RNA sequence is available on…. _____ **GENE BANK**
67. Martine algorithm predict for structure _____**2 RNA.**
68. In extraction of ORF, 1st step is obtained the _____**3 codons.**
69. Example of energy based algorithm _____**Zukar algorithm.**
70. Dynamic programming help us to……. _____ **reduce computational cost.**
71. When unpaired bases in 2 region binds is result form_____ **Buldges.**
72. which one of following is not objective based method of phylogenic …_____ **UPGMA**
73. protein sequencing can be done by _____**Swiss prot & UniProt.**
74. Drugs can be better developed after understanding molecular basis of … _____**Disease**
75. Online tools are available for _____**homology modeling.**
76. Insertion and deletion in alignment cause…… _____ **gaps.**
77. Alligment of insertion and deletion can be performed by …. _____**dynamic programming**
78. Exons in proteins tend to be more conserved than ……. _____ **intron.**
79. Blast confined sequences from—— _____ **ID,name,spcies**
80. ribosmes read RNA and collect amino acid from…… _____**cell cytosol.**
81. For exact maching …….required. _____**Traceback**
82. Motif protein chain fold to take . _____**3D structure**
83. Next generation sequencing is used to known about unknown sequences of ___**nucleotide.**
84. RNA fold to form _____**2 structure.**
85. Which has less computational cost ……_____ **Dynamic programming**
86. We compare all length of nucleotides in ….._____ **Global alignment.**
87. Destabilized RNA molecule have _____**(+)value.**

88. In Zukar algorithm, we need construct _____all possible combination.
89. We use to sum up energy to find lowest matrix. _____Zukar algorithm
90. In comparing RNA structure, we used to select which having_____ lowest energy.
91. Addition of nucleotide is done by ….. _____hydrogen bondi
92. Adman degradationis used for _____protein sequencing.
93. We can do repeated sequences alignment by slight modification in_____ waterman model.
94. Length of hairpin loop is _____4 bases.
95. ‘’T" Thrionine is an basic element..._____.true
96. Structural information of a molecule can be obtained by_____ X.ray crystallography , nuclear magnetic sequence.
97. Compare a protein query sequence against a nucleotide sequence database…._____.tBlastn
98. Protein formation of abnormal folding called _____Pseudoknots.
99. By using uniprot we can find sequence of….._____.(protein)
100. The 5´ end of molecule caped with _____7 Methyl Guanosine Triphosphate.
101. DNA get modified by_____ mutation, substitution, insertion, deletion.
102. UPGMA is used to calculate the _____distance of each cluster.
103. scoring scheme has an extremely crucial role in producing _____optical alignment.
104. why is there need to predict RNA structures _____to know about its function
105. DP recombines the nucleotide recombination through a process of _____Traceback.
106. Which of the following is not an advantage of Needleman-Wunsch algorithm?
_____ This does not help in determining evolutionary relationship
107. .how many open reading frame exist for each DNA sequence._____6ORFs
108. tblastn search translated nucleotide using.._____ _____.Protein query
109. Blast can find sequencing from protein or nucleotide database_____ it can find known and unknown sequencing.
110. name any two frequently used scoring matrix _____Blosum and pam.
111. From Fasta we can obtain the information o_____
_____the parent organism, function and evolution history.
112. The correct order of progressive alignment for multiple sequencing alignment is
_____Pair wise alignment, Guide tree, branching order in tree.
113. CLUSTALW can run in two modes _____slow/accurate, fast/appropriate.
114. Gene bank and UniProt have _____Large number of sequencing publically.
115. The _____ refers to a structure with two ends of a single-stranded region (loop) connecting a base-paired region (stem)._____ hairpin loop
116. In multiple sequencing, we have _____Global alignment.
117. Ubiquitin play important role recycling _____protein.
118. Faster active alignment by using local alignment and protein nucleotide. pairwise alignment
119. Expasy was develop by _____Swiss Bioinformatics Institute (SIB)
120. When unpaired bases in 2 region binds itself form _____Buldges.
121. In Nussinov Jacob algorithm which turn is added to compare to score for diagonal position set _____diagonal and lower tri diagonal to zero.
122. Trackback begins from the position in Nussinov Jacobson score matrix. _____
_____ _ 4rth possition
123. Genomics, evolutionary studies and system biology are application of_____bioinformatics
124. Exact matching require which type of nucleotide-_____same number of nucleotide
125. Accurate solution of protein structure is one of _____the toughest problems
126. …is overlap matches can start from any position in scoring matrix._____ _Tract back
127. The matrix has .....scores. _____Positive and negative
128. ….._can search people mass finger printing and shotgun proteomics data set____ MASCOT
129. Proteomics is the study of _____ set of proteins

130. When we ionize the protein, it can be deflected by a _____ in proportion to its mass and the mass of _____ can be measured by _____ **Magnetic field, protein, spectrometry**

131. Spectrometer measures the protein by_____ **Mass/charge**

132. Simulation of the folding process depends _____**Energy function**

133. Information about …..is available on gene bank _____ **DNA/RNA**

134. Rna is a genetic material in . _____ _____**in some viruses and some prokaryotes and rerely in eukaryotes**

135. Salient ionization techniques include Matrix Assisted Laser Desorption Ionization(MALDI) _____**& Electro Spray Ionization (ESI)**

136. **Mass/Charge helps us to calculate _____.**_____ **Mass of protein**

137. Which algorithm is used by local alignment ? …. _____**Smith waterman**

138. In eukaryotes, which RNA polymerase makes rRNA?_____ **RNA polymerase II**

139. originally the fold program of ….predicted …… having the minimum free energy. _____**M.Zuker.two structure**

140. In gene Bank specific gene can be searched by using.. _____**all of the above**

141. Removal and addition of amino acid in protein and nucleotide in DNA RNA by using gap named as…_____ **indels**

142. Dot plot of repeating elements would be small cross on plant…_____**false**

143. Blast uses a…to find matching words whereas faster identify these identical matching words using the…. _____**Substitution matrix blocks**

144. Gene assembly can be performed using:_____ Bioinformatics tools

145. Which is the molecular structure visualisation tool…_____**cn3d**

146. Progressive alignment method used the dynamic programming method to build MSA starting with the most related sequences and then progressively adding less related sequences or groups of sequences to the initial alignment…_____**true**

147. Messenger RNA directs the building of protein through a sequence of….. _____**exons**

148. Which of the following Rna characteristically contains unusual purines and pyrimidine…..__ _____**mRNA**

149. Capping occurs after the messenger rna and synthesise is complete…_____**false**

150. ……. Helps in calculating the mass of the protein…_____**mass/charge ratio**

151. Which of the following is not an ionization technique in protein…. _____**X-ray crystallography**

152. Which of the following is not a benefit or a factual of fasta over over blast…_____ _____ _____ **it gives more sensitive results**

153. Blast might not find matches for very short sequences…. _____**False**

154. Single letter code of tyrosine is…_____**Y**

155. …Is the free energy of molecule available for reaction…. _____**Gibbs free energy**

156. M fold gives out output several structures along with their…. _____**Energetic stability**

157. Base pairing in rna is A-G and C-U…. _____**False**

158. Building blocks of nucleic acid are…_____**nucleotides**

159. Which of the following is not a correct about the blast…. _____ _____**Blast is not based on a heuristic searching methods**

160. Translation is the process in which**…..** _____**protein is formed on Rna a template**

161. Because there is no indication of which node at present in ancestor there is no direction of an evolutionary path in an unrooted the tree..._____**false**

162. One of the drawback of edman degradation methods is that restricted to chain only ….. reduces……_____**60**

163. Which of the following is untrue about the modification of pam matrices**…..** _____ _____ **this took place in 1976**

164. XP Pasy is is a comprehensive protomice web server with a suite of programs for searching peptides information from the Swiss prot and TREMBL database…. _____**False**

165. Which are two types of protomice protocols that usually employed**….** _____ _____**Bottom-up and top down**

166. Energy is a released when .. nucleotide make Bond…_____**unstable**

167. Which of the following does not describe dynamic programming…_____ _____**the method can be e useful in in aligning protein sequences to protein sequences only**

168. Which of the following is not in advantages off needleman wunsch algorithm….. _____**this does not help in in determining revolutionary relationship**

169. Amino acids are attached to the**….** _____**Codon arm of tRNA**

170. WHICH OF THE FOLLOWING STATEMENT IS TRUE…_____ _____ _____**the sugar molecule of the nucleotide is an l -configuration**

171. Which is the default scoring matrix using the blast…_____**BLOSUM62**

172. IN THE NEIGHBOUR JOINING STEP. The UPGAMA method uses unweighted distance and assume that texa have constant evolutionary rate… _____**false**

173. Which of the virus has double standard RNA as genetic material…_____**Rio viruses**

174. Dynamic programming uses a…. To deal with matches mismatches and gaps…. _____**Scoring function**

175. Function and Structure of genome is _____._____ **Genomics**

176. The role of…is giving genetic information from DNA to ribosomes where protein are beinassembled..**.**_____**mRNA**

GRAND QUIZ

1. which of the following statements about amino acids is correct? _____**acids are classified according to the structures and properties of their side chains**

2. Which of the following methods are a broad class of computational algorithms that rely on repeated random sampling to obtain numerical results. _____**Monte Carlo methods**

3. Zuker s Algorithm involves energy .**_____Minimization**

4. FASTA format start with symbol._____**>**

5. how many types of secondary structures can be formed possibly by folding of RNA?_____**4**

6. All of the following are protein sequence database, except. _____**DDJB**

7. DOt plot of repeating elements would be small crosses on plot._ _____**FALSE**

8. RNA is found in the nucleolus of the cell and DNA is found in _____**Chromosomes**

9. Which one of the following are Archaea?_____**Thermococci**

10. Single letter code of Tyrosine is _____**Y**

11. Which algorithm is used by global alignment?_ _____**Needleman-Wunsch**

12. In Bottom up proteomics. Enzyme ….is used to cleave the proteins _____**Trypsin**

13. RNA is the genetic material in _____**Only In some viruses**

14. Mfold gives out (output) several structures along with their_____ **energetic stability**

15. rRNA is synthesized by:_ _____**Nucleolus**

16. Mfold is computationally _____ and can give result for _____ structures that have sequences _____._____ **Expensive, 1', 2', 8000**

17. What are 2 things required in order to successfully use the dynamic programming technique? _____**Optimal substructure and overlapping subprablems**

18. Score in protein fragmentation technique is awarded on the basis of the closeness between_____ Both A and B

19. There are 64 codons in the genetic dictionary as _____ _____ ____ _____ _____**There are 3 nonsense codons and 61 sense codons**

20. Nussinov-Jacobson (NJ) Algorithm ts a dynamic Programming (DP) strategy to predict _____**Optimal RNA 2" structures**

21. Sum of stabilizing and destabilizing energies can help determine the _____ _____**Quality of 2RNA structure**

22. Block analysis methods use substitution matrices such as the PAM and BLOSUM matrices to score matches. _____**TRUE**

23. A problem exists when comparing _____ sequences using the dot matrix method, namely, the _____._____ **large, high noise level**

24. TBLASTX queries protein sequences to a nucleotide sequence database with the sequences translated in all six reading frames._____ **TRUE**

25. In multiple sequence alignment..._____ _____ _____**doing so many progressive alignments can be slow**
26. Which of the following is the smallest of the RNAs? _____**Transfer RNAs**
27. Dynamic Programming uses a __" to deal with matches, mismatches and gaps. _____**Dynamic Algorithms**
28. Which one of the following issues are addressed by using computational methods, data processing techniques and algorithms? _____**ALL of the above**
29. GeneBank and UniProt have _____**large number of sequences publically**
30. The sugar in RNA is_____ **Ribose**
31. The amino acid sequences of thousands of different proteins from many species have been determined using principles first developed by_____ **Sanger**
32. Elongation of transcription and RNA processing are all interconnected? _____**TRUE**
33. What is the length of a motif, in terms of amino acids residues?_____ **10-20**
34. Zuker's Algorithm involves computing stabilizing and destabilizing energies of a _____ RNA structure._____**2**
35. Bioinformatics not only organizes, stores and analyzes biological data, but can also validate novel? _____ **Hypotheses**
36. Which portion of DNA is highly conserved?_____ **Exon**
37. ExPASY is a comprehensive proteomics web server with a suite of programs for searching peptide information from the SWISS-PROT and TrEMBL databases. _____**TRUE**
38. Score can be negative in Smith-Waterman algorithm. _____**FALSE**
39. Edman reagent cleaves polypeptides _____**At the N-terminus**
40. Phylogenetics is the study of the evolutionary history of living organisms using treelike diagrams to represent pedigrees of these organisms. _____**TRUE**
41. ECD, a protein fragmentation technique stands for _____**Electron Capture Dissociation**
42. The Alignment method suitable for finding out conserved patterns in DNA or Protein sequences is _____**Pair wise alignment**
43. During transcription, if the nucleotide sequence of the DNA strand that is being coded is ATACG, then the nucleotide sequence in the mRNA would be _____**UAUGC**
44. Which of the following RNA serves as adaptor molecule during protein synthesis_____**tRNA**
45. The tool …..compares translated nucleotide query sequence against protein databases. _____**blastx**
46. Which of the following is a database of RNA 3D structure motifs and their contact. both with themselves and with proteins. _____**RNA Bricks**
47. Which statement best represents the meaning of term evolution _____ _____**Changes in species toward greater complexity overtime**
48. Screening the predicted protein sequences against _____ _____**expressed sequence tag (EST)**
49. Because there is no indication of which node represents an ancestor, there is no direction of an evolutionary path in an unrooted tree. _____**FALSE**
50. To find out the unknown sequences of nucleotides and proteins we use. _____**BLAST**
51. Laboratory work using computers and computer-generated models generally offline is referred as: _____ **Both A& B**
52. The procedure of aligning two sequences by searching of patterns that is in the same order in the sequences?_____**Pairwise Alignment**
53. Which algorithm is used by local alignment? _____**Smith-Waterman**
54. Which one of the following are not the components of RNA? _____**Thymine**
55. The …….refers to two single-stranded regions on opposite strands connecting two adjacent base-paired segments. _____**helical junctions**
56. Which of the following main component of mass spectroscopy deal with resolving the ions into their characteristics mass components according to their mass-ta-charge ratio? _____**Analyzer**
57. Zukers Algorithm involves computing stabilizing and destabilizing energies of a …RNA structure _____**2**
58. Which of the following is untrue about dot plot method and its applications? _____ ___**This method gives a direct visual statement of the relationship between two sequences**

59. More than one codon can code for same amino acids as there are 20 amino acids involved in protein synthesis. _____TRUE
60. Messenger RNA is _____single stranded
61. Genetic information is transferred from nucleus to cytoplasm through _____RNA
62. What ere the structural units of nucleic acids? _____Nucleotides
63. What is characteristics of the coding capacity of codons? _____B and c
64. Which is a molecule structure visualization tool? _____Cn3D
65. Isotopes that contain an…. number of…. have an intrinsic magnetic moment and angular momentum _____ odd….protein/neutron
66. Watson and Crick suggestion of the complementary strand synthesis taking one of the parent strand as template was proposed in - _____1953
67. Who were the first to suggest that one strand of DNA might act as a template for the synthesis of its complementary strand?_____ Watson and Crick
68. Most vertebrate gene use as the translation start codon and have a uniquely conserved flanking sequence call a Kozak sequence (CCGCCATGG). _____UAA
69. In a DNA strand the nucleotide are held together by _____phosphodiester Bond
70. Which of the following is not a correct about BLAST? _____ _____BLAST is not based on heuristic searching methods
71. Edman degradation. developed by Pehr Edman. is a method of sequencing amino acids in a ribosome. _____FALSE
72. Building blocks of nucleic acids are _____Nucleosides
73. In Smith-Waterman algorithm. in initialization Step, the …row and…. column are subject to gap penalty. _____first, first
74. Intact masses can measure the intact _____Both A& B
75. Which trees are computationally expensive? _____Rooted
76. Which of the following occurs a wide variety in RNA and it mostly forms the stable hairpin loop _____ CUUCGG
77. Human genome contains about _____25,000 genes
78. Which of the following is not a correct about FASTA? _____FASTA uses ˜hashing strategy to find matches for a short stretch of identical residues with a length of k
79. The actual genetic code used by cells is_____ Triplet code
80. Protein Data bank is a_____-all
81. Which of the following is untrue about the modification of PAM matrices? _____ _____This took place in 1976
82. Which of the following is untrue about distance based methods?__ Clustering is the only method among the algorithms for the distance-based tree-building method
83.
84. Which of the following are characteristic of the genetic code? _____Both A&B
85. What is the composition of nucleotide? _____A base+A sugar + a phosphate
86. The most abundant type of RNA in the cell is _____rRNA
87. ……alignment procedure perform end to end sequence alignment. _____Global
88. Which one of the following can identify coding portions in a DNA?_____ local Alignment
89. Identify the purine base of nucleic acids in the following _____Adenine
90. Laboratory work using chemical, drugs and water is refered as _____Wet lab
91. Like other alignment programs, CLUSTAL uses a null score for opening a gap in 6 sequence slignment and a penalty for extending the gap by one residue. _____FALSE
92. Which of the following are the main strategies for RNA structure predictions algorithms. _____Both A&B
93. Mass spectrometry helps us to measure the molecular weight of _____ Both A&B
94. In computer science. algorithm refers to a special method usable by a computer for the solution to a problem. _____TRUE
95. Capping occurs after the mRNA synthesis is complete. _____TRUE
96. Capping is done by the addition of _____._____ Methylated G
97. DNA makes RNA and RNA makes protein, this process is termed as_____Central dogms

98. Choose the incorrect statement out of the followings? _____ _____ _____**Only L-amino acids are found in the biological system**

99. Codon select the amino acids and… make the protein by polymerization process _____ _____ _____**RNA Sequences**

100. Thymine in DNA is replaced by _____**Uracil in RNA**

101. Messenger RNA is synthesized during _____**Transcription**

102. Amino acids and nucleotide are more easily substituted because, _____ _____**They have similar chemical nature**

103. Which of the following codons is the mRNA start codon that initiates translations? _____**AUG**

104. Ribozymes are _____**RNAs with catalytic activity**

105. Name the technique in which one proton is added to protein or peptide and the molecular weight is increases by one and Mass spectrometry reports the molecule at +1_____**MAIDI**

106. The tree construction process is somewhat similar to that used UPGMA,_____**TRUE**

107. Which is the default scoring matrix used in BLAST?_____**BLOSUM 62**

108. In molecular genetics, an open reading frame (ORF) is the part of a reading frame that has the potential to code for a protein or peptide._ _____**TRUE**

109. Which of the following is not the objective to perform sequence comparison? _____ _____ _____**To study evolutionary relationships**

110. The web based FASTA program is offered by the European Bioinformatics Institute ___**TRUE**

111. Structural information of a molecule can be obtained from _____ _____**X ray crystallography & Nuclear magnetic resonance (NMR)**

112. mRNA directs the building of proteins through a sequence of _____**Exons**

113. FASTA the online software can perform genome and proteome similarity search. _____**true**

114. BLAST can find sequences from protein or nucleotide database. _____ _____**it can find only unknown sequences**

115. FASTX, which compares a protein query sequence to 4 translated DNA database. _____**FALSE**

116. Which one of the following statements is true with respect to the DNA double helix? _____**The base pairs have opposite polarity**

117. Fragmentation techniques include ECD. CID etc. intact molecule fragmentation splits the molecule into parts _____**Two parts**

118. UPGMA produces ultrametric trees. _____**FALSE**

119. Crystallized RNA diffracts X-rays which helps estimate _____.__ Atomic Positions

120. Proteins can be sequenced by _____.__ _____**Edman's degradation**

121. In double helix of DNA, the two DNA strands show which type of characteristics?_____ _____**coiled around a common axis**

QUIZ NO 2

1. Each nucleotide in RNA has …..bond to backbone sugar. _____**covalent**

2. - ……."helps in calculating the mass of protein. _____ **Mass/charge ratio**

3. PST acronym stands for -…...._____**peptide sequence tags**

4. How many types of secondary structure can be formed possibly by folding of RNA. _____**5**

5. Which are two types of proteomics protocol that are usually employed. _____ _____**bottom up, Top down**

6. Messenger RNA is only …..----- of RNA. _____**5-10%**

7. RNA is ....._____ _____**Single stranded**

8. How we can denote a 50 amino acid protein in position 11. _____**c11, 239**

9. The structure of RNA sample can be measured by…….-. _____**all of above**

10. Bioinformatics is -……..- _____**applied science**

11. Which of the following is not protein fragmentation technique _____**Microarray**

12. - is the free energy of RNA molecule available of reaction. _____**Gibbs free energy**

13. What does means of MS/MS measurement of molecular weight of protein. _____ **ionized peptides will be analyzed thrice after fragmentation**
14. can search peptide mass fingerprinting and shotgun proteomic dataset. _____ **MASCOT**
15. The 3end of has a poly A tail. _____ **mRNA**
16. The alignment procedure that tries to align region with high level of matches without considering the alignment of rest of the sequence is _____ **Local Alignment**
17. The role of ……”- is carrying genetic information from DNA to ribosome where protein are being assembled. _____ **tRNA**
18. RNA contains……_- sugar. _____ **ribose**
19. Which of the following is not an ionization technique in protein. ___ **X-ray crystallography**
20. How many types of peptide may be injected in Mass Spectrometer chamber. _____ **three hundred thousand-four hundred thousands**
21. RNA molecule are readily degraded due to their …….-shelf life. _____ **short**
22. Which trees are computationaly expensive. _____ **Rooted**
23. Thymine is replaced by ……”- in RNA. _____ **Uracil**
24. RNA can play role its structure is/are …….. _____ **all of above**
25. Ribosomes are composed of_____ **RNA and Proteins**
26. What will be weight of V amino acid 376.14 and Q amino acid 277.11 ay _____ **99.03**
27. RNA molecule have -……quantity of free energy. _____ **lowest**

FINAL TERM

1. Genomics, evolutionary studies and system biology are application of _____ **bioinformatics**
2. Exact matching require which type of nucleotide-_____ **same number of nucleotide**
3. Accurate solution of protein structure is one of the _____ **toughest problems**
4. …is overlap matches can start from any position in scoring matrix_____ **Tract back**
5. The matrix has .....scores. _____ **Positive and negative**
6. ___ can search people mass finger printing and shotgun proteomics data set_____ **MASCOT**
7. Spectrometer measures the protein by _____ **Mass/charge**
8. Simulation of the folding process depends _____ **Energy function**
9. Information about …..is available on gene bank_____ **DNA/RNA**
10. …is used to measure the molecules weight of intact proteins._____ **TDP**
11. Coordinates in the protein backbone can be used for structural visualization._____ _____ **of alpha carbon**
12. in structure visualization c-alpha atoms are traced to recreate a ____ **3D protein structure**
13. in UPGMA distance is calculated btw _____ **two clusters and between two trees**
14. pk value of Aspartic acid _____ **3.9**
15. Largest amino acid ..... _____ **Tryptophan**
16. 2 STRUCTURE of proteins can be obtained by using_____ **Chou ndFasman algorithm**
17. NJ Algorithm strategy is used to predict __ _____ **RNA 2` structure**
18. The loop of hair pin must be at least _____ **in length 2 aminoacids**
19. Fold recognition is also called .....**threading**
20. Ab initio method in contrast ,base their prediction on _____ **low energy model**.
21. Bioinformatics require.....smart mind, and connected to internet,___ _____ **both**,
22. Sequence alignment tool is.....**PROSIGHT and MASCOT**.
23. MS1,MS2 provide us data identifying unknown.....**Proteins sequence.**
24. MGF files develop as an open standard for..... **proteomics data.**
25. ..coordinates alpha carbon in protein back bone can be used for comparison ._____ **PDB**
26. Alpha carbon atom can be obtained from.....**PDB**

27. in formation of 2nd structure of protein C & N can make....._____**Hydrogen bonds**
28. information for protein folding is into its native structure is in..... _____ _____ **protein's amino acid sequence**
29. Energy …during bond formation._____ **released**
30. hydrogen bonding occur in .....Protein structure_____ secondary
31. Bond between C and H in alpha helices_ _____Hydrogen bonding
32. Amino acid having 3 codons_____ isoleucine
33. MALDI typically adds a ..... to protein or peptide. _____ Proton
34. low quality match gets____ _____lower score
35. ____ are sequences of amino acids produced during MS2. _PST (peptide sequence tags)
36. MS2 data can be used to extract....._____peptide sequence tags.
37. The alpha helices propensity should be more than_____(1.0)
38. How many types are of protein sequencing techniques__ _____(2)
39. Blast can find sequences of _____nucleotides amino acid
40. If a protein sequence of 26 amino acids is fragmented at 11 amino _____acid C11, Z25
41. To find out unknown sequence of nucleotide we use . _____NGS or Mass spectrometry
42. How many types of peptide May be injected in mass spectrometry chamber.. . _____ _____ three hundred thousand to four hundred thousand
43. …..can search peptide mass finger printing and shotgun proteomics dataset._____ MASCOT
44. Stabilizing and destabilizing energy give us ..... _____quality of 2nd structure
45. The Fasman algorithm is based on ..of Amino Acids in known structures._____ _____ _____ statistical occurrence
46. DP recombinant the nucleotide recombination through process of ..... _____Traceback
47. .....add proton to a protein and a peptide.._____ MALDI
48. Positively charged amino acids are..... _____**3**
49. Aromatic amino acid include ..... . _____**phenylalanine, tyrosine tryptophan**
50. Homology modelling fail to predict ....._____ **quality structure**
51. How many forces are involved in protein folding. _____**4**
52. Pairwise alignments tells the similarity between sequence... _____ _____ **by maximizing the matches.**
53. Dot plots employ dot matrix with two sequence plotted represents....._____ _____ **pairwise alignment and comparison.**
54. Differentiate b/w DNA and RNA sequence.....**RNA has more variety of sequence**
55. Cell molecule are produced after transformation of ..... _____**DNA to protein**
56. Dot plot cannot deal with..... _____**Insertion, deletion and gaps.**
57. Exons are may be more ..... _____**conserved**.
58. In dynamic algorithm we can do comparison ..... of more sequence. _____**Three**
59. Mass/Charge ratio helps calculate the mass of the .... _____**Protein**
60. In scale tree branches lengths are equal to the magnitude of change in the ..... ____**nodes**.
61. How many types of peptides mix in MS chamber..... _____**300,000 – 400,000**
62. mRNA is a ..... structure.___ _____**Planer**
63. When unpaired bases of 2´ stricture join to form..... _____**3´ structure**
64. N Jackobson use to pridict _____**2nd ´structure**
65. Aromatic amino acid ..... _____**Tyrosine**
66. Positive charge amino acids..... _____**Lysine, arginine, histidine**
67. MS begins with the measure of ..... _____**intact protein**
68. Amount of amino acids in alpha helix....._____(**4**)
69. Intact mass of protein can be found by..... _____**MS1**
70. MSA can be done by..... _____**CLUSTAL**
71. Which is used as sequence alignment tool..... _____**PROSIT**
72. MS1 and MS2 help to identify ..... _____**(PST)**
73. MGF is used for ..... _____**proteomics**

74. X,Y,Z positions of alpha carbon available online on ..... _____(**PDB**)
75. Method to determine RNA structure ..... _____**NMR, X-Ray crystallography**
76. CLUSTLA runs....._____ **Slow accurate/ fast appropriate**
77. Which give info about precursor protein..... _____**PSTS**
78. Raw data files format can be converted to ..... _____**Open format**
79. Which of the following is cartoonic figure....._____ **Ribbon diagram**
80. In CATH protein recognize according to their..... _____**structural similarity**
81. Method for obtaining 1´ structure....._____ _____ **(Edman Degradation Tandem Mass Spectrometry)**
82. In dot plot matrix, if one nucleotide with one but not in diagonal fashion called _____**Noise**
83. Dot plot employ dot matrix with two sequences.... _____**top and left matrix**
84. Match, mismatch and gap penalty is used to compute score from**.._____ each position**
85. **Functional domain** in proteins tend to be more conserved than the**_hydrophobic core**
86. Trackback in overlap matches, can start from any position.._____ **Yes**
87. FASTA the online software can perform genome and proteome similarity search..___ **_True**
88. Is thrionine(T) is a hydrophobic amino acid.._____ **No**
89. The main difference between fast and approximately and slow and accurate method in CLUSTULAW is that there is no need to mention is slow and accurate method....._____ _____ _____**Window size**
90. What does meant by MS/MS measured of molecular weight of protein.._____ _____ **Peptide will be analyzed two times after fragmentation**
91. ...reverse the chemical direction of RNA molecules.._____ **hairpin**
92. Mass of peptide search in...._____ **bottom up proteomics**
93. Amino acid are building block of..._____ **protein**
94. Aromatic amino acid include... _____ **Phenylalanice, Tyrosine and Tryptophan**
95. Protein are assembled at which organelles in a cell _____ **Ribosome**
96. Why do we create all possible combination of diagonals in Zukar.._____ _____ **to minimize stacking destabilizing energy**
97. Score is awarded on the basis of closeness B/W.._____**theoretical and experimental**
98. Which is not required in protein structure threading...__ **structure of entire backbone**
99. Chu fasman used to predict ....structure of protein_ _____**Secondary**
100. The structure classification of protein is carried out to _____ _____ _____**to check hierarchy of classification**
101. Multiple sequence alignment can be done through the following alignment software..._____**CLUSTAL**
102. Threading involves.. _____**Passing nucleotide sequence through each fold in database**
103. How many strategies for structural modeling_ _____**Three**
104. ...are sequence for peptide produce during MS2_ --_____ **PST**
105. Composite scoring scheme are needed to combine scores coming in from _____ _____ _____ **Multiple criteria**
106. ....can make the MS comprise of m/z ratio and intensities of each molecules that is measured..._____ **Data output**
107. ...coordinate of alpha carbon is protein backbone can be used for comparison_ _____**PDB**
108. The(x,y,z) position of alpha carbon atoms can be obtained from.. _____**PDB**
109. If we are given an amino acid sequence, then we can check the..table. _____ **Propensity**
110. ....are formed OH between Ci and Ni + 4.._____**Alpha helices**
111. Secondary structure of protein formed by H bond formation between N terminal and **C terminal** portion backbone
112. 1 angstrom is equal to __.. _____**0.1**
113. Structure is represented by secondary structures fold using simple ____**cartoon figures.**

114. . How many types of RNA present in a cell....._____5
115. Amino acid have unique structural properties ....._____protein folding
116. Alpha helix are formed by ....._____hydrogen bonding
117. Hydrogen atoms of N-Term may come together with **Oxygen** atoms
118. Atomic distances tell us about atomic interactions with neighboring _____atoms
119. Fold recognition called ....._____Threading technique
120. In structure visualization alpha helix carbon are traced to form ...._3Dprotein structure
121. Which is the largest amino acid ....._____Tryptophan
122. 3D-10 methods convert structure and environment information into "_____profiles"
123. Unique, stable & kinetically accessible minimum free energy_____ Anfinsen's dogma
124. Each component of structure protein has unique and precise role in construction ..._____hydrophobic and hydrophilic
125. Beta Sheets which are stabilized by _____H-bonds!
126. Using the masses reported by Ms1, protein databases can be filtered for _____ _____ candidate protein
127. PST stands ....._____Peptide sequence tags
128. Mass spectrometer is used to measure mass/charge ratio of **ionized** proteins and peptides
129. Mass Spectrometry proteomics identifies as well as determines the protein __**sequence**
130. The backbone of **tyrosine** strongly prefers two **rotamers**
131. Threading technique useful when homology modeling fail to predict___ quality structure
132. Bio informatics is _____new developing field

151-250

1. ----- provide mass of intact molecules and its fragments _____MS1 and MS2
2. theoratical data is usually ----- as compared to exp. Data, _____ larger
3. Three scoring schemes can be applied to ------_____score the match at each
4. In the above scoring method, MW and PST stabs for ____ & ___ respectively_____MS1 & MS2
5. _____ are assembled by Ribosomes in cellular cytoplasm _____Proteins
6. Amino acids have characteristics like polarity, hydrophobicity, and charge states. these characteristics are governed by _____Elemental composition of an amino acid's side chain (R group)
7. Hydrophobic amino acids are non-polar due to _____ _____very little dipole moments between H and C
8. Core of amino acids is _____ and surface is _____ Hydrophobic & hydrophilic
9. At pH ____ Five amino acids are charged, 2 negatively and 3 positively _____7
10. pK values for an amino acid is the pH at which: _____ _____ Exactly half of the chargeable group is charged
11. The pK of aspartic and glutamic acids are ___-& ____ respectively. _____3.9 & 4.2
12. If pH < pK for an amino acid, the amine side chains_____Gain proton and hence basic
13. If pH > pK for an amino acid, _____ loses a proton and hence acidic _____ _____The carboxyl side chain
14. Lysine and arginine have pK ____ &___ respectively. _____10.5 & 12.5
15. Glycine residues increase backbone flexibility because :_____-- They have no R group_____(only an H)
16. _____ reduce the flexibility of polypeptide chains. _____Proteins residues
17. ___ Cement together by making _____ to stabilize 3-D protein structures _____Cysteines, disulfide bonds
18. Chemically _____ remain in the core of protein._____ _____Inactive amino acids

19. Backbone is highly polar (hydrophilic) due to polar ___ and ___ in each peptide unit_____-NH & C=O
20. A-helices & B-sheets are stabilized by_____ H-bonds
21. The very few hydrophobic "patches" on protein surface are involved in: _____ _____Protein-protein interactions
22. Alpha Helix is Stabilized by H-bonds between every ____ in backbone _____~ 4th residue
23. Proteins fold spontaneously to achieve: _____Thermodynamic stability
24. If a protein is ____, then it can lead to a lack of function in the protein _____Misfolded
25. Given algorithms and procedures to fold a protein, we can: _____ _____Fold amino acid chains to form 3D proteins
26. Each amino acid can fold into____,__ & ____-Alpha Helices, Beta Sheets and Loops
27. If computations of protein folding take 1/10^{th} of a nano-second(10^{-10}), then to compute all the folding possibilities will take_____1.6 x 10^{30} years
28. All the information required for folding a protein into its native structure is present within the: _____-Protein's amino acid sequence
29. Primary sequence can also be referred to as ____ _____1' structure
30. Protein structures are organized into _____ modular conformations_____1', 2', 3' and 4'
31. Edman Degradation & Tandem Mass Spectrometry are methods for obtaining_1' structures
32. 2' structures are also referred to as _____---Secondary structures
33. C- Terminus is _____ charged _____Negatively
34. N-terminus is _____ charged _____Positively
35. 2' structures or secondary protein structures are formed as a result of H-Bond formation between ___ in a protein backbone _____N and C termini
36. _____ connect helices and sheets _____Loops
37. _____ are Secondary structures that are not helices, sheets, or recognizable turns____Coils
38. ____ are also secondary structure which form the first structures after folding of protein's amino acids _____Loops and Coils
39. Combinations of Alpha helices, Beta sheets, coils and loops help form _____ 3' structures
40. ___, ____ & _____ interactions enforce the 3' structure _____Covalent bonds, Hydrogen _____bonds and hydrophobic
41. The stability of all the structures of protein is as: _____4' > 3' > 2' > 1'
42. The protein folding into 3D structures leads to: _____Reduction in bond angles
43. Peptide bond between two amino acids is _____ _____Planar & rigid
44. The angle between the 1st, overlapped and the 4th points forms a _____ _Dihedral angle
45. ____ can be used to construct the backbone of a protein towards its visualization_____ _____C-Alphas
46. The _____ is used to express the size of atoms, molecules and extremely small biological structures, the lengths of chemical bonds, the arrangement of atoms in crystals. _____ _____Angstrom
47. Atoms of phosphorus, sulfur, and chlorine are _____ in covalent radius, while a hydrogen atom is _____ respectively. _____~1 Å & 0.25 Å
48. Crystallography data gives _____ of atomic coordinates _____Relative positions
49. ____ Proteins are used to determine protein structures. _____Crystallized
50. _____ contains protein structure information _____PDB
51. PDB has the coordinates of C-Alphas for over _____ proteins _____ 50,000
52. To view a protein, we need to evaluate the: _____physical location of its atoms
53. To trace the backbone of a protein, _____ trace can be used_____CA atoms
54. Coordinates of CA atoms can be obtained from the ____ _____PDB
55. Protein structures are visualized using several online tools. These tools include:_____ _____ Rasmol, CHIME, Swiss PDB Viewer and Cn3D.
56. CPK stands for: _____Corey-Paulin-Koltun Diagrams
57. In CPK diagrams, each atom is represented by a _____ _____Solid sphere

58. Spheres are equal to atomic _____ **Van der Waal radius**
59. Ribbon Diagrams represent the protein secondary structures by using: _____ _____**Simple cartoon figures**
60. Balls & Stick (BS) Models have atoms as _____ and intermediate bonds as _____ respectively _____**colored balls, sticks**
61. "Proteins fold for a unique, stable and minimum free kinetic energy structure". The given statement is called: _____**--Anfinsen's thermodynamic hypothesis**
62. There is _____ free energy accessible to each atom for further interactions. _____**Lesser**
63. The greater the number of bonds, the more ____ a protein becomes. _____**Stable**
64. The basic idea of thermodynamic stability is to _____ bonding in order to _____ the free energy _____**Maximize, minimize**
65. We can calculate energy of a folded protein based on the _____ of atomic interactions _____**number and types**
66. To determine the number of each type of interaction within a protein, we need to find its _____ **inter-atomic distances**
67. Based on specific atomic distances in protein, we can guess the _____ ___ _____**type of atomic interaction**
68. To determine the structure of protein, we use_____ _____ **X-Ray or NMR**
69. If two atoms are participating in a covalent bond, their distance is: _____**-~0.96A**
70. Incase of hydrogen bond formation between atoms, the inter-atomic distance is :_____ _____ **~1.97A**
71. X-Ray data should have a minimum of _____ resolution _____**1.97A**
72. **"**Helix Formers" are generally _____ amino acids (M, A, L…) _____**Hydrophobic**
73. Alpha Helices are formed by hydrogen bonding (O-H) between _____atoms in the protein backbone _____**Ci and Ni+4**
74. _____ resides are needed to make a Beta Strand _____**5 to 10**
75. _____is made of a single beta sheet that twists and coils upon itself. _____-**Beta-barrel**
76. Beta Sandwiches are made of two beta sheets which are usually _____so their strands are aligned _____**twisted and packed**
77. Loops are formed by amino acids present in the middle of _____ in a protein backbone_ _____ _____ **the Alpha Helices and Beta Sheets**
78. Variability in _____ allows loops to join Alpha Helices and Beta Sheets in a variety of ways._____**length and conformation**
79. Loops are mostly comprised of _____ amino acids _____**charged and polar**
80. Hairpin loops are two amino acids long and join _____ Beta strands _____ **anti-parallel**
81. Essentially, a secondary structure which is not a helix, sheet or loop, is a_____ **coil**
82. Coils are apparently_____ regions _____**disordered**
83. _____ are semi-independent functional structures in a protein _____**Domains**
84. Domains have over _____ residues _____**~40**
85. _____interact (H-bonds) more internally than externally **_Locally Compact – Domains**
86. Domains have a _____ core _____-- **hydrophobic**
87. Domains can be classified into structural classes. Classes can be further classified into _____**Architecture and Topologies**
88. _____ classifies proteins by their structural similarity _____**CATH**
89. FSSP stands for _____ based on the DALI algorithm _____ _____**Family of Structurally Similar Proteins**
90. SCOP stands for:– _____**Structural Classification of Proteins**
91. So for two different proteins, sharing the same domain, we may want to compare: _____ _____**only a portion of the overall structure i.e. a domain**
92. Whole protein structures can be compared by calculating the _____ between their Alpha Carbons positions _____**root mean squared difference (RMSD)**
93. The _____ the RMSD, the similar are the proteins _____**Lower**

94. Full protein structures can be compared and ranked by the overall _____ in positions between their Alpha Carbons_____ _____**differences**

95. ____ tells us about the quality of the matches _____**RMSD**

96. _____ mostly found in Alpha Helices _____**Alanine**

97. The first such algorithm to predict 2' given an amino acid sequence was the: _____ _____**Chou-Fasman Algorithm**

98. we only know _____ 3D protein structures, but 10 times more sequences_____ **100,000**

99. For a primary sequence, and a tentative 2' structure, _____ can help us compute the overall propensity___ _____**propensity table**

100. _ amino acids are needed to start an Alpha Helix and ___ amino acids for Beta Shee_____**4,5**

101. Loops are small amino acids _____**~ 3-4**

102. Alpha Helices are formed from 4 contiguous amino acids having an Alpha-Helix propensity over _____-__ **1.0**

103. Chou Fasman Algorithm helps predict _____**Alpha Helices, Beta Sheets and Turns**

104. Chou Fasman can be improved to better predict secondary structures by incorporating___ _____ __**biochemical factors**

105. Structure visualization, classification and prediction are important for: _____ _____ _____ ___**Understanding disease and designing drugs for treating them**

106. There are_____ salient steps in any Homology Modelling pipeline _____**seven**

107. The backbone of _____ strongly prefers two rotamers and the real side-chain may fit one of them _____**Tyrosine**

108. _____is a software for homology modelling _____ **Modeller**

109. A _____ is defined by the way the secondary structure elements of the structure are arranged relative to each other in space _____-**protein fold**

110. Common folds include _____ bundle and the TIM barrel. _____--**4-helix**

111. There are _____ stable folds in nature _____**5,000**

112. Fold recognition or Threading is a technique for predicting : _____**protein structures**

113. Threading involves _____ the amino acid sequence through each fold in the database _____**"passing"**

114. Software for automated protein structure &function prediction based on the sequence-to-structure-tofunction. _____**Iterative threading assembly refinement (I-TASSER) server**

115. _____ helps thread amino acid sequences on fold and secondary structure databases_____ **iTASSER**

116. Fold recognition or threading is useful in cases where _____ fails to predict quality structures ____ _____**homology modeling**

117. 3D-1D Bowie Algorithm was Proposed by _____ **Bowie et al in 1991**

118. 3D-1D Bowie Algorithm Converts____structure into a ____ string profile for each structure in the fold library _____**3D, 1-D**

119. 3D-1D Bowie Algorithm Identify amino acids based on: _____**protein core, side**

Q.1. In mass electrometer chamber protein is deflected according to its ……

a) charge b)velocity c)charge, mass d) volume

Q.2 in MS chamber mass/charge ratio can be measured by …… of protein molecules.

a) volume b) mass c) charge d) speed

Q.3 while scoring protein mass score is awarded on the base of ……. b/w experimental and theoretical mass.

a) Distance b) closeness c) equality d) all of these

Q.4 Tandem MS is used to …… of protein fragments.

a) Ionization b) fragmentation c) measure the mass d) sequence

Q.5 Fragmentation techniques split the molecules of protein into……..

a) Two part b) three part c) four part d) multiple part

Q.6 Tandem MS is typically called as,

a) MS b) MS1 c) MS2 d) ALL

Q.7 Each fragmentation technique differ in their work on bases of ,

a) Having different splitting sites

Q.8 Mass spectrometer use o measure the mass of protein and ……

A) Amino acids b) lipids c) peptides d) DNA

Q.9 Protien ionization is the …… step in MS based proteomics.

a) first b) second c) third d) fourth

Q.10 Ionization process involve gaining and removal of …….

a) electron b) proton c) ion d) atom

Q.11 molecular weight increases by ……. Of proton

a) removing b) adding c) transfer d) All of these

Q.12 We can score each protein hit in away that the better match gets a ……

a) higher score b) lower score c) equal score d) opposite score

Q.13 ECD gives out "C" and …….. ions,

a) B b) Y c) Z d) A

Q.14 CID gives out "B" and …….. ions,

a) Y b) Z c) B d) C

Q.15 PST is abbreviation of…..

a) Protein sequence tags b) protein sequence tools

c) Peptide sequence tags d) peptide sequence tools

Q.16 PST is sequence of peptidesproduced during…..

a) MS b) MS1 c) MS2 d) ALL

Q.17 Fragmentation at consecutive sits lead to a mass difference equal to that of a single …..

a) peptide b) amino acid c) protein d) DNA

Q.18 The data base protein which have matching PSTs to invitro protein are known as ……

a) Sample protein b) candidate protein c) data base protein d) PST protein

Q.19 Experimental peptides also known as ……

a) data base peptide b) observational peptide

c) invitro peptide d) sample peptide

Q.20 Errors in peak matching also considered by…….

a) Sequence b) fragmentation c) ionization d) scoring scheme

Q.21 Which one is the right sequence of steps to create a composite protein search engine.

a) PST – MW – In silico comparison b) MW – PST – In silico comparison

c) In silico comparison – MW – PST c) MW – in silico comparison – PST

Q. 22 We can´t add these three scoring comparison until we divide them by their……

a) length b) error c) numbers d) distance

Q.23 Typtic peptides may reach upto an order of …….

a) 200,000 – 30,000 b) 30,00 – 40,00 c) 300,000 – 400,000 d) 400,00 – 500,00

Q.24 large scale proteomics is a developed form of

a) mass spectrometry b) BUP technique c) TDP technique d) uniprot technique

Q.25 Mass spectrometry use to measure ……. Ratio of ionized protein or peptide.

a) mass/distance b) charge/distance c) mass/charge d) distance/charge

Q.26 Raw file in a format are often found in ……. That we have to change in an open format so as to make it readable.

a) Theoretical form b) decimal form c) percentile form d) binary form

Q.27 mzData was developed by ……

a) mzML b) HUPO-PSI c) RAW format d) MGF

Q.28 MASCOT is an online ……. Search engine,

a) Bottom up proteomic b) top down proteomic c) both d) none

Q.29 …….. is a simple human readable format for MS/MS developed by Matrix Science.

a) MASCO b) ABI c) MGF d) RAW format

Q.30 theoretical vs experimental fragments comparison helps us as the …….. Stage for short listing candidate protein list.

a) 1st b) 2nd c) 3rd d) 4rth

Q.31 To ensure that all fragments of precursor molecules are also charged, we can use,

a) ESI b) RAW c) BOT d) NONE

Q.32 Monoisotopic mass is the mass of ……..

a) single isotope b) average isotope c) least abundant isotope d) most abundant isotope

Q. 33 Proteins are assembled by ..... in cellular cytoplasm

a.golgi apparatus b.mitochndria **c.ribosomes** d.nucleaus **Q.**

34.Each protein is synthesized by polymerization of .....

a.peptide bond **b.aminoacids** c.nucleotide sequance d.all

Q.35.The characteristics are governed by the elemental composition of an amino acid's .....

a.alpha carbon b.H group **c. R group** d.amino group

Q.36 Since ....and ..... introduce very little dipole moments in hydrophobic amino acids, these amino acids are non polar

a.H &C b.H & Y c.Y & H d.A & Z

Q. 37 Mostly ..... aminoacids found at the surface of folded proteins

a.polar b.nonpolar c.acidic d.basic

Q.38. At pH=7, five amino acids are charged, ... negatively and ..... positively

a.3 & 2 b.3 & 3 **c.2 & 3** d.2 & 2

Q. 39.The charge state depends on the acceptance or donation of ..... from the R group side chains

a.electron **b.proton** c.neutron d.both a & b **Q. 40.**

Amino acids have property such as .....

a.charge state b.polarity c.hydrophobicity **d.all**

Q. 41 Tendem MS helps to measure the moleculer weight of ……

a) isotops b) precursor protein c) ionized fragments d) invitro protien

Q. 42.If pH >pK for aminoacids carboxyle side losses the proton and bocome negative which is

.....

a.basic b.neutral c.less acidic **d.acidic**

Q. 43. ..... are key factors participating in protein folding

a.solublity b.reactivity **c.both** d.none

Q. 44.The charge state depends on the acceptance or donation of ..... from the R group side chains

a.electron **b.proton** c.neutron d.both a & b

Q. 45. Amno acid with the side chain containing organic acids are soluble in …….

a) ethanol b) water c) lipids d) a and c

Q. 46 …….. residue has no R group.

a) cystine b) proline c) glycine d) adinine

Q. 47 proline residue …… the flexibility of polypeptide chain.

a) increase b) decrease c) stabilize d) no effect

Q. 48 A peptide unit has a backbone of polar NH and C=O which is highly……..

a) hydrophilic b) hydrophobic c) neutral d) acidic

Q.49 Functional group of protein always attach to its hydrophobic ……..

a) surface b) coat region c) core region d) backbone

Q. 50 After polymerization ……. are formed.

a) 3D structure b) alpha helix c) beta sheets d) All of these

Q. 51 The force involve in protein folding ……

a) electrostatic interactions b) van der waals interactions c) hydrogen bonding d)all

Q. 52 ……folded form of protein is thermodynamically more stable than others.

a) Combined b) native c) partially d) a and b

Q. 53 3´ structural can combine with other polypeptide units to form …….

a) secondary structure b) tertiary structure c) quaternary structure d) all

Q. 54 There is a …… variety of protein structure exist.

a) rare b) restrict c) limited d) huge

Q. 55 Folded sub structures are called …….

a) primary structure b) secondary structure c) tertiary structure d) All

Q. 56 The protein super structure are called ……

a) primary structure b) secondary structure c) tertiary structure d) All

Q. 57 Lysine and tryptophan have respectively …….. 1-letter codes.

a) L – T b) S – F c) K – W d) N - A

Q. 58 C- terminus is …… charge.

a) negatively b) positively c) neutral d) none

Q. 59 …… bonds are the reason of 2' structure formation.

a) covalent b) hydrogen c) ionic d) All

Q. 60 Loops and Coils are very important 2' structures in that they form …… of proteins.

a) active site b) inactive site c) acidic site d) basic site

Q. 61 …….. Enforce the 3' structure.

a) Covalent bonds b) Hydrogen bonds c) hydrophobic interaction d) all

Q. 62 Proteins with ……. chain can form 4' structures.

a) single b) double c) linear d) multiple

Q.62 Dihedral Angles always form between ……

a) two points b) three points c) four points d) five points

Q.63 ……. angles are taken up as a result of folding of protein.

a) phi b) psi c) both d) none

Q .64 …… can be used to construct the backbone of a protein towards its visualization.

a) N b) C c) C Beta d) C Alpha

Q. 65 Atoms of phosphorus,sulfur, and chlorine are …… in covalent radius.

a) ~1 Å b) ~2 Å c) ~3 Å d) ~4 Å

Q. 66 The angstrom is used to express ……

a) small biological structures b) lengths of chemical bonds,
c) arrangement of atoms in crystals. d) All

Q. 67 Distances between C-Alphas are measured in the unit "Angstrom" by the process of ……..

a) adman degradation b) X Ray crystallography c) Blossom matrix d) PAM Matrix

Q. 68 1 A resolution is better than ……..

a) 9 A b) 8 A c) 7A d) 10 A

Q. 67 PDB contains protein …… information

a) structure b) size c) distance d) function

Q. 68 PyMOL and Cn3D is a python-script based tool for visualizing the ……..

a) protein size b) protein structure c) amino acids size d) a and b

Q.69 BS Models have atoms as colored balls and intermediate bonds as ……..

a) straw b) spindle c) stick d) small line

Q. 70 In CPK diagrams, each atom is represented by a ……..

a) dot sign b) Solid Ball c) doted circle d) solid sphere

Q.71 According to "Anfinsen's thermodynamic hypothesis" protein folds for making a ……..

a) stable structure b) structure having minimum free energy c) structure having maximum free energy d) a and b

Q.72 Hence, the basic idea of thermodynamic stability is to …….. bonding in order to …… the free energy.

a) minimizing, maximizing b) maximizing, minimizing c) maximizing, stable d) b and c

Q.73 **To determine the number of each type of interaction within a protein, we need to find its ……**

a) type of amino acid b) types of bonds c) inter-atomic distances d) atomic mass

Q.74 By looking up at the bond/energy table, we can compute the …….

a) overall energy b) molecular mass c) atomic mass d) bond sequence **Q.75 If two atoms are participating in a covalent bond, their distance is ……**

a) ~0.95 A b) ~0.96A c) ~0.97 A d) ~0.98 A

Q.76 "Helix Formers" are generally…….. amino acids (M, A, L…)

a) acidic b) basic c) hydrophilic d) hydrophobic

Q.78 Alpha Helices are formed by hydrogen bonding (O-H)between …… and ……. atoms in the protein backbone

a) C – H b) H – N c) C – N d) P - N

Q.79 …… resides are needed to make a Beta Strand,typically

a) 3 – 4 b) 5 – 10 c) 6 – 7 d) 5 – 6

Q.80 Loops are the …… of secondary structure

a) 1st b) 2nd c) 3rd d) 4rth

Q.81 what is made by a single beta sheet when it twists and coils upon itself.

a) Beta Barrel b) beta sandwich c) loops d) coils

Q.82 ……. are flexible & can adopt multiple conformations.

a) coils b) helices c) loops d) All

Q.83 …… are apparently disordered regions with random structure and random length.

a) coils b) helices c) loops d) beta sheets

Q.84 …… are Non-functional Combinations of 2' structures)

a) loops b) coils c) beta sheets d) motifs

Q.85 Domains are semi-independent functional structures in a protein having over ……..

a) ~30 residues b) ~40 residues c) ~50 residues d) ~60 residues

Q. 86 The structural classes of Domains can be further classified into ……

a) Architecture b) Topologies c) both d) none **Q. 87**

Architecture also called ……..

a) domain b) motifs c) coil d) fold

Q.88 …….. is one such system in which proteins are organized into classes, architecture, topology & homology.

a) CATH b) FSSP c) LOCK d) DOLI

Q.89 By comparing …….. can help us evaluate if the proteins do similar function.

a) sequence b) free energy c) structure d) All

Q.90 For comparing the complete or partial protein structures, the position of ……. can be used.

a) N terminal b) C terminal c) Alpha Carbon d) functional group

Q.91 Whole protein structures can be compared by calculating the ……..difference b/w their Alpha Carbons positions.

a) distance b) root mean squared c) energy d) All

Q.92 The ……. the RMSD, the similar are the proteins.

a) Equal b) higher c) lower d) random

Q.93 RMSD tells us about the quality of the matches

a) matches b) mismatches c) sequence d) bonding

Q.94 There are over a million protein sequences but only about …….. protein structures.

a) 10,000 b) 100,000 c) 10,0000 d) 1,000

Q.95 if we have several …… in the sequence, then we can anticipate that a helix may be formed by them.

a) Proline b) Thiamin c) Alanines d) Adenine

Q.96 Product of propensity values is computed for ……. for each 2' structure.

a) individual propensity b) overall propensity c) greatest propensity d) All

Q.97 For Alpha Helices ….. contiguous amino acids are required.

a) 6 b) 5 c) 8 d) 4

Q. 98 Alpha-Helix propensity should be more than ……. Once this propensity falls, AlphaHelix stops.

a) 1.0 b) 1.5 c) 2.0 d) 2.5

Q. 99 …… amino acids are needed to start an Alpha Helix and ……. amino acids for Beta Sheet.

a) 3 , 4 b) 4 , 5 c) 5 , 6 d) 4 , 6

Q. 100 Regions where overlapping alpha-helices and betasheets occur are declared helices.

a) loops b) coil c) helix d) helices

Q.101 In chou fasman algorithm Alpha Helices can be finalized if their propensity is ……. to the propensity for Beta Sheets.

a) equal b) less c) higher d) different

Q.102 Chou Fasman Algorithm helps us to predict,

a) Alpha Helices b) Beta Sheets c) Turns d) All

Q.103 Beta sheets can be predicted from …… amino acid sequences

a) primary b) secondary c) tertiary d) quaternary

Q.104Chou Fasman Algorithm is based on statistical occurrence of ……. in known structures.

a) protein b) Amino Acids c) nucleotide d) a and b

Q. 105 Functional evaluation of proteins can be performed by……

a) Structure visualization b) , classification c) prediction d) All

Q.106 Functional evaluation of proteins is important for understanding …….

a) disease b) designing drugs c) treatment d) ALL

Q.107 X-Ray Crystallography and NMR Spectroscopy are used to find the structures of ……..

a) DNA b) Amino acid c) proteins d) nucleotides

Q.108it is then possible to identify unknown protein structures by just examining the …….. protein sequences

a) heterogeneous b) homologous c) multiple d) binary

Q.109In homology modelling, proteins with similar …….. are considered.

a) 1' sequences b) 2' sequences c) 3' sequences d) 4' sequences

Q.110Homology modelling is used to predict structures of proteins having high sequence similarity with other proteins with ……..!

a) known structures b) unknown sequence c) similar sequence d) a and c

Q111 Protein structure can be predicted by…… methods: a).

Homology Modelling b). Threading/Fold Recognition

c). Ab Initio Modelling d). All

Q.112 in homo logy modeling,Definition of Template (……..) & Target (……)

a) known – unknown b) unknown - known c) known – similar d) unknown - similar

Q.113 One way to handle loops is to take some residues before and after the insertion as …….. residues.

a) sample b) predicted c) anchor d) similar

Q.114 The backbone of tyrosine strongly prefers ……. rotamers & the real side-chain may fit one of them.

a) 1 b)2 c)3 d)4

Q.115 In homo logy modeling when protein backbone is ready,then next, ……. were modelled and used to bridge gaps.

a) α helices b) β sheets c) coils d) loops

Q.116 Loop modeling is ……step of Homology modeling.

a) 2nd b) 3rd c) 4th d) 5th

Q.117 Homology Modelling …….. study conformational changes.

a) can b) cannot c) may be d) some times can

Q.188 We can overcome the limitations of homology modeling by ……..

a) Threading b) Bowie Algorithm c) Ab Initio modeling d) a and c

Q.119 Q……. is a software for homology modelling.

a) Homolog b) Modeller c) iTASSER d) a and b

Q.120 There are …….. stable folds of protein in nature.

a) 30,000 b) 4,000 c) 40,000 d) 5,000

Q.121 Several tools are available to perform homology modelling in a programmatic or automated way such as…..

a) Swiss Model b) Robetta c) 3D Jigsaw d)all

Q.122 Fold recognition is also called …….

a) scoring sequencing c) Threading d) modeling

Q.123 Homology modelling fails to predict quality structures ……. is a technique for predicting these structures.

a) Fold recognition b) Swiss Model c) Threading d) a and c

Q.124 Threading Scoring typically involves using a …….. function based on energy of a molecule

a) Z-Score b) C-score c) A-score d) Y-score

Q.125 …… cannot be predicted using threading.

a) invtro protein b) primary protein c) 2ndary protein d) Novel proteins

Q.126 3D-1D Bowie Algorithm Proposed by Bowie et al in ……

A) 1997 B) 1991 C) 1995 D) 1998

Q.127 3D-1D Identify amino acids based on ……..

a) protein core b) side chain positioning c) solubility d) All

Q.128 3D-1D methods convert structure and environment information into ……..

a) software b) packets c) profiles d) subunits

Q.129 Ab initio modeling is Computationally ……. Suitable for proteins with less than …… residues

a) cheap – 500 b) expensive – 100 c) cheap – 100 d) expensive - 500

Q.130 Ab Initio Modelling methods attempt to identify the structure with minimum …….

a) charge ratio b) distance c) molecular weight d) free energy

Q.131 Energy released during the …….. process is computed for predicting structure.

a) Sequencing b) folding c) breakdown d) bonding

Q.132 Ab initio methods base their predictions on ……. models of folding mechanisms

a) Chemical b) statistical c) physical d) all

Q.133 Global optimization finds the most …….

a) matching sequence b) stable structure c) similar pattern d) a and c

Q.134 ……structure not always at the global minimum.

a) Native b) invitro c) insilico d) stable

Q.135 Ab initio methods are the very …….. to design.

a) easy b) difficult c) suitable d) accessible

Q.136 In homology modeling first we perform ……. and then ……..

a) Loop modeling - Backbone generation b) Backbone generation - Loop modeling

Q.137 There are ……. Steps of Homology Modelling.

A) 5 b) 6 c) 7 d) 8

Q.138 Ab initio modeling is Computationally ……. And …… process.

a) easy and expensive b) Expensive and complex c) cheap and easy
d) difficult and cheap

Q.139 Ab initio methods predict protein structures by folding proteins based on each constituent atom's ……..

a) volume b) charge c) mass d) all

Q.140 UPGMA is a …… method.

a) clustering b) non clustering c) both

Q.141 Homology modelling is performed in cases of …… identity and alignment score • For the "Twilight zone", other strategies are employed

a) high b) low c) equal d) overall

Q.142 Maximum Parsimony etc are……. methods (not included in this course).

a) clustring b) nonclustering c) mapping d) visualizing

Q.143 Online tools: ……..

a) mascot b) sequest c) prosight PC d) all

Q.144 Number of known protein sequences is …….. as compared to known proteins structures

a) larger b) much larger c) smaller d) mech smaller

MUHAMMAD IMRAN	BIF401-Bioinformatics-I
PAST PAPER SOLVED	MID TERM

Rooted and Un-rooted phylogentic tree?

Rooted and Unrooted trees can be used to show phylogenetic relationships between sequences • Several types of algorithms exist which are divided into two classes.

Purpose of bowie algorithm?

Fold Recognition/Threading Online Tools for Fold Recognition GOR Algorithm Homology Modelling 3D-1D Bowie Algorithm Machine Learning Approaches to Structure Prediction Neural Networks for Structure Prediction PSIPRED Introduction to Hidden Markov Models Ab initio modeling. **(PPT)**

Complete protein sequence? F-x(3)-X-R-F-K-X (4-5) –D-E-R

FXXXXRFKXXXXXDER (not sure)

Enlist salient Feature of block diagram of mass spectrum?

A mass spectrum is a plot of the ion signal as a function of the mass-to-charge ratio. These spectra are used to determine the elemental or isotopic signature of a sample, the masses of particles and of molecules, and to elucidate the chemical structures of molecules and other chemical compounds.

What does uni port sequence give information about protein query?

Types and Function of RNA?

Three major types of RNA are mRNA, or messenger RNA, that serve as temporary copies of the information found in DNA; rRNA, or ribosomal RNA, that serve as structural components of proteinmaking structures known as ribosomes; and finally,tRNA, or transfer RNA, that ferry amino acids to the ribosome to be assembled.

Protein folding?

Protein folding is the physical process by which a protein chain acquires its native 3-dimensional structure, a conformation that is usually biologically functional, in an expeditious and reproducible manner.

Chou Fasman Algorithm?

The Chou–Fasman method is an empirical technique for the prediction of secondary structures in proteins, originally developed in the 1970s by Peter Y. Chou and Gerald D. Fasman.

Protein sequence and database name?

Protein sequencing is the practical process of determining the amino acid sequence of all or part of a protein or peptide. The Protein Data Bank (PDB) is a database for the three-dimensional structural data of large biological molecules, such as proteins and nucleic acids. ... The PDB is overseen by an organization called the Worldwide Protein Data Bank, wwPDB.

pH less than pK?

If the pH is less than the pK_a, then the acid form of the compound predominates. If the pH is greater than the pK_a, then the conjugate base predominates. This is shown graphically here: Most ionizable groups fallinto two patterns depending on the charge found on the acidic form.

Difference between RNA and DNA with ref. to nucleotide?

Nucleotide[edit] A nucleotide is composed of a nucleobase (nitrogenous base), a five-carbon sugar (either ribose or 2'-deoxyribose), and one to three phosphate groups. ... In DNA, the purine bases are adenine and guanine, while the pyrimidines are thymine and cytosine. RNA uses uracil in place of thymine.

Redundancy codon, name an amino acids which coded for three different codon?

Phylogenetic tree types?

Scaled Trees: Branch lengths are equal to the magnitude of change in the nodes.
Unscaled Trees: Only representing the relationship between sequences.

Purpose of uni port and their feature?

UniProt is a freely accessible database of protein sequence and functional information, many entries being derived from genome sequencing projects. It contains a large amount of information about the biological function of proteins derived from the research literature.

What do you know about rammananchandrah plot?

Ramachandran Plot is a way to visualize dihedral angles ψ against φ of amino acid residues in protein structure.Ramachandran recognized that many combinations of angles in a polypeptide chain are forbidden because of steric collisions between atoms.

Blastx and tblastx work?

Blastx: Search protein database using translated nucleotide query.
Tblastx: Search translated nucleotide database using translated nucleotide query.

How we can modify dot plot to consider and mis match in alignment?

In the Alignment View locate the first region of mismatch in your alignment. Turn on the ... Youshould realign these sequences with modified settings. Rather than ... In the Dotplot there is also an inverted region (you will need to activate the. Reverse ... sequences and do you think it is a fair representation of their sequence.

Why do we need to use dynamic programming in predicting RNA structure?

RNA sequences comprises of 4 types of nucleotides • G/C, G/U & A/U are complementary & can form H-bonds • RNA sequences may contain hundreds of nucleotides, hence many combinations are possible. • So, given an RNA sequence, there is a large number of possible 2' structures • This presents us with an extremely complex and large problem

Few challenges in field of biotechnology?

Bioinformatics is full of challenges and opportunities. • Amongst other frontiers in bioinformatics, there are protein structures, systems biology and personalized medicine!

Step to use FASTA?

Step 1: Specify the tool input (sequence and database).
Step 2: Entering of input sequence.
Step 3: Set up the parameters.
Step 4: Submit the query for processing.

Difference between local and global alignment?

Global alignment - maximizes the number of matches between the query and source sequences along the entire length of both the sequences.
Local alignment - gives the highest scoring local match between both query and sequences

Domain of protein?

Domains are semi-independent functional structures in a protein • Protein may contain multiple domains • Hence, we can try to classify proteins by their domains.
Examples of Protein Domains 1. Alpha Domains 2. Beta Domains 3. Alpha/Beta Domains 4. Alpha + Beta Domains 5. Alpha & Beta Multi-Domains 6. Membrane & cell-surface proteins

Uni port three main functions?

Scoring matrix in detail?

Scoring Matrix. Scoring matrices are used to determine the relative score made by matching two characters in a sequence alignment. ... There are many flavors of scoring matrices for amino acid sequences, nucleotide sequences, and codon sequences, and each is derived from the alignment of "known" homologous sequences.

PAM matrix in detail?

Alignment scoring matrices are very useful in giving suitable scores to matches and mismatches • There are 2 types of scoring matrices i.e. PAM and BLOSUM • PAM means "Point Accepted Mutations" • Point accepted

mutation is a substitution of one amino acid by another such that the protein functions stays conserved. PAM unit is a time in which about 1% of amino acids in a sequence undergo accepted mutations.

Bioinformatics Promise?
???????????

Draw the structure of RNA and DNA?

Newick notation example?

?????????
??
Complete protein sequence formula 3 role of 5'cap and 3'cap?
???????????

Dot plot matrix?
Dot plots employ dot matrix with two sequences plotted on top & left of the matrix • Matches are represented by dots • Dots on diagonals are connected and represent alignments.

Formula of sequence?
?????????

Write field of bioinformatics?

Bioinformatics is an interdisciplinary field that develops methods and software tools for understanding biological data. As an interdisciplinaryfield of science, bioinformatics combines computer science, statistics, mathematics, and engineering to analyze and interpret biological data.

How bulge formed?

Bulges, are formed when a double-stranded region cannot form base pairs perfectly. • Bulges can be asymmetric with varying number of base pairs on one side of the loop.

Why need dynamic programming?

Dynamic programming is both a mathematical optimization method and a computer programmingmethod. ... Likewise, in computer science, if a problem can be solved optimally by breaking it into sub-problems and then recursively finding the optimal solutions to the sub-problems, then it is said to haveoptimal substructure.

Why diagonal form energy matrix of zuker algorithm?
?????????
?
Find out the mis match match and gap final score?
????????
?
Code of protein sequence ATCATCCATAC?
???????????

Basic principle for progressive alignment for MSA?

Progressive alignment methods are efficient enough to implement on a large scale for many (100s to 1000s) sequences. ... They recommend Clustal Omega which performs based on seeded guide trees and HMM profile-profile techniques for protein alignments. They offer different MSA tools forprogressive DNA alignments.

How pseudo knot effect RNA?

The pseudoknot is a potentially important tertiary structural motif of RNA.

How many type of data base in bioinformatics and What type of information they contain?
These formats include text, sequence data, protein structure and links. Each of these can be found from certain sources, for example: Text formats are provided by PubMed and OMIM. Sequence data is provided by GenBank, in terms of DNA, and UniProt, in terms of protein.

Application of bioinformatics?

Genomics • Transcriptomics • Proteomics • Metabolomics • Structural Proteomics • Drug Design • Systems Biology •
Personalized Medicine

Difference between rooted and un-rooted tree?

Rooted trees shows the most basal ancestor of the tree.Unrooted phylogenetic tree does not show an ancestralroot. ... Unrooted trees represents the branching order but do not indicate the root or location of the last common ancestor. Unrooted trees shows the relatedness oforganisms without indicating ancestry.

Define mutation?

the changing of the structure of a gene, resulting in a variant form which may be transmitted to subsequent generations, caused by the alteration of single base units in DNA, or the deletion, insertion, or rearrangement of larger sections of genes or chromosomes.

Purine and Pyramidne Bases?

Purines and Pyrimidines are nitrogenous bases that make up the two different kinds of nucleotide bases in DNA and RNA. The two-carbon nitrogen ring bases (adenine and guanine) are purines, while the onecarbon nitrogen ringbases (thymine and cytosine) are pyrimidines.

Diffrence acidic and basic amino acids?

Acidic amino acids have acidic side chains at neutral pH while basic amino acids have basic side chains at neutral pH. carboxylic acid is the side chain foracidic amino acids and basic amino acids contain nitrogen containing groups. ... Lysine, arginine and histidine are basic amino acids.

ORF and FASTA stand for?

ORF (Open Reading Frame)
FASTA (Fast Alignment)

Conserved sequence?

Conserved sequence: A basesequence in a DNA molecule (or an amino acid sequence in a protein) that has remained essentially unchanged, and so has been conserved, throughout evolution.

BLAST function?

BLAST (basic local alignment search tool) is an algorithm for comparing primary biological sequence information, such as the amino-acid sequences of proteins or the nucleotides of DNA and/or RNA sequences.

BLOSUM steps?

- 1 Collecting sample blocks. ...
- 2 Computing probabilities. ...
- 3 Computing the BLOSUM matrix.

Way of constructing phylogenetic tree?

Several methods exist for constructing phylogenetic trees. Broadly, they belong to objective methods or clustering methods. • We will study UPGMA and Distance Methods.

Genetic mutation?

Genetic mutation is the basis of species diversity among beetles, or any other organism. Mutationsare changes in the genetic sequence, and they are a main cause of diversity among organisms. These changes occur at many different levels, and they can have widely differing consequences.

Dot plot benefits?

Data points may be labelled if there are few of them.Dot plots are one of the simplest statistical plots, and are suitable for small to moderate sized data sets. They are useful for highlighting clusters and gaps, as well as outliers. Their other advantage is the conservation of numerical information.

Clustring method?

The 5 Clustering Algorithms Data Scientists Need to Know. Clustering is a Machine learning technique that involves the grouping of data points. ... Clustering is a method of unsupervised learning and is a common technique for statistical data analysis used in many fields.

Protein sequence contain?

Protein sequences contain A, R, N, D, C, E, Q, G, H, I, L, K, M, F, P, S, T, W, Y & P

Gibbs Free energy?

A thermodynamic quantity equal to the enthalpy (of a system or process) minus the product of the entropy and the absolute temperature.

Categories of RNA?

- mRNA - Messenger RNA: Encodes amino acid sequence of a polypeptide.
- tRNA - Transfer RNA: Brings amino acids to ribosomes during translation.
- rRNA - Ribosomal RNA: With ribosomal proteins, makes up the ribosomes, the organelles that translate the mRNA.

Diffrence between Blast and FASTA?

BLAST is the most widely used tool for the local alignment of nucleotide and amino acid sequences. FASTA is a fine similarity searching tool which uses sequence patterns or words.

Why RNA is less stable than DNA?

Unlike DNA, RNA in biological cells is predominantly a single-stranded molecule. While DNA contains deoxyribose, RNA contains ribose, characterised by the presence of the 2'-hydroxyl group on the pentose ring. This hydroxyl group make RNA less stable than DNA because it is more susceptible to hydrolysis.

Domian of protein?

A protein domain is a conserved part of a given protein sequence and (tertiary) structure that can evolve, function, and exist independently of the rest of the protein chain. Each domain forms a compact threedimensional structure and often can be independently stable and folded.

Step of compute PAM?

Steps to compute PAM matrices

Step 1: Align proteins sequences which are 1-PAM unit diverged

Step 2: Let Ai,j be the number of times Ai is substituted by Aj

Step 3: Compute the frequency fi of amino acid Ai Then, PAM1 = pij = PAM'n' = (PAM1) n

Step of FASTA algorithm?

FASTA can search sequence databases and identify unknown sequences by comparing them to the known sequence databases. • This can help obtain information on the parent organism, function and evolutionary history.

Gibbs energy what role play in RNA folding?

Gibbs Free Energy" is the free energy of an RNA molecule available for reaction. The smaller, the better!. RNA structure formation lowers the free energy.

To quantify the similarity achieved by an alignment, scoring matrices are used: they contain a value for each possible substitution, and the alignment score is the sum of the matrix's entries for each aligned amino acid pair.

How biology simulation benefits to society?

- Genomics:
- Evolutionary Studies: Systems Biology:

Conclusion: Bioinformatics not only organizes, stores and analyzes biological data, but can also validate novel hypotheses. Modern day bioinformatics also helps predict disease outcomes as well as drugs to treat them!

Mathematical relationship between phlyogenetic tree?

A phylogenetic tree is a diagram that representsevolutionary relationships among organisms. ... Intrees, two species are more related if they have a more recent common ancestor and less related if they have a less recent common ancestor.Phylogenetic trees can be drawn in various equivalent styles.

How hydroxyl group of RNA make it unstable?

This hydroxyl group make RNA less stable than DNA because it is more susceptible to hydrolysis. RNA contains the unmethylated form of the base thymine called uracil (U), which gives the nucleotide uridine.

Given sequence find scoring matrix?

Wobble hypothesis?

The Wobble Hypothesis explains why multiple codons can code for a single amino acid. One tRNA molecule (with one amino acid attached) can recognise and bind to more than one codon, due to the lessprecise base pairs that can arise between the 3rd base of the codon and the base at the 1st position on the anticodon.

Which tool or databases use for nucleotide sequence?

Sequence Translation is used to translate nucleic acid sequence to corresponding peptide sequences. Backtranslation is used to predict the possiblenucleic acid sequence that a specified peptide sequence has originated from.
NUCLEOTIDE DATABASE – contains sequencedata from GenBank, EMBL, DDBJ as well as from the Genome Sequence Database and the US Patent and Trademark Office. It includes STSs and ESTs.

Sequence alignment by using matrix?

Seconday structure of RNA?

The secondary structures of biological DNA's and RNA's tend to be different: biological DNA mostly exists as fully base paired double helices, while biological RNA is single stranded and often forms complex and intricate base-pairing interactions due to its increased ability to form hydrogen bonds stemming from the ...

Enlist the field of Bioinformatics that can be explored under expasy?

Developed by Bioinformatics Institute (SIB) • Website provides access to databases and tools • Proteomics, Genomics, Phylogeny, Systems biology, Population genetics, transcriptomics etc.
Expasy provides access to a variety of online databases and tools. • Depending upon your requirement, you find sequence information from Expasy.

How pesudoknot alter RNA?

A 1’ RNA structure can fold into 2’ structure • 2’ structures can then form 3’ structures but avoid pseudoknots!

Why uni port query is used?

Batch search with UniProt IDs or convert them to another type of database ID (or vice versa) ... Find sequences that exactly match a query peptide sequence ...

Protien sequence?

Protein sequencing is the practical process of determining the amino acid sequence of all or part of a protein or peptide.

Matrix formula?

A determinant is just a special number that is used to describe matrices and finding solutions to systems of linear equations. The formula for calculating a determinant differs according to the size of thematrix. For example, a 2×2 matrix, the formula is ad-bc.

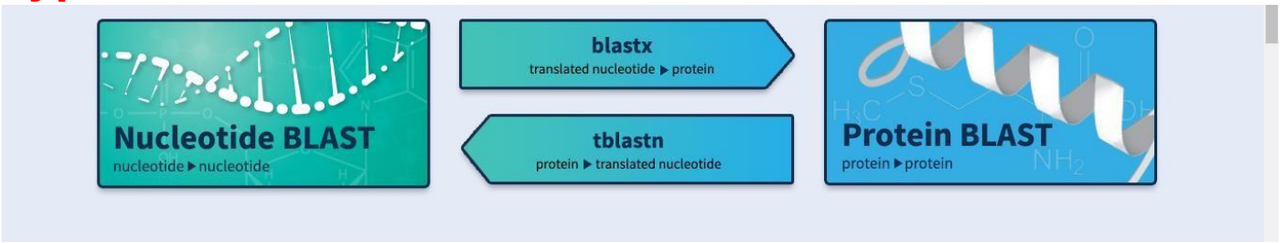
How tertiary structure of protein is less stable than secondary structure?

All proteins have primary, secondary and tertiary structures but quaternarystructures only arise when a protein is made up of two or more polypeptide chains. ... Primary structure is when amino acids are linked together by peptide bonds to form polypeptide chains.

Uses of CLUSTULW?

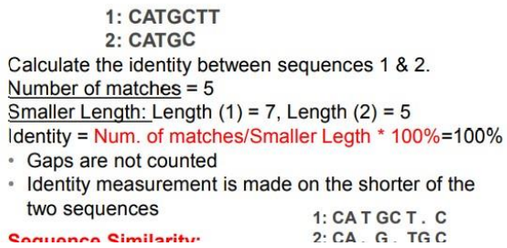
ClustalW is a widely used system for aligning any number of homologous nucleotide or protein sequences. For multi-sequence alignments, ClustalW uses progressive alignment methods.

Types of Blast and Uses?



Identity and Similarty and the formula of identity alignemts?

• Identity is the count of exact matches between two sequences. • Gaps are excluded • Similarity is the comparison between sequences calculated by using alignment approach.



4 Scoring possibility of Nussinov and Jacob algorthem?

4 positions are considered to calculate the NJ scoring matrix
• At every step, we can see score contributions from the 4 possible locations

Scope of bioinofrmatics?

Use of computational algorithms and techniques for: 1. Storage, 2. Organization, 3. Analysis, and 4. Representation of biological information

Why use castalw?

ClustalW is a widely used system for aligning any number of homologous nucleotide or protein sequences. For multi-sequence alignments, ClustalW uses progressive alignment methods. ... These scores are computed using the pairwise alignment parameters for DNA and protein sequences.

BIF401

Solved past and current papers (plz correct if u see any mistake) solved by Zareen Fatima

Define mutation (2) Ans:

Mutations are:

- Replacement of amino acids with certain other amino acids in proteins
- Replacement of nucleotides with certain other nucleotides in DNA/RNA

Databases use for proteins (2)

Ans: uniport and swisport

Generally used databases and which type of information they store (3) Ans:

Genebank. Pdb, uniport swisport

- Locus
- Accession number

- Sequence
- Molecular mass
- Authors
- Journal etc.

Types of RNA and their functions (5) Ans:

RNA can be divided into two categories

- Coding RNAs
- Non-Coding RNAs
- Coding RNAs as is obvious from their name, code for Proteins
- Non-Coding RNAs regulate/assist in the process of translation

Types of RNA

1. Messenger RNA (mRNA) Messenger RNA (mRNA) carries the genetic information copied from DNA in the form of a series of three-**base**code "words," each of which specifies a particular amino acid
2. Transfer RNA (tRNA) is the key to converting the code words in mRNA.
3. Ribosomal RNA (rRNA) associates with a set of proteins to form ribosomes. These complex structures, which physically move along an **mRNA** molecule, catalyze the assembly of **amino acids** into protein chains. They also bind tRNAs and various accessory molecules necessary for **protein synthesis**.
4. Micro RNAs (miRNA) functions in RNA silencing and post-transcriptional regulation of gene expression.
5. Small Interfering RNA (siRNA)

Small (or short) interfering **RNA** (**siRNA**) is the most commonly used **RNA** interference (**RNAi**) tool for inducing short-term silencing of protein coding genes.**siRNA** is a synthetic **RNA** duplex designed to specifically target a particular mRNA for degradation.

Uses of clustalw 2

Ans: CLUSTALW can:

• create multiple alignments,

• optimize existing alignments,

• profile analysis &

• create phylogenetic trees

Pseudonots 3

Ans: • Tertiary or 3' structures may form pseudoknots

• A simple graph-based approach "Circular Plot" can help us detect the pseudoknots

A **pseudoknot** is a nucleic acid secondary structure containing at least two stemloop structures in which half of one stem is intercalated between the two halves of another stem. Pseudoknots fold into knot-shaped three-dimensional conformations but are not true topological knot

Uniprot swisprot Ans:

- Protein Sequences are reported from sequencing experiments

- This data is stored in protein sequence databases
- The famous ones include UniProt & SwissProt

Scoringatrix formula 5

$$s(a,b) = \frac{1}{\lambda} \log \frac{p_{ab}}{f_a f_b}$$

$s(a,b)=1/\lambda \log p_{ab}/f_a f_b$

Pam matric steps5" Ans:

Step 1: Align proteins sequences which are 1-PAM unit diverged

Step 2: Let $A_{i,j}$ be the number of times A_i is suabstituted by A_j

Step 3: Compute the frequency f_i of amino acid A_i

$$\frac{A_{ij}}{\sum_k A_{ik}}$$

Then, PAM1 = p_{ij} =

PAM'n' = (PAM1) ^{n} **STEP OF**

BLOSUM?

Ans:

Step 1: Eliminate sequences that are identical in x% positions

Step 2: Compute observed frequency $f_{i,j}$ of aligned pair A_i to A_j. Hence, $f_{i,j}$ becomes the probability of aligning A_i and A_j in the selected blocks.

Step 3: Compute f_i which is the frequency of observing A_i in the entire block

$$s_{ij} = \log_2 \frac{f_{ij}}{p_{ij}}, \quad p_{ij} = \begin{cases} f_i f_i & i = j \\ 2 f_i f_j & i \neq j. \end{cases}$$

Name three hydrophobic amino acid?

Ans: glycine (Gly), alanine (Ala), valine (Val), leucine (Leu), isoleucine (Ile), proline (Pro), phenylalanine (Phe), methionine (Met), and tryptophan (Trp).

2.rooted and unrooted tree difference?

Ans: Rooted and Unrooted trees can be used to show phylogenetic relationships between sequences

Rooted tree shows the phylogeny of ancestors

Phylogenetic trees take several forms: They can be rooted or **unrooted**, binary or general, and may show, or not show, edge lengths. A rooted**tree** is a **tree** in which one of the nodes is stipulated to be the root, and thus the direction of ancestral relationships is determined

- Rooted trees reflect the most basal ancestor of the tree in question.
- Unrooted trees do not imply a known ancestral root.

dynammic programming to create scoring matrix?r 1 series sequence dia hua tha scoring matrix maloom krny k liy?

ans: From the large number of possible nucleotide combinations, it is hard to find the optimal one

- Dynamic Programming (DP) helps break the problem into smaller problems

Select a nucleotide from sequence

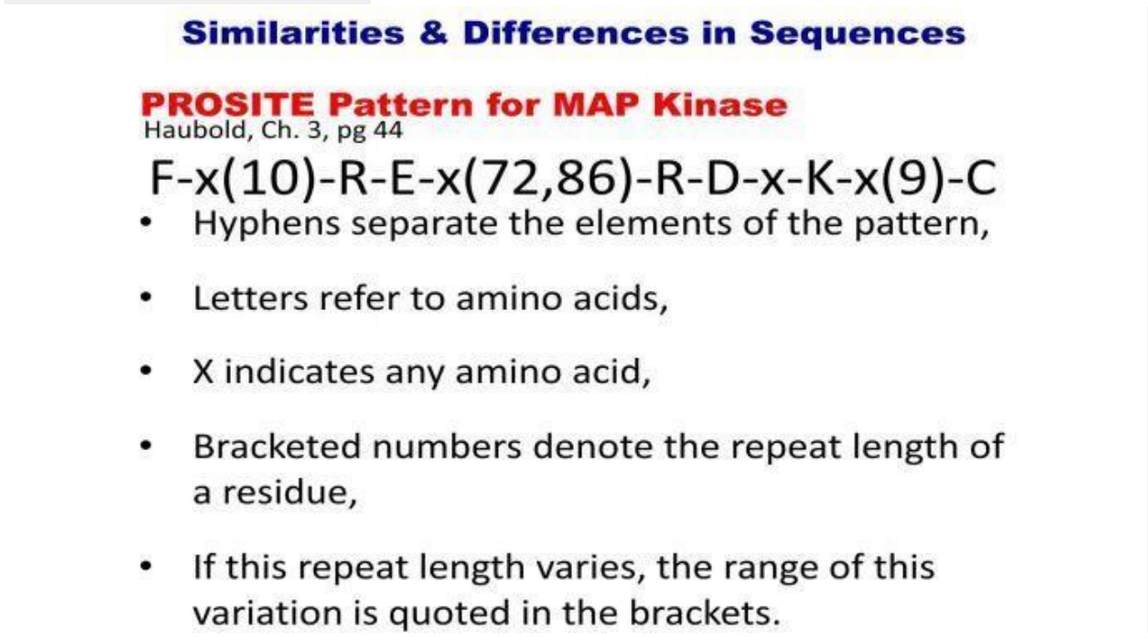
List all possible complementary position for the selected nucleotide in the complete sequence.

How DP works?

- DP exhaustively makes all possible nucleotide combinations
- DP then recombines such combinations in a process called “Traceback” to ensure that the highest coupled 2’ structure is reported

Complete the protein sequence .F-x(3)-x-R-F-K-x(4-5)-D-E-R?

FXXXXRFKXXXXDER
FXXXXRFKXXXXXDER



Clustering? clustering is used to build groups of genes with related expression patterns **Define system biology and its application?**

. Systems Biology

Systems biology is the computational and mathematical modeling of complex biological systems

Application

- Model protein and gene interactions
- Dynamical analysis of such models

- Understand system properties • Predict system level behaviors **Need for bioinformatics ?**

1 An interdisciplinary field
2. New but rapidly developing field

3. Low requirement on infrastructure and research equipment

4. Vast opportunities for scientific discovery

Life, evolution and disease can be better understood
• Drugs can be better developed after understanding molecular basis of life

Possible contributions
• Solve important problems related to life

• Handle mammoth quantities of experimental data

• Develop efficient solutions towards these goals **Role of 5' cap and poly a tail?**

1. The 5' cap helps identify mRNAs at the Ribosomes

2. The 5' cap also acts as a shield against 5' exonuclease, thus leading to an increase in mRNA stability

The 3'-end of mRNA has a polyA tail (around 30-200 adenylate residues) which help shield against 3' exonucleases

Mid BIF401 Solve

1. **What is the bioinformatics promise to the society? (2 marks)**

Bioinformatics is a multidisciplinary science that solves and analyzes biological problems. With the quantum explosion in biomedical data, the demand of bioinformatics has increased gradually. Present paper provides an overview of various ways through which the biologists or biological researchers in the domain of neurology, structural and functional biology, evolutionary biology, clinical science, etc., use bioinformatics applications for data analysis to summarise their research. A new perspective is used to classify the knowledge available in the field thus will help general audience to understand the application of bioinformatics.

2. **What dose role of 5' cap and 3' role A poly tail? (2)**

The 5′ cap protects the nascent **mRNA** from degradation and assists in ribosome binding during translation. A poly (A) tail is added to the 3′ end of the pre-**mRNA**once elongation is complete.

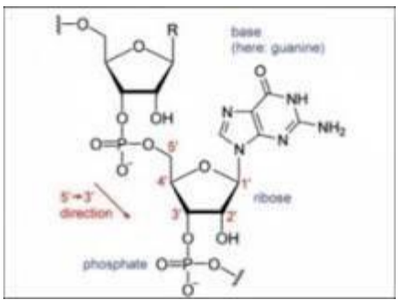
3. **What is dot plots?**

In bioinformatics a dot plot is a graphical method for comparing two biological sequences and identifying regions of close similarity. It is a type of recurrence plot.

4. **What does a Uniprot sequences give any information about protein query?**

5. **Write down step of BLOSUM?**

Step 1: Eliminate sequences that are identical in x% positions Step 2: Compute observed frequency f i,j of aligned pair Ai to Aj . Hence, fi,j becomes the probability of aligning Ai and Aj in the selected blocks. Step 3: Compute fi which is the frequency of observing Ai in the entire block



6. Hydroxyl group is less stable in RNA as compared to DNA? Explain it with a figure?

RNA. ... Unlike **DNA**, **RNA** in biological cells is predominantly a singlestranded molecule. While**DNA** contains deoxyribose, **RNA** contains ribose, characterised by the presence of the 2'-**hydroxyl group** on the pentose ring (Figure 5). This **hydroxyl group** make **RNA less stable** than **DNA** because it is more susceptible to hydrolysis ...

7. Diffrentiate between acid and basic amino acids?

There are three **amino acids** that have **basic** side chains at neutral pH. These are arginine (Arg), lysine (Lys), and histidine (His). Their side chains contain nitrogen and resemble ammonia, which is a**base**. Their pKa's are high enough that they tend to bind protons, gaining a positive charge in the process.

8. Differentiate between mass spectrometry and finger printing spectrometer?

Mass spectrometry (MS) is an analytical technique that ionizes chemical species and sorts the ions based on their**mass**-to-charge ratio. In simpler terms, a **mass** spectrum measures the masses within a sample. **Mass spectrometry** is used in many different fields and is applied to pure samples as well as complex mixtures.

Fingerprint refers to the characteristic **spectrum** or image of a test material which can be related to its properties and thus to its authenticity in the same way as a human **fingerprint** is specific of a certain person and unequivocally identify him/her.Oct 7, 2017

9. How tertiary structure of RNA formed?

Nucleic acid tertiary structure is the three-dimensional shape of a **nucleic acid polymer**. RNA and **DNA molecules** are capable of diverse functions ranging from molecular recognition to catalysis. Such functions require a precise three-dimensional tertiary structure.

10. What are pseudoknots?

A **pseudoknot** is a nucleic acid secondary structure containing at least two stem-loop structures in which half of one stem is intercalated between the two halves of another stem.

12. What is the purpose of 3D-1D bowle algorithm? PPT

Algorithm And Examples Fold Recognition/Threading Online Tools for Fold Recognition GOR Algorithm Homology Modelling 3D-1D Bowie Algorithm Machine Learning Approaches to Structure Prediction Neural Networks for Structure Prediction PSIPRED Introduction to Hidden Markov Models Ab initio modeling

13. What is needs of bioinformatics?

The primary goal of **bioinformatics** is to increase the understanding of biological processes. ... Common activities in **bioinformatics** include mapping and analyzing DNA and protein sequences, aligning DNA and protein sequences to compare them, and creating and viewing 3-D models of protein structures.

14. Uses of blastx, tblastx and tblastn?

• blastn (nucleotide BLAST) • blastp (protein BLAST) • blastx (translated BLAST) • tblastn (translated BLAST) • tblastx (translated BLAST)

15. Steps involved in FASTA algorithm?

Step 1: Local regions of identity are found. Step 2: Rescore the local regions using PAM or BLOSUM matrix. Step 3: Eliminate short diagonals below a cutoff score. Step 4: Create a gapped alignment in a narrow segment and then perform Smith Watermann alignment

16. Formula of domain and descibeit's method?

In mathematics, and more specifically in naive set theory, **the domain** of **definition** (or simply **the domain**) of a **function** is **the** set of "input" or argument values for ...

17. How to show phylogenetic tree describe the types to visuilaizae it?

A phylogenetic tree or evolutionary tree is a branching diagram or "tree" showing the .... This typeof tree only represents a branching pattern; i.e., its branch spans do not represent time or relative amount of character change. .... Exploration This is a programming library to analyze, manipulate andvisualize phylogenetic trees.

18. Difference between .... Peptide and unshot protein

Structurally, **proteins** and **peptides** are very similar, being made up of chains of amino acids that are held together by **peptide** bonds (also called amide bonds). So, what distinguishes a **peptide** from a**protein**? The basic distinguishing factors are size and structure. **Peptides** are smaller than **proteins**.

19. Why we use FASTA algorithem?

FASTA takes a given nucleotide or amino acid sequence and searches a corresponding sequence database by **using** local sequence alignment to find matches of similar database sequences. The **FASTA** program follows a largely heuristic **method** which contributes to the high speed of its execution.

20. What is the acronym meaning/definition of BLAST?

BLAST (basic local alignment search tool) is an algorithm for comparing primary biological sequence information, such as the amino-acid sequences of proteins or the nucleotides of DNA and/or RNA sequences.

21. Threonine is an hydrophobic amino acid?

Polar amino acids include serine, **threonine**, **asparagine**, glutamine, histidine and tyrosine. The **hydrophobic** amino acids include alanine, valine, leucine, **isoleucine**,**proline**, **phenylalanine**, tryptophane, cysteine and **methionine**.

22. How does protein is sequenced?

Protein sequencing is the practical process of determining the amino acid **sequence** of all or part of a **protein** or peptide. ... The two major direct methods of **protein sequencing are** mass spectrometry and Edman degradation using a**protein** sequenator (sequencer).

23. What is the uses and functions of blast?

In bioinformatics, BLAST (basic local alignment search tool) is an **algorithm** for comparing primary biological sequence information, such as the amino-acid sequences of proteins or the nucleotides of DNA and/or RNA sequences.

24. How does protein sequenced is identify?

Predicted **protein sequences are** an important resource for **protein identification**by mass spectrometry. ... The **sequence** of the cloned DNA was then determined and used to deduce the full amino-acid **sequence** of the **protein**.

25. What are Substitutions & Indel?

Indel. ... **Indels** can also be contrasted with Tandem Base Mutations (TBM), which may result from fundamentally different mechanisms. A TBM is **defined** as a**substitution** at adjacent nucleotides (primarily **substitutions** at two adjacent nucleotides, but **substitutions** at three adjacent nucleotides have been observed.

26. Why unpaired nucleotides should make the Structure of RNA destabilized?

27. Write formula for finding distance between two Sequences and elaborate it.

28. Top down proteomics?

Measures intact proteins followed by their peptides after fragmentation.

29. 7 Computed each possible alignment?

30. Usage of sequence data?

Sequence data can be used to obtain: • Similarity of sequences • Evolutionary History • Predict the function of molecules

31. Challenges in the field of bioinformatics.

Bioinformatics is full of challenges and opportunities. Amongst other frontiers in bioinformatics, there are protein structures, systems biology and personalized medicine.

32. Give diff b/w Rooted and uprooted also write its uses 5 marks

In a phylogenetic tree, every node represents a species. ... A **rooted tree** is a **tree in** which one of the nodes is stipulated to be the **root**, and thus the direction of ancestral relationships is determined. An **unrooted tree**, as could be imagined, has no pre-determined **root** and therefore induces no hierarchy.Jan 1, 2001

33. Difference b/w uniprot & EMBL?

UniProt is the Universal Protein resource, a central repository of protein data created by combining the Swiss-Prot, **TrEMBL** and PIR-PSD databases. ... UniProt is a freely accessible database of protein sequence and functional information, many entries being derived from genome sequencing projects.

34. About amino acids and protein which has 3 codons?

The **three** stop **codons** have **names**: UAG is amber, UGA is opal (sometimes also called umber), and UAA is ochre. Stop **codons** are also called "termination" or "nonsense" **codons**.

35. Scoring Matrix & NJ algorithm?

Scoring matrices are used to determine the relative**score** made by matching two characters in a sequence alignment. ... There are many flavors of **scoring matrices**for amino acid sequences, nucleotide sequences, and codon sequences, and each is derived from the alignment of "known" homologous sequences.

The neighbor-joining **method** (**NJ**) is a distance based **method** (requires a distance matrix) and uses the star decomposition **method**. **Algorithm**. Neighbor-joining is a recursive **algorithm**.

36. About bioinformatics tools?

Tool	Description
BLAST	It is a search tool, used for DNA or protein sequence search based on identity.
HMMER	Homologous protein sequences may be searched from the respective databases using this tool.
Clustal Omega	Multiple sequence alignments may be performed using this program.
Sequerome	Used for sequence profiling.
ProtParam	Used to predict the physico-chemical properties of proteins.
JIGSAW	To find genes, and to predict the splicing sites in the selected DNA sequences.
novoSNP	Used to find the single nucleotide variation in the DNA sequence.
ORF Finder	The putative genes may be subjected to this tool to find Open Reading Frame (ORF).
PPP	Prokaryotic promoter prediction tool used to predict the promoter sequences present upstream the gene
Virtual Foorprint	Whole prokaryotic genome (with one regular pattern) may be analysed using this program along with promoter regions with several regulator patterns.
WebGeSTer	This is a database containing sequences of transcription terminator sequences and is used to predict the termination sites of the genes during transcription.
Genscan	Used to predict the exon-intron sites in genomic sequences.
Softberry Tools	Several tools are specialized in annotation of animal, plant, and bacterial genomes along with the structure and function prediction of RNA and proteins.

37. What is the use of blast & its performance?

It demands high computational resources and this demand keeps increasing.**BLAST uses** a similarity searching heuristic that determines sequences from a database that are most similar to a query

sequence. We performed all experiments using a set of queries containing genes or amino acid sequences.

38. How Gibbs energy fold RNA Secondary structure?

39. Difference between global and local alignment?

If you think your sequence is a subsequence of the reference, do a local alignment. But if you think your entire sequence should match your entire reference, you would do a global alignment.

40. Progressive alignment?

Progressive alignment (Feng and Doolittle, 1987) is a heuristic for multiple sequence **alignment** that does not optimize any obvious **alignment** score. The idea is to do a succession of pairwise alignments, starting with the most similar pairs of sequences and proceeding to less similar ones.

41. MSA?

A multiple sequence alignment is a sequence alignment of three or more biological sequences, generally protein, DNA, or RNA. In many cases, the input set of query sequences are assumed to have an evolutionary relationship by which they share a linkage and are descended from a common ancestor.

43. PSA?

Prediction of probable secondary structures and fold-class; good for visualizing amphipathic helices, where present.

44. Define system biology and its application?

Systems biology is **the computational** and mathematical modeling of complex **biological systems**. **It** is a **biology**-based interdisciplinary field of study that focuses on complex interactions within**biological systems**, using a holistic approach (holism instead of **the** more traditional reductionism) to **biological** research.

45. Write the methods of clustering?

Cluster analysis or **clustering** is the task of grouping a set of objects in such a way that objects in the same group (called a **cluster**) are more similar (in some sense) to each other than to those in other groups (**clusters**). ...**Clustering** can therefore be formulated as a multi-objective optimization problem.

46. Why RNA has short life span? Illustrateur answer with diagram?

This kind of **RNA** is considered **short** lived. In contrast, **RNA** that is crucial to basic cellular processes like cell structure should stick around in cells for a long time to carry out its function

47. Write the four objectives of the comparing sequencing?

Sequence homology is the biological homology between DNA, RNA, or protein sequences, defined in terms of shared ancestry in the evolutionary history of life. Several conclusions can be drawn on a set of sequences in terms of: • Similarity • Specific Differences • Relationship • Evolutionary Insight

48. Purines and pyrimidins structures?

Purines vs. **Pyrimidines**. **Purines and Pyrimidines** are nitrogenous bases that make up the two different kinds of nucleotide bases in DNA and RNA. The two-carbon nitrogen ring bases (adenine and guanine) are **purines**, while the one-carbon nitrogen ring bases (thymine and cytosine) are**pyrimidines**.

49. Benefits of dot plot?

Data points may be labelled if there are few of them.**Dot plots** are one of the simplest statistical **plots**, and are suitable for small to moderate sized data sets. They are useful for highlighting clusters and gaps, as well as outliers. Their other **advantage** is the conservation of numerical information.

50. EMBL and EBI stands for?

The European Bioinformatics Institute (**EMBL**-**EBI**) is an IGO which as part of the European Molecular Biology Laboratory (**EMBL**) family focuses on research and services in bioinformatics.

51. Different field of expasy?

Expasy provides access to a variety of online databases and tools. • Depending upon your requirement, you find sequence information from Expasy.

52. Describe the steps of PAM?

Steps to compute PAM matrices

Step 1: Align proteins sequences which are 1-PAM unit diverged

Step 2: Let $A_{i,j}$ be the number of times A_i is substituted by A_j

Step 3: Compute the frequency f_i of amino acid A_i

Then, PAM1 $= p_{ij} = \frac{A_{ij}}{\sum_k A_{ik}}$

PAM'n' = (PAM1) ^{n}

53. Hairpin structure of RNA?

hairpin loop (mRNA) A hairpin loop is an **unpaired**loop of messenger RNA (mRNA) that is created when an mRNA strand folds and forms base pairs with another section of the same strand. The resulting structure looks like a loop or a U-shape. Hairpins are a common type of secondary structure in RNA molecules.

54. Application of bioinformatics?

Genomics • Transcriptomics • Proteomics • Metabolomics • Structural Proteomics • Drug Design • Systems Biology • Personalized Medicine

55. PCR, Electrophoresis and Dynamic programming?

PCR (**polymerase chain reaction**): **PCR** (**polymerase chain reaction**) is a technique in molecular genetics that permits the analysis of any short sequence of DNA (or RNA) even in samples containing only minute quantities of DNA or RNA.**PCR** is used to reproduce (amplify) selected sections of DNA or RNA for analysis.

Gel **electrophoresis** is a technique commonly **used in** laboratories to separate charged molecules like DNA?, RNA? and proteins? according to their size. ... The movement of charged molecules is called migration. Molecules migrate towards the opposite charge.

Dynamic programming

is both a mathematical optimization method and a computer programming method. The method was developed by Richard Bellman in the 1950s and has found applications in numerous fields, from aerospace engineering to economics.

56. DNA, RNA and Protein sequences can be aligned using Needleman Wunsh algorithm? DNA, RNA and Protein sequences can be aligned using Needleman Wunsh algorithm

57. Local alignment focuses on the sub regions within the sequences. Global alignment can full length?

58. Modification in trace back strategy can help find overlap matches containing leading or trailing sequence?

Yes: Trace back can also be started from high score on the right column and bottom row. • Such trace backs helps find overlap matches containing leading or trailing sequence.

59. CLUSTALW can run Fast and slow modes?

Slow

60. BLAST can search sequence databases and identify unknown sequences by comparing them to the known sequences?

61. Multiple types of FASTA exist which assist in aligning DNA/RNA/Protein sequences (Fasta)

62. Structural information can be get by X-ray Crystallography and NMR?

Structures are obtained from X-Ray Crystallography, Atomic Force Microscopy & Nuclear Magnetic Resonance Spectroscopy. Protein Databank 73.

63. Protein Sequencing can be done b?

Next Generation Sequencers

64. Object based method is?

Least Square Distances

65. Not Clusturing Approch Other than?

UPGMA MPGMA, Nieghbor Joining Single Linkage Complete Linkage

66. Define mutation

Mutations are:

- Replacement of amino acids with certain other amino acids in proteins
- Replacement of nucleotides with certain other nucleotides in DNA/RNA

67. Databases use for proteins?

uniport and swisport

68. Generally used databases and which type of information they store? Genebank. Pdb, uniport swisport

- • Locus
- Accession number
- • Sequence
- • Molecular mass
- Authors
- Journal etc.

69. Types of RNA and their functions?

RNA can be divided into two categories

- Coding rnas
- Non-Coding rnas
- Coding rnas as is obvious from their name, code for Proteins
- Non-Coding rnas regulate/assist in the process of translation

Types of RNA

1. Messenger RNA (mrna) Messenger RNA (mrna) carries the genetic information copied from DNA in the form of a series of three-**base**code "words," each of which specifies a particular amino acid 2. Transfer RNA (trna) is the key to converting the code words in mrna.

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4. Micro rnas (mirna)

Functions in RNA silencing and post-transcriptional regulation of gene expression.

5. Small Interfering RNA (sirna)

Small (or short) interfering **RNA** (**sirna**) is the most commonly used **RNA** interference (**rnai**) tool for inducing short-term silencing of protein coding genes.**sirna** is a synthetic **RNA** duplex designed to specifically target a particular mrna for degradation.

70. **Uses of clustalw?**

CLUSTALW can:

- create multiple alignments,
- optimize existing alignments,
- profile analysis &
- create phylogenetic trees

71. Uniprot swisprot

- • Protein Sequences are reported from sequencing experiments
- This data is stored in protein sequence databases
- The famous ones include uniprot & swissprot

72. Scoringatrix formula?

$S(a,b)=1/\lambda \log pab /fafb$

73. Pam matric steps5"

Step 1: Align proteins sequences which are 1-PAM unit diverged

Step 2: Let Ai,j be the number of times Ai is suabstituted by Aj

Step 3: Compute the frequency fi of amino acid Ai

Then, PAM1 = pij =
PAM'n' = (PAM1) n

74. **Name three hydrophobic amino acid?**

glycine (Gly), alanine (Ala), valine (Val), leucine (Leu), isoleucine (Ile), proline (Pro), phenylalanine (Phe), methionine (Met), and tryptophan (Trp).

75. **Dynammic programming to create scoring matrix?R 1 series sequence dia hua tha scoring matrix maloom krny k liy?** Ans: From the large number of possible nucleotide combinations, it is hard to find the optimal one

- Dynamic Programming (DP) helps break the problem into smaller problems

Select a nucleotide from sequence
List all possible complementary position for the selected nucleotide in the complete sequence.
How DP works?

- DP exhaustively makes all possible nucleotide combinations
- DP then recombines such combinations in a process called "Traceback" to ensure that the highest coupled 2' structure is reported

76. **Complete the protein sequence .F-x(3)-x-R-F-K-x(4-5)-D-E-R?**

FXXXXRFKXXXXDER FXXXXRFKXXXXXDER

77. **Clustering?**

Clustering is used to build groups of genes with related expression patterns

MUHAMMAD IMRAN BIF401 -Bioinformatics I

MCQs&Questions final term Past papers /2019/2020

1. What Are Uses Of Mass Spectrometry? 2

Mass spectrometry (MS) is an analytical technique that ionizes chemical species and sorts the ions based on their mass-to-charge ratio. In simpler terms, a mass spectrum measures the masses within a sample. Mass spectrometry is used in many different fields and is applied to pure samples as well as complex mixtures.

2. Need For Chou Fasman Algorithms? 2

The Chou-Fasman algorithm is simple in principle.The conformational parameters for each amino acid were calculated by considering the relative frequency of a given amino acid within a protein, its occurrence in a given type of secondary structure, and the fraction of residues occurring in that type of structure.

3. Three individual scores Integrative Scoring Scheme? 2

Three individual scores can be obtained including

1. MW Match Score
2. PST Match Score
3. In vitro & In silico Match Score.

4. **Write benefits of phylogenetic studies marks • 03**
5. **Advantage of Folded Protein over Denatured Protein. 5**

Difference Between Folding Protein and Denature Protein, / What Factors Are Used For Folding Proceeds?

Protein folding is the physical process by which a protein chain acquires its native 3-dimensional structure, a conformation that is usually biologically functional, in an expeditious and reproducible manner.

Denaturation of proteins involves the disruption and possible destruction of both the secondary and tertiary structures. Since denaturation reactions are not strong enough to break the peptide bonds, the primary structure (sequence of amino acids) remains the same after a denaturation process.

Protein folding is a process by which a polypeptide chain folds to become a biologically active protein in its native 3D structure. Protein structure is crucial to its function. Folded proteins are held together by various molecular interactions. ... The amino acid sequence of a protein determines its 3D structure. **The forces involved in protein folding include**: • Electrostatic interactions, • van der Waals interactions, • Hydrogen bonds, • Hydrophobic interactions

6. Score in Silico fragment scoring?.5

- Count the matches between in silico and in vitro peaks.
- Give an equivalent score to the candidate protein.
- Weigh each of the aforementioned match by the mass error.
- Accumulate the score

7. How Loops Are Found After The Finding Of Alpha And Beta Sheets?5

Alpha helices and beta strands are connected by these turns and loops. ... Loops and turns generally lie on the surfaces of proteins so they often participate in interactions between proteins and other molecules. In a loop, there are no regular structures as can be found in helices or beta strands.

8. Tandem mass spectrometry

Tandem mass spectrometry (MS-MS) is a related technology in which compounds are separated by molecular weight by one **mass spectrometer**, fragmented as they exit, and identified on the basis of their fragments by a second **mass spectrometer**

9. Advantage and disadvantage of Ab initio method?

Advantages:

Ab Initio methods can fold any target sequence using only physical atomic properties. Predictions are mostly accurate and correctly describe the natural folding process. **Disadvantages:**

Ab initio methods are the very difficult to design (energy function). These methods are slow due to the huge possibilities.

10. Types of secondary structures of RNA:

1. Single stranded:

3'end may fold on to the 5' end.

2. Helices:

Double stranded RNA helix of stacked base pairs **3.**

Hairpin loop:

The loop of the hairpin must at least four bases long to avoid steric hindrance with base-pairing in the stem part of the structure **4. Bulge Loops:**

Bulges, are formed when a double-stranded region cannot form base pairs perfectly.

5. Interior loop:

Interior loops are formed by an asymmetric number of unpaired bases on each side of the loop.

6. Junctions or intersections:

Junctions include two or more double-stranded regions converging to form a closed structure. The unpaired bases appear as a bulge

11. How homology modeling is used for knowing the sequence of unknown protein sequence?

If another protein which has a similar sequence also has its structure known, the structure of an unknown protein can be predicted based on that similar protein. So, it is then possible to identify unknown protein structures by just examining the homologous protein sequences. Good sequence alignment and identity ensures that homology modelling will give accurate results. Thus, Homology modeling is used to predict structures of proteins having high sequence similarity with other proteins with known structures:

12. Role of Amino Acid?

Protein plays a crucial role in almost all biological processes and amino acids are the building blocks of it. A large proportion of our cells, muscles and tissue is made up of amino acids, meaning they carry out many important bodily functions, such as giving cells their structure.

13. Write Down The Following Terms Of PDB (3)

Title (Title of the structure)
Source (Identify which organism structure from)
COMPND (Brief detail of the structure)
REVEDAT (The data of last revision

14. What Happend When Ph Is Less Than The Pk Value?

If the pH is higher than the pKa, then the compound will be deprotonated. A further consideration is the charge on the compound. Acids are neutral when protonated and negatively charged (ionized) when deprotonated. Bases are neutral when deprotonated and positively charged (ionized) when protonated.

15. Chou Fasman Algorithm?

The Chou–Fasman method is an empirical technique for the prediction of secondary structures in proteins, originally developed in the 1970s by Peter Y. Chou and Gerald D. Fasman.

16. Function of Tandem MS?

Tandem MS helps in measurement of mass to the fragments as well. This process provides another step in further scoring and ranking and Protein identification thus becomes easier.

17. Factors That Participate In Folding Protein?

The external factors involved in protein denaturation or disruption of the native state include temperature, external fields (electric, magnetic), molecular crowding, and even the limitation of space, which can have a big influence on the folding of proteins

18. Need For Chou Fasman Algorithms?

The Chou-Fasman algorithm is simple in principle.The conformational parameters for each amino acid were calculated by considering the relative frequency of a given amino acid within a protein, its occurrence in a given type of secondary structure, and the fraction of residues occurring in that type of structure.

19. Optimal Energy Function In Protein?

Results: For a single protein sequence, the probability of success (i.e. the probability that the folded state is the lowest energy state) is derived. We then maximize the average probability of success for a set of proteins to obtain the optimal potential energy function

20. Silico Fragments Comparison?

Matching experimental fragments with in silico fragments is the final resort in protein search & identification

21. Chou – Fasman method?

It is a technique for the prediction of secondary structures in proteins i.e. Alpha Helices, Beta Sheets and Turns is Chou – Fasman technique. The method is based on analyses of the relative frequencies of each amino acid in alpha helices, beta sheets, and turns based on known protein structures solved with X-ray crystallography. From these frequencies a set of probability parameters (in our handouts, it is propensity table) were derived

For the appearance of each amino acid in each secondary structure type. To predict the probability that a given sequence of amino acids would form a helix, a beta strand, or a turn in a protein.

Chou-fasman algorithm (alpha helix):

For Alpha Helices, 4 contiguous amino acids are required.

- Their Alpha-Helix propensity should be more than 1.0 Once
- this propensity falls below 1.0, Alpha-Helix stops.

22. Improvements Of Chou Fasman Algorithm?

The Chou-Fasman algorithm, one of the earliest methods, has been successfully applied to the prediction. ... We improved Chou-Fasman method in three aspects. (a) Replace the nucleation regions with extreme values of coefficients calculated by the continuous wavelet transform.

23. What is hydrophobicity of amino acid?

In chemistry, hydrophobicity is the physical property of a molecule that is seemingly repelled from a mass of water. In contrast, hydrophiles are attracted to water. Hydrophobic molecules tend to be nonpolar and, thus, prefer other neutral molecules and nonpolar solvents.

24. how propensity table can helps us?

25. Seven Steps present in flow chart of homology modeling? 10

1. Template recognition and initial alignment.
2. Alignment correction.
3. Backbone generation.
4. Loop modeling.
5. Side-chain modeling.
6. Model optimization.
7. Model validation

26. Role of Amino Acid?

Protein plays a crucial role in almost all biological processes and amino acids are the building blocks of it. A large proportion of our cells, muscles and tissue is made up of amino acids, meaning they carry out many important bodily functions, such as giving cells their structure.

27. Role of Global Protein?

Global protein function prediction from protein-protein interaction networks. ... The availability of entire genome sequences and of high-throughput capabilities to determine gene coexpression patterns has shifted the research focus from the study of single proteins or small complexes to that of the entire proteome.

28. Acidic and Basic Amino Acid?

There are three amino acids that have basic side chains at neutral pH. These are arginine (Arg), lysine (Lys), and histidine (His). Their side chains contain nitrogen and resemble ammonia, which is a base. Their pKa's are high enough that they tend to bind protons, gaining a positive charge in the process.

29. Points of Bottom up Proteomic?

The major alternative workflow used in high-throughput proteomics is called top-down proteomics and does not use proteolytic digestion. Essentially, bottom-up proteomics is a means of determining the protein make- up of a given sample of cells, tissues, etc. (Internet)

30. Varies RNA Secondary Structure?

Types of RNA Secondary Structures - I

- RNA Complementary bases in primary structure bond together to fold RNA into a secondary structure

Types of RNA Secondary Structures – II

- RNA 1' sequence (5' – 3') can fold onto itself to make 2' structures.
- Two previous examples of 2' structures include Helix and Hairpin structures. **Types of RNA Secondary Structures - III**
- RNA can be fold to form helices, bulge loops, and interior loops. An other 2' RNA structure is the Junction or Intersection.

31. Objective Base Method and Clustering Approach?

Clustering Approach UPGMA WPGMA Neighbor Joining Single Linkage Complete Linkage Objective based Methods Least Square Distances Maximum Likelihood Maximum Parsimony

32. Shot Gun and Peptide Mass Finger Printing?

Shotgun proteomics digests the entire protein mix 1st followed by peptide analysis & protein database search Peptide mass fingerprinting involves protein separation followed by a single protein's peptide analysis.

33. substitution and Indels?

Substitution at adjacent nucleotides (primarily substitutions at two adjacent nucleotides, but substitutions at three adjacent nucleotides have been observed.

Indels, being either insertions, or deletions, can be used as genetic markers in natural populations, especially in phylogenetic studies.

34. Prediction of Protein Structure?

Protein structure prediction is the inference of the three-dimensional structure of a protein from its amino acid sequence—that is, the prediction of its folding and its secondary and tertiary structure from its primary structure.

35. How Can We Predict Amino Acid In Alpha Helix And Beta Sheets?

Proteins that convert from alpha helix to beta sheet: implications for folding and disease. The sequence of a protein normally determines which amino acid residues will form alpha helices, and which one beta sheets, to an extent that allows secondary structure prediction to be made with a reasonable reliability..

36. Rooted Un Rooted Tree?

Rooted Tree:
Each node with descendants represents the inferred most recent common ancestor of the descendants.

- The edge lengths in some trees may be interpreted as time estimates.
- Rooted trees can show temporal evolutionary direction.
- Expensive.

Unrooted:

- Only the relatedness of the leaf nodes.
- Do not require the ancestral root to be known or inferred.
- Less expensive.

Uses: In bioinformatics, such as
Rooted and unrooted trees can be used to show phylogenetic relationships between sequences..

37. Molecular biology?

It is the branch of biology that deals with the structure and function of the macromolecules (e.g. proteins and nucleic acids) essential to life. Molecular biology is the study of biology at a molecular level.

38. Protein Fragmentation techniques?

- Electron Capture Dissociation (ECD).
- Electron Transfer Dissociation (ETD).
- Collision Induced Dissociated (CID).

39. Protein structure prediction?

There are three different strategies for structure prediction
1. Homology Modelling
2. Threading/Fold Recognition
3. Ab Initio Modelling

MCQS final term

1) Fasta Was Develop In_____(**Ans1988)**
2) In Corey Paulin Modal Of The Protein Each Atom Is Represented By____(**Solid Sphere)**
3) How Many Storages Use For Structural Modling _____(**3)**
4) Modeler Software Use For Homology Modling Which Use(Phython File Format,Not Confrm)
5) Mass Of The Based Proteomic Begin With Measure Of(**Intact Protein Amino Acid)**
6) Protein Sequence Obtained From(**Uni Port**)
7) During The Formation Of The 5 Nucleotide Hydrogen Bond Energy Release(-**12kcal/Mol**)
8) Which Method Is Not Use For Making Polygenatic???**Msa**
9) Rna Dont Have Structer???**Quatnaryy**
10) Large Numbr Of Sequence Availble??**Publically**
11) Coding Region Of Dna Is???**Exon**
12) Blast Use For???**Unknown And Known Protein**
13) Expasy Develop By???**Sib**
14) Variation In Sequence Develop By??? **Insertion,Deletion And Substituin**

15) Bond Form Between C And N Terminas??? **Hydrogen Bond**

16) X.Y.Z Position Of Protein Are Available By??? **Pdb**

17) Technique Use To Measure The Mass Of Protein??? **Mass Spactrometry**

18) Large Amino Acid??? **Tryptophan**

19) Positive Charge Amino Acid??? **Lysine And Arginine**

20) Mrna Has??? **Single Strand**

21) C.Alpha Atom Are Traced To Recreat??? **3d Structer Of Protein**

22) Bond Form Between Two Atom??? **Energy Release**

23) Provide Clue Of Precurser Protein Peptide Sequence?**?Pst**

24) Amino Acid Requird For Betasheet??? **3-5**

25) .....Are Formed By Bonding Betweenci And Ni+4n Protein Backbone??? **Alpha Halix** **26)** Blasta Stand For??? **Basic Local Alignment Search Tool**

27) Amino Acid Have Uniwe Trait Which Can Help In??? **Protein Folding** **28)**

Align Comparison Betweendna/Dna Anddna/Protein??? **Fasta.**

29) Ratio Use To Calculate Mass Of Protein??? **M/Q.**

30) Treading Not Requied For??? **Structer Of Entire Backbone.**

31) Needle Man Wounch Man Or Smith Water Man Algorithm Use For??? **Local And Global Alignmnt.**

32) Dianamic Programing Recombinant Nucleotide By??? **Traseback.**

33) Insertion And Deletion Cause??? **Gap.**

34) Information Required For Follding Of Amino Acid Is Present In??? **Amino Acid.**

35) Mgf Use For??? **Proteomic.**

36) Nj Algorith Use To Pridict The??? **2nd Structer Of Rna..**

37) Genomics, evolutionary studies and system biology are application of **bioinformatics**

38) Exact matching require which type of nucleotide-**same number of nucleotide**

39) Accurate solution of protein structure is one of the ___ **toughest problems**

40) **Tract back** is overlap matches can start from any position in scoring matrix

41) The matrix has ……scores. **Positive and negative**

42) **MASCOT___**can search people mass finger printing and shotgun proteomics data set

43) Spectrometer measures the protein by _____ **Mass/charge**

44) Simulation of the folding process depends _____ **Energy function**

45) Information about **DNA/RNA** is available on gene bank

46) **TDP** is used to measure the molecules weight of intact proteins.

47) Coordinates of **alpha carbon** in the protein backbone can be used for structural visualization.

48) in structure visualization c-alpha atoms are traced to recreate a _____**3D protein structure**

49) in UPGMA distance is calculated btw _____**_ two clusters and between two trees**

50) pk value of Aspartic acid **3.9**

51) Largest amino acid …… **Tryptophan**

52) 2nd STRUCTURE of proteins can be obtained by using_____ **Chou Fasman algorithm**

53) NJ Algorithm strategy is used to predict __ **RNA 2` structure**

54) The loop of hair pin must be at least ___ in length **2 aminoacids**

55) Fold recognition is also called …….**threading**

56) Ab initio method in contrast ,base their prediction on **low energy model**.

57) Bioinformatics require…….smart mind, and connected to internet, **both**,

58) Sequence alignment tool is…….**PROSIGHT and MASCOT**

59) MS1,MS2 provide us data identifying unknown…….**Proteins sequence**.

60) **MGF** files develop as an open standard for……. **proteomics data**.

61) **PDB** coordinates alpha carbon in protein back bone can be used for comparison .

62) Alpha carbon atom can be obtained from…**…PDB**

63) in formation of 2nd structure of protein C & N can make……**Hydrogen bonds**

64) information for protein folding is into its native structure is in…….. **protein's amino acid sequence**!

65) Energy **released** during bond formation.

66) hydrogen bonding occur in …**secondary** ..Protein structure

67) Bond between C and H in alpha helices____**Hydrogen bonding**

68) Amino acid having 3 codons_____ **isoleucine**

69) MALDI typically adds a …… to protein or peptide. **Proton**

70) low quality match gets_____**lower score**

71) ____ are sequences of amino acids produced during MS2. **PST (peptide sequence tags)**

72) MS2 data can be used to extract…… **peptide sequence tags**.

73) The alpha helices propensity should be more than____(**1.0)**

74) How many types are of protein sequencing techniques__ **(2)**

75) Blast can find sequences of _____**nucleotides amino acid**

76) If a protein sequence of 26 amino acids is fragmented at 11 amino acid **C11, Z25**

77) To find out unknown sequence of nucleotide we use . **NGS or Mass spectrometry**

78) How many types of peptide May be injected in mass spectrometry chamber.. . **three hundred thousand to four hundred thousand**

79) .**MASCOT**.can search peptide mass finger printing and shotgun proteomics dataset.

80) Stabilizing and destabilizing energy give us **…….. quality of 2nd structure** .

81) The Fasman algorithm is based on **statistical occurrence** of Amino Acids in known structures.

82) DP recombinant the nucleotide recombination through process of ..... **Traceback**

83) **83) MALDI**.....add proton to a protein and a peptide..

84) Positively charged amino acids are..... **3**

85) Aromatic amino acid include ..... . **phenylalanine, tyrosine tryptophan**

86) Homology modelling fail to predict …… **quality structure**

87) How many forces are involved in protein folding.

88) Pairwise alignments tells the similarity between sequence**... by maximizing the matches.**

89) Dot plots employ dot matrix with two sequence plotted represents.....**pairwise alignment and comparison**.

90) Differentiate b/w DNA and RNA sequence…….. **RNA has more variety of sequence**

91) Cell molecule are produced after transformation of …… **DNA to protein**

92) Dot plot cannot deal with……. **Insertion, deletion and gaps.**

93) Exons are may be more …… **conserved**.

94) In dynamic algorithm we can do comparison …… of more sequence. **Three**

95) Mass/Charge ratio helps calculate the mass of the …. **Protein**

96) In scale tree branches lengths are equal to the magnitude of change in the …….. **nodes**.

97) How many types of peptides mix in MS chamber…….. **300,000 – 400,000**

98) mRNA is a …… structure. **Planer**

99) When unpaired bases of 2´ stricture join to form…….. **3´ structure**

100) N Jackobson use to pridict **2´structure.**

101) Aromatic amino acid ……. **Tyrosine**

102) Positive charge amino acids……. **Lysine, arginine, histidin**

103) MS begins with the measure of …….. **intact protein**

104) Amount of amino acids in alpha helix…….(**4**)

105) Intact mass of protein can be found by….. **MS1**

106) MSA can be done by……. **CLUSTAL**

107) Which is used as sequence alignment tool……. **PROSIT**

108) MS1 and MS2 help to identify …… (**PST**)

109) MGF is used for …….. **proteomics**

110) X,Y,Z positions of alpha carbon available online on …… (**PDB**)

111) Method to determine RNA structure …….. **NMR, X-Ray crystallography**

112) CLUSTLA runs……. **Slow accurate/ fast appropriate**

113) Which give info about precursor protein……. **PSTS**

114) Raw data files format can be converted to ……. **Open format**

115) Which of the following is cartoonic figure……. **Ribbon diagram**

116) In CATH protein recognize according to their…….. **structural similarity**

117) Method for obtaining 1´ structure**…… (Edman Degradation Tandem Mass Spectrometry).**

118) FASTA achieves alignment by using **short lengths of exact matches.**

119) Dot plots cannot capture **insertions, deletions and gap indications**

1. Dot plots employ dot matrix with two sequences plotted on **top and left of the matrix**
2. In multiple sequencing, we have **both**
3. CLUSTALW can run in two modes **slow/accurate, fast/appropriate.**
4. The correct order of progressive alignment for multiple sequencing alignment is **Pair wise alignment, Guide tree, branching order in tree.**

5. Blast can find sequencing from protein or nucleotide database **it can find known and unknown sequencing.**
6. : **Next generation sequencing** is used to known about unknown sequences of nucleotide.
7. In multiple sequence algorithm **doing so many progressive alignment can be slow.**
8. : Structural information of a molecule obtained by **X ray crystallography & Nuclear magnetic resonance.**
9. Before starting to mach nucleotide or amino acid are need to find the **Location of conserved residues.**
10. Bioinformatics is an interdisciplinary relative new development of **biology, mathematics, computer science, chemistry and physics.**
11. Fasta online software Can also perform genome and proteome similarity search .**true**
12. From Fasta we can obtain the information **on the parent organism, function and evolution history.**
13. name any two frequently used scoring matrix **Blosum and pam**.
14. Amino acids are chained together by **peptide bonds**
15. To find out the unknown sequence of nucleotide and protein we use **blast.**
16. GenBank and Uniprot, have **A large number of sequences are available in publically .**
17. Protein databank is a **data base**
18. In dynamic programming we can do multiple sequence alignment upto **21 sequence** .
19. How many types of secondary structure can be formed possible of RNA? **5.**
20. RNA structures have the **lowest (or close)** quantity of free energy.
21. RNA can play role due to its structure . **all of above.**
22. Rna contains **ribose** sugar?
23. Thymine is replaced by **uracil** in rna.
24. RNA molecules readily degrade due to their **short** shelf life .
25. Each Nucleotide in rna has **covalently bonded** to the backbone .
26. **Gibbs Free Energy"** is the free energy of an RNA molecule available for reaction .
27. The structure of rna sampled can be measured by.**Atomic Force Microscopy**
28. The 3'-end of **mRNA** has a poly A tail
29. The role of **mRNA** Carry genetic information from DNA to Ribosomes where proteins are being assembled .
30. Messenger rna is only **5-10%** of rna content in the cell.

33.

FOR MID TERM PREPARATION

1: Gaps are produced by insertion or deletion.

2: Example of energy base Zuker algorithm.

3: Two forms of DNA rending sequence.

4: Translation involves coding of protein by RNA at ribosome.

5: some medicine have side effect on certain patient Frontiers in personalized medicine.

6: By comparing sequence we can get similarity, specific difference, relationship, evolutionary insight.

7: Non clustering method include Maximum parsimony.

8: Destabilizing energy of a nucleotide has (+) value.

9: Comparison of protein sequence can be done with…Uniprot

10: Function is a property of the RNA structure,which is formed from the RNA sequences.

11: Structural information of a molecule obtained by X ray crystallography & Nuclear magnetic resonance.

12: Protein have 20 amino acid.

13: If sequence are 9 then according formula it will be 9^2.

14: Fast is available online.

15 Fasta was developed in 1988.

16: Bioinformatics is the inter disciplinary relative new developing field.
17: Genbank can be reached by…online website or www.ncbi.nim.nih.gov/genebank.

18: Before starting to mach nucleotide or amino acid are need to find the Location of conserved residues.

19: some medicine have side effect on certain patient Frontiers in personalized medicine.

20: Identity is the number of nucleotide or amino acid which match exactly between two biological sequence.

21: Alignment are represented by diagonals in dot matrix.

22: Needleman wunsch man algorithm we use for…… Global alignment.

23: Exons in protein tend to be more conserved then the intron.

24: In MSA the distance between 2 sequences in equal to NO of identical residue/number of aligned residue.

25: All the changes which accrued in our cell …….. metabolism

26: The loop of the hairpin must be least 4 bases lo

28: NJ algorithm is used to produc ……. Optimal RNA 2´ sturucture.

29: energy base algorithm type is Zuker algorithm.

30: Types of RNA…present in organism. (5)

31: The online search is for DNA and RNA is…….. genebank

32: Rooted tree is computational expensive.

33: In multiple sequence algorithm doing so many progressive alignment can be slow.

34: DNA dictates the production of cell protein and carbohydrate

35: protein have long sequences …in no?

36: In Exact partial matching of residue and order with room for variation in both.

37: DNA/RNA sequence is available on…. GENE BANK

38: Martine algorithm predict for structure 2 RNA.

39: In extraction of ORF, 1st step is obtained the 3 codons.

40: Example of energy based algorithm Zukar algorithm.

41: Dynamic programming help us to……. reduce computational cost.

43: When unpaired bases in 2 region binds is result form Buldges.

44: which one of following is not objective based method of phylogenic … UPGMA

45: protein sequencing can be done by Swiss prot & UniProt.

46: Drugs can be better developed after understanding molecular basis of …… Disease

47: Online tools are available for homology modeling.

48: Insertion and deletion in alignment cause…… gaps.

49: Alligment of insertion and deletion can be performed by ……. dynamic programming

50: Exons in proteins tend to be more conserved than ……. intron.

51: Blast confined sequences from—— ID,name,spcies

52: ribosmes read RNA and collect amino acid from…… cell cytosol.

53: Simulation can predict diseases progression & outcomes.

54: For exact maching …….required. Traceback

55: trackback is start from high score on couple nucleotide.

56: Motif protein chain fold to take . 3D structure

57: Next generation sequencing is used to known about unknown sequences of nucleotide.

58: RNA fold to form 2 structure.

59: MFOLD implement the upgraded version of Zuker.

60: Which has less computational cost …… Dynamic programming

61: In scoring matrics, we score with maximum 4 matrix.

62: We compare all length of nucleotides in …. Global alignment.

63: we compare smaller fragment Local alignment.

64: Destabilized RNA molecule have (+)value.

65: In Zukar algorithm, we need construct all possible combination.

66: We use Zukar algorithm to sum up energy to find lowest matrix.

67:Swiss prot.is a box in uniprot.

68: In comparing RNA structure, we used to select which having lowest energy.

69: Addition of nucleotide is done by ….. hydrogen bondi

71: Adman degradationis used for protein sequencing.

72: Why do we create all possible combination of diagonals in zukar algorithm with overall lowest energy is selected.

73: We can do repeated sequences alignment by slight modification in waterman model.

74: Length of hairpin loop is 4 bases.

75: ‘‘T" Thrionine is an basic element....true

76: To align sequence what will be necessary firstly..... . to find conserved rigion

77: Sequence of biomolecules can be known by using ..... . uniprot or gene bank

78: Defination of Expasy: Expasy provide access to variety of online database and tools.

79: Structural information of a molecule can be obtained by X.ray crystallography , nuclear magnetic sequence.

80: Compare a protein query sequence against a nucleotide sequence database…..tBlastn

81: Protein formation of abnormal folding called Pseudoknots.

82: By using uniprot we can find sequence of…..(protein)

83: The 5´ end of molecule caped with 7 Methyl Guanosine Triphosphate.

84: DNA provide blue print for building.

85: DNA get modified by mutation, substitution, insertion, deletion.

86: UPGMA is used to calculate the distance of each cluster.

87: Global alignment used to determin overall conversation.

88: if the sequence is multiple there can be several possible alignment.

89: can we identify the post translational modifications by . Expacy

90: scoring scheme has an extremely crucial role in producing optical al[illegible]

91: why is there need to predict RNA structures to know about its function

92: DP recombines the nucleotide recombination through a process of Traceback.

93: .how many open reading frame exist for each DNA sequence.6ORFs

94: tblastn search translated nucleotide using...Protein query

95: Blast can find sequencing from protein or nucleotide database it can find known and unknown sequencing.

96: name any two frequently used scoring matrix Blosum and pam.

97: From Fasta we can obtain the information on the parent organism, function and evolution history.

98: The correct order of progressive alignment for multiple sequencing alignment is Pair wise alignment, Guide tree, branching order in tree.

99: CLUSTALW can run in two modes slow/accurate, fast/appropriate.

100: Gene bank and UniProt have Large number of sequencing publically.

101: In multiple sequencing, we have Global alignment.

102: Ubiquitin play important role recycling protein.

103: 6 form of DNA ending sequencing.

104: Non-clustering method such as maximum parsimony may be used for making trees.

105: If in needleman and Smith watermark algorithm along ‘0’ place is in……relationship. 1st row and 1st column.

107: Faster active alignment by using local alignment and protein nucleotide. pairwise alignment

109: Expasy was develop by Swiss Bioinformatics Institute (SIB)

110: .In phylogenetics tree , ancestor are placed at Root/bott[illegible]

111: .In UPQMA two sequence are .

112: .When unpaired bases in 2 region binds itself form Buldges.

113: Zukers algo[illegible]

114: In Nussinov Jacob algorithm which turn is added to compare to score for diagonal position set diagonal and lower tri diagonal to zero.

115: Trackback begins from the position in Nussinov Jacobson score matrix. 4rth possition

1.Genomics, evolutionary studies and system biology are application of **bioinformatics.**

1) Exact matching require which type of nucleotide-**same number of nucleotide.**

2) Accurate solution of protein structure is one of **the toughest problems**

3) **Tract back** is overlap matches can start from any position in scoring matrix.

5) The matrix has .....scores. **Positive and negative.**

6) **MASCOT**….._can search people mass finger printing and shotgun proteomics data set

7) Spectrometer measures the protein by_____ **Mass/charge**

8) Simulation of the folding process depends_____**Energy function**

9) Information about **DNA/RNA** is available on gene bank

1.Domain are----

2RNA brick database-----

3.overcome homology modelling limitations by-----

4.homology predicts structure of protein-------

5.aromatic amino acids-----

- 6.DNA/RNA sequence----database

7.mRNA does not have------

- 8. BLAST can find sequences----------

- 9.MS based protemics begins measurement of ----------

10. alpha carbon atom can be obtain by---

11. Ab Initio methods can predict the sytructure on the bases of _____, charg volume mass.

12. The value of one angstrom _____.

13. _____- amino acid are required for alpha helices.

14. Blosum aberration_____.

15. The pka value of valin_____-.

16. The sequences that are similar we _____ modeling.

17. Exact matching require wchich type of nucleotide_____-

18. _____- cordinats the alpha carbon.

19. We start from ___**top right**__ in global alignment.

19. In structr visualization the sequence are visibal as cartoon_____.

20. pk value of Aspartic acid **3.9**

21. expasy was made by _____-.

22. Dot plots employ dot matrix with two sequence plotted represents.....**pairwise**

23. Which give info about precursor protein……. **PSTS**

24. mgf files format can be converted to …….

25. Molecular evolution can be caused by ….insertion deletion substitution.

26. Xyz position of position is available in _____
27. We can predict better by _____.
28. The amino acid join togather by _____bond.
29. One and 4 amino in alpha helix joind by ____bond.
30. Cn3d is an online tool used for_____.
31. Dot plots employ dot matrix with two sequences plotted on_____-

40. Fasta Was Develop In(**Ans1988)**
41. In Corey Paulin Modal Of The Protein Each Atom Is Represented By(**Solid Sphere**)
42. How Many Storages Use For Structural Modling **(3)**
43. Modeler Software Use For Homology Modling Which Use(Phython File Format,Not Confrm)
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48. Rna Dont Have Structer???**Quatnaryy**
49. Large Numbr Of Sequence
50. Coding Region Of Dna
51. Blast Use For???**Unknown And Known Protein**
52. Expasy Develop
53. Bond Form Between C And N Terminas??? **Hydrogen Bond**
54. X.Y.Z Position Of Protein Are Available By??? **Pdb**
55. Technique Use To Measure The Mass Of Protein???**Mass Spactrometry**
56. Large Amino Acid??? **Tryptophan**
57. Positive Charge Amino Acid??? **Lysine And Arginine**
58. Mrna Has???Single Strand
59. C.Alpha Atom Are Traced To Recreat??? **3d Structer Of Protein Bond**
60. Form Between Two Atom??? **Energy Release**
61. Provide Clue Of Precurser Protein Peptide Sequence??**Pst**
62. Amino Acid Requird For Betasheet??? **3-5**
63. Are Formed By Bonding Betweenci And Ni+4n Protein Backbone??? **Alpha Halix**
64. Blasta Stand For??? **Basic Local Alignment Search Tool Amino Acid**
65. Have Uniwe Trait Which Can Help In??? **Protein Folding Align**
66. Comparison Betweendna/Dna Anddna/Protein??? **Fasta.**
67. Ratio Use To Calculate Mass Of Protein**??? M/Q**.
68. Treading Not Requied For??? **Structer Of Entire Backbone.**

Needle Man Wounch Man Or Smith Water Man Algorithm Use For**??? Local And Global Alignmnt.**

1. Electro poration technique?

Ans. It is a micro biology techniquein which an electrical field is applied to cells in order to increase the permeability of the cell membrane , allowing chemical drugs or DNA to be introduced into the cell(also called electrotransfer)

2. Short note on DNA profiling?

DNA profiling or DNA finger printing is a forensic technique used to identify individuals by the characteristics of their DNA

3. Criteria for the synthesis of probe?

A great advantages of hybridization for library screening is that it is extremely versatile. Condition can be used in which hybridization is very stringent, so that only sequence identical to the probe are identified.

4. Benton lift procedure?

Benton and Davis (1977) devised a method called plaque lift,in which nitrocellulose filter is applied at the upper side of agar plate making direct contact between plaques and filter

Number of alternative labeling methods ae available that avoid the use of radioactivity.it included digoxogenin or biotin

5. Various factors which effects the mobility of DNA in the gel?

1.Concentration of gel either high concentration or low concentratio**n**

2. Ionic strength of buffer

3.Voltage

4.Ethidium bromide(a chemical used as a stain to dye gel)

6. Side effects of ethidium bromide if contact with body?

Ethidium bromide may be a mutagen,carcinogen.or a teratogen

Health effects: because ethidium bromide can bind with DNA it is highly toxic as mutagen. It may potentially cause carcinogenic or teratogenic effects.exposour routes of ethidium bromide are inhalation,ingestion and skin absorption.

7. What is chain termination procedure?

Chain termination or dideoxy procedure

This method was developed by ssanger et al.(1977) .it capitalizes on the ability of DNA polymerase to incorporate dideoxy nucleoside triphosphates as substrate.

Chain termination or dideoxy procedure

DNA synthesis is carried out in the presence of four deoxynucleoside triphosphates one or more of which is labeled with P32 and in four separate incubation mixes

8. How fragments of desired size can be obtained?

Average fragments size can be controlled by mechanical shearing

The more commonly used procedure is restriction endonuclease**.**

9. How PCR helps to improve our environment?

PCR has also facilitate research in detection of pathogens in plants ,animals and environment.

10. Write the application of AFLP?

AFLP is a PCR based tool used in genetic engineering

11. Define biotechnology and name its branches?

The exploition of biological processes for industrial and other purposes especially the genetic manipulation of micro-organisms for the production of anti-biotics and harmones etc.

12. Why PCR is used as alternative mean in cloning?

PCR with specific primer could be used to isolate genes directly from genomic DNA, obviating the need for the production of genomic libraries.

13. Difference between linker and adopter?

The main difference between linker and adaptore is that linker have bulnt end while adopter have one cohesive end

14. Linkage mapping?

Genetic linkage maps illustrate the order of gene or genetic markeron a chromosome and the relative distance between those genes

15. Inverse PCR?

Inverse PCR is used to identify the flanking sequences around genomic inserts

16. Insertional inactivation?

Insertional inactivation is the inactivation of a gene upon insertion of another gene inside in its place or within its coding sequence

17. Sequencing of genome?

The process of determining the exact order of nucleotide within DNA or RNA molecule is called sequencing genome.

18. Green biotechnology?

It is concerned witht the modification of the genetic composition of plants to enhance existing traits or add new ones.

Example Bt corn from syngenta.Bt cotton from monsanto

19. Cloning strategies?

Any cell based cloning procedure has 4 essential parts

i). a method of generating DNA fragments for cloning

ii).a reaction which insert that fragment into the chosen cloning vector

iii).a mean for introducing that recombinant vector into a host cell where it is replicated

iv).a method of selecting recipient cell that have acquired the recombinant

20. Griffths experiment?

Grifth experiment was an experiment done in 1928 by Frederick Griffith. It was one of the experiment showing that bacteria can get DNA through a process called transformation. Griffith used two strains of streptococcus pneumonia. These bacteria infect mice.

21. Cosmids?

Plasmids have been constructed that contain a fragment of lambda DNA including *cos* sites

22. Degenerated primer?

Degenerate primer is a mixture of primer , all of similar sequence but with variation at one or more positions

23. One difference between hetrochromatin and euchromatin?

Difference Between Heterochromatin and Euchromatin. The major difference between heterochromatin and euchromatin is that heterochromatin is such part of the chromosomes, which is a firmly packed form and are genetically inactive, while euchromatin is an uncoiled (loosely) packed form of chromatin and are genetically active.

24. Importance of plasmid copy?

The number of ribosomes in a cell far exceeds any one class of mRNA

One way of increasing the expression of a cloned gene is to increase the number of the corresponding transcripts.

25. Note on RAPD?

RAPD- Random Amplification of Polymorphism DNA is a type of PCR reaction but the segments od DNA that are amplified are random.

26. Stability of plasmids?

Having maximized the expression of a particular gene it is important to consider what effect this will have on the bacterium harbouring the recombinant plasmid.

27. Chromosomes jumping?

In chromosome jumping, the DNA of interest is identified , cut into fragments with restriction enzymes and circularized.

28. RFLP?

Restriction fragments length polymorphism exploit variation in homologous DNA sequences .

In RFLP, DNA sample is digested by restriction enzymes and separated by gel electrophoresis.

29. RH mapping?

RH mapping used X-ray breakage of chromosomesto determine the distance between DNA marker as well as their order in chromosome.

30. Where is s-d located?

Shine dalgarno(s-d) sequence

It is a complimentary base pair with 3` end of 6s rRNA

31. Real time and multiple PCR?

Real time quantitative PCR is used to measure the quantity of target sequence. It is used to measure starting amount of DNA, cDNA ,RNa

Multiple PCR is used to amplify several DNA simultaneously

32. Difference between chimpanzee and human genome?

Human and chimpanzee genomes differ by 1.2%, at single base-pairs, and by

2.7% because of insertions and deletions

33. Charguff rule?

Charguff rule state that DNA from any cell of any organism shoul have 1:1 ratio of pyrimidine and purine bases and more specifically that the amount of guanine should be equal to cytosine and the amount of adenine should be equal to thymine. This pattern is found in both strands of the DNA.

34. As MS chamber is only be used for ionization then, how can we make possible further fragmentation of protein in it?

ANS: For this sac there will be an addition of fragmentation instrument in MS chamber, which can perform this job nicely.

35 Name the techniques used for protein fragmentation?

ANS: 1. Electron capture dissociation

2. Electron transfer dissociation

3. Collision induced dissociation

36 Define PSTs? No option

Ans: peptide sequence tags (PSTs) are actually small amino acid sequence of peptides producing during MS2.

37what are three major types of scoring scheme which obtain from protein sequence data base search algorithm?

Ans: 1) MW match score 2) PST match score 3) invitro and in silico match score

Q. Name two types of open formats?

Ans: (1) mzXML (2) MGF

Q. Name some companies that are making mass spectrometer in market? Ans:

Agilent ----- Bruker ----- ABI ----- Micromass ----- Waters ----- Shimadzu

Q. How many types of masses are there? Mention their names.

Ans: 1) Nominal Mass 2)Monoisotopic Mass 3) Average Mass

Q. Differentiate between polar and non polar amino acids?

Ans: **Non polar amino acids:**

Non polar amino acids are hydrophobic in nature. They are mostly found inside of folded protein.H and C introduce very little dipole movement. e.g Phe, Ala, Leo, Val etc.

Polar amino acids:

These amino acids are polar but are not charged. Prefer to reside or interact to aqueous environment. Mostly found at surface of folded protein. E.g Cys, Gly, Ser, Thr, Tyr etc.

Q. What is meant by “PK” value of an amino acid?

Ans: PK is the value for an amino acid is that PH at which exactly half of amino acids are charged and half are not charged.

Q. what are the key factors participating in protein folding?

Ans: solubility and reactivity are the key factors that are participating in protein folding.

Q. Mention some unique properties of amino acids? No option

Ans: amino acids has unique set of properties such as charge state, polarity and hydrophobicity.

Q. write two methods which we use to obtain 1’ structure? No option

Ans: 1. Edman degradation

2. tandem mass spectrometry

Q. briefly describe any three properties of Loops? NO option

Ans: • Loops connect helices and sheets • Loops vary in length and 3-D configurations • Loops are mostly located on the surface of proteins • Loops are more “acceptable" of mutations • Loops are flexible & can adopt multiple conformations • Loops tend to have charged and polar amino acids • Loops are frequently components of active sites

Q. Define angstrom. No options

Ans: Angstrom is a unit of length equal to 10^{-10} m (one tenbillionth of a meter) or 0.1nm .

Q. Name some atomic representation techniques which are helpful in protein structure visualization? No options

Ans: These include CPK, Ribbon and Balls & Stick diagrams.

Q. Name any two Techniques for structure determination. No options

Ans: 1. X-Ray Crystallography

2. Nuclear Magnetic Resonance (NMR) Spectroscopy

Q. What are the other possible structures which beta strands can give rise to?

ANS: 1. Beta Sheet 2. Beta Barrel 3. Beta Sandwiches

Q.. How many types of Domains are there?

ANS: 1). Alpha Domains 2). Beta Domains

3). Alpha/Beta Domains 4). Alpha + Beta Domains

5). Alpha & Beta Multi-Domains 6). Membrane & cell-surface proteins

Q..Name the Algorithms use for Structure Classification? No option

Ans:1. CATH 2.CSOP 3. FSSP 4. Pclass

Q.What is the abbreviation of I-TASSER?

Ans: Iterative threading assembly refinement (I-TASSER)

Q. How many Strategies are there for Structural Modelling?

Ans: There are three basic Strategies,

1.Homology Modelling

2.Fold Recognition

3.Ab Initio Modelling

Q. How many basic types of Alignments, mention their names ?

Ans: there are two types of alignment,

Global Alignment (Needle Wunsch)

Local Alignment (Smith Waterman)

Q. How many types of Phylogenetic Trees are there?

Ans:

There are two types of phylogenetic trees:

• Rooted Trees

• Un rooted Trees

Q. ExPASY stand for ?

Ans: ExPASY is one of the best collection of data and tools for proteomics. It stand for “Expert protein analysis system”

Q. differentiate between bottom up and top down proteomics?

Ans: There are two major types of MS-based proteomics e.g mass spectrometry proteomics,

“Bottom up proteomics deals with peptides”.

“Top down proteomics can handle whole proteins”.

Bottom up proteomics measures the peptide masses produced after protein enzymatic digestion. And Top down proteomics measures the intact proteins followed by peptides after fragmentation.

Q. how loops are found after the finding of alpha and beta

Ans: After computing the propensity of alpha helices and beta sheets, we need to settle for loops.

- Let’s see how can we find out the loops using Chou Fasman Algorithm. For any jth residue in sequence, we calculate f (Total) = f(j) f(j+1) f(j+2) f(j+3) (tetra peptide).
- If,
- f (Total) > 0.000075.
- The average value for P (turn) > 1.00 in the tetra peptide.
- The averages for the tetra peptide are such P(a-helix) < P(turn) > P(b-sheet), It is a Turn!

Q. . How protein structure identified?

Ans: If another protein which has a similar sequence also has its structure known, the structure of an unknown protein can be predicted based on that similar protein. So, it is then possible to identify unknown protein structures by just examining the homologous protein sequences. Good sequence alignment and identity ensures that homology modelling will give accurate results

Thus, Homology modeling is used to predict structures of proteins having high sequence similarity with other proteins with known structures:

Q. What happens when pH is < or > than pK?

Ans: If pH<pK for an amino acid the amine side chain gain a proton (H+) and become a positively charged, hence basic.

If pH > pK for an amino acid, the carboxyl side chains loses a proton (H+) and become negatively charged, hence acidic.

Q. What are the properties and role of amino acids?

Ans: Properties:

Amino acids have several properties such as charge

state, polarity and hydrophobicity Amino acids not only have physical and chemical properties, but also structural properties

Role:

These structural properties are equally important in

giving rise to protein structures Since some amino acids are hydrophobic, they may be employed in forming a stable core in a protein

• Also, chemically inactive amino acids reduce chances of destabilizing reactions in core

Q. Why folding is important?

Ans: Proteins fold spontaneously. Proteins fold to achieve thermodynamic stability. Proteins fold to organize themselves for performing functions in cells.

Q. What is ab inition modelling describe the limitations of this model?

Ans: Ab initio methods have Anfinsen's thermodynamic hypothesis at the center

- These methods attempt to identify the structure with minimum free energy

Limitation:

- Computationally expensive
- Suitable for proteins with less than 100 residues

Q. write down the terms of PDB file format.

Ans: **HEADER**: contain the brief description of the structure, the date and PDB ID code.

TITLE: title of the structure.

SOURCE: identifies which organism the structure came from.

COMPND: brief deail of the structure.

REVDAT: the date of the last revision.

JRNL: one or more literature reference thet describe the structure.

KEYWDS: list the set of useful words/phrases that describe the structure.

AUTHOR: the scientist depositing the structure.

REMARKS: detail of the experimental method used to determine the structure are contain in this subsection.

Q. define ionization in terms of proteins?

Ans: ionization of protein means to ionize a protein molecule to make it charged by removing an electron from it. Protein ionization is the first step in MS-based proteomics protocols

- Ionization process involves loading of a proton onto a protein or the removal of a proton from the protein
- Ionization results in an increase or decrease of the protein/peptide mass Two primary techniques used for ionization of protein,

1. Electrospray ionization (ESI)
2. Matrix - assisted laser desorption/ionization (MALDI)

3. Q. Briefly describe chou fasman algorithm?

Ans: **Chou – Fasman method:**
It is a technique for the prediction of secondary structures in proteins i.e. Alpha Helices, Beta Sheets and Turns is Chou – Fasman technique. The method is based on analyses of the relative frequencies of each amino acid in alpha helices, beta sheets, and turns based on known protein structures solved with X-ray crystallography. From these frequencies a set of probability parameters (in our handouts, it is propensity table) were derived

- **For the appearance** of each amino acid in each secondary structure type.
- **To predict the probability** that a given sequence of amino acids would form a helix, a beta strand, or a turn in a protein.

• Q. How can alpha helix expressed?

Ans: Chou-fasman algorithm (alpha helix):

- For Alpha Helices, 4 contiguous amino acids are required. Their Alpha-Helix propensity should be more than 1.0 Once this propensity falls below 1.0, Alpha-Helix stops.

• Q. Describe scores in silico fragmentation?

Ans: Silico fragmentation scoring:

- Count the matches between in silico and in vitro peaks.
- Give an equivalent score to the candidate protein.
- Weigh each of the aforementioned match by the mass error. Accumulate the score

Q. What are hydrophobic amino acids?

Ans: Hydrophobic Amino Acids:
Since H and C introduce very little dipole moments in hydrophobic amino acids, these amino acids are non polar. Hydrophobic amino acids are mostly found at the inside of folded proteins.

Q. Explain summary of alpha helix and beeta sheets?

Ans: Alpha Helices are formed by hydrogen bonding (O-H) between Ci and Ni+4 atoms in the protein backbone. X-Ray Crystallography of protein shows that Hydrogen atoms of N-Term come together with Oxygen atoms of C-term amino acid at 4th neighboring position to make Hydrogen bonds!

Beta Sheets are an other common secondary structure

- They are constituted by several Beta Strands which come together
- 5 to 10 resides are needed to make a Beta Strand, typically

Q. what is 3D_1D bowie algorithem?

Ans: Proposed by Bowie et al in 1991

- Converts 3D structure into a 1-D string profile for each structure in the fold library
- Align the target sequence to these profiles 3D-1D methods convert structure and environment information into “profiles”
- Score for each amino acid is computed for each profile

Q. public database that commenly used for protein?

Ans: Examples include Uniprot, Swissprot amongst several others.

Write steps of protein sequence identification? OR Enlist protein sequence method.

Ans: Mass spectrometers are used to measure the molecular weight of proteins and peptides.

Following steps involve in protein sequence identification,

1. complex protein
2. separation
3. ionization by mass spectrometery
4. fragmentation MS2
5. mass spectra
6. EST Determination
7. filter protein database
8. in silico fragmentation of candidate protein

9.matching of experimental and insilco peak list

10. post translational modification
11. protein score.

Define Peptide mass finger printing nd shotgun?

Ans: Shotgun proteomics digests the entire protein mix 1st followed by peptide analysis & protein database search

• Peptide mass fingerprinting involves protein separation followed by a single protein's peptide analysis.

Q. Principal of dynamic programming?

"Dynamic programming is an algorithm technique used commonly in sequence analysis".

Select a nucleotide from RNA sequence

AUGCUGUCAUCG

- List all possible complementary positions for selected nucleotide in the complete sequence

AUGCUGUCAUCG
AUGCUGUCAUCG
AUGCUGUCAUCG
AUGCUGUCAUCG and so on…

DP exhaustively makes all possible nucleotide combinations

- DP then recombines such combinations in a process called "Traceback" to ensure that the highest coupled 2' structure is reported.

Q. write types of RNA ?

ans: **Coding RNAs**

• Coding RNAs as is obvious from their name , code for proetein
• **Non-coding RNAs**
• Non-coding RNAs regulate/assist in the process of translation.Q

Other types of RNA are,

1= m RNA

2=t RNA

3= r RNA

4= miRNA 5= si RNA

Q.: Explain different databases and their specificity.

Ans: There are two basic types of data bases, ***1. Gene Bank:***
It is an online public sequence data base. In Gene Bank we go to online NCBI

Gene Bank to find

sequence of DNA and RNA using following Website,
www.ncbi.nim.nih.gov/genebank Gene Bank can be research by typing,

1. Sequence name 2. Species 3. ID 4. Locus 5. Author 6. Journal 7. Accession number 8. Name

2.Uniport:
It is also an online public data base. By using Uniport we can find sequence of proteins. If

we want to find a protein sequence we must go an online Website given bellow, www.uniport.org

In home page there is a box named **"Swiss Prot"** which contains human curated protein information, molecular mass, observed and predicted modifications etc. uniport can be research by typing. 1. Amino acid 2. Name 3. ID 4. Sequence

Further Data Bases and their specificity is mention bellow,
Taxonomic Database …… For classification

Genomic Database …… For Gene level information

Q1 ;Optimal energy function in structural prediction?

Ans: Energy based methods involve evaluating the free energy structures. To compute the RNA sequence for 1' or 2' optimal structure prediction we use Zuker's Algorithm. RNA Secondary Structure Prediction. Zuker's Algorithm helps us to compute the stabilizing energies (-ve) and also destabilizing energies (+ve values). And also compute the sum of +ve and –ve energies. All possible 2' structures are generated. The best 2' structure is selected

Q2; RAW file Formet (5)

is a format in which an instrument outputs data in binary form. Several software exist for converting RAW file formats into open software formats. Each open format has its own unique advantages. mzXML and MGF formats are most frequently used

Q;3 Types of secondary structures of RNA:

1. Single stranded: 3'end may fold on to the 5' end

2. Helices: Double stranded RNA helix of stacked base pairs

3. Hairpin loop: The loop of the hairpin must at least four bases long to avoid steric hindrance with base-pairing in the stem part of the structure

4**. Bulge Loops**: Bulges, are formed when a double-stranded region cannot form base pairs perfectly

5. Interior loop: Interior loops are formed by an asymmetric number of unpaired bases on each side of the loop.

Q4Advantage and disadvantage of Ab initio:

Advantages:

- Ab Initio methods can fold any target sequence using only physical atomic properties.
- Predictions are mostly accurate and correctly describe the natural folding process.

Disadvantages:

- Ab initio methods are the very difficult to design (energy function).
- - These methods are slow due to the huge possibilities

.Q5 Describe scores in silico fragmentation?

Ans: Silico fragmentation scoring:

- Count the matches between in silico and in vitro peaks.
- Give an equivalent score to the candidate protein.
- Weigh each of the aforementioned match by the mass error.
- Accumulate the score\
- With "all possible" fragments in in silico spectrum, and "reported" fragments in experimental spectrum, we can match and rank.
- Scoring scheme should also consider the errors in peak matching

Q6';Function of Tandem Ms:

Tandem MS helps in measurement of mass to the fragments as well. This process provides another step in further scoring and ranking and Protein identification thus becomes easier.

Q7Write steps of protein sequence identification? \

Ans: Mass spectrometers are used to measure the molecular weight of proteins and peptides. Following steps involve in protein sequence identification,

1. complex protein
2. separation
3. . ionization by mass spectrometery
4. . fragmentation MS2
5. . mass spectra
6. . EST Determination
7. . filter protein database
8. . in silico fragmentation of candidate protein
9. .matching of experimental and insilco peak list
10. . post translational modification
11. protein score.

Q8Silico fragment scoring?

- Count the matches between in silico and in vitro peaks.
- Give an equivalent score to the candidate protein
- Weigh each of the aforementioned match by the mass error
- .Accumulate the score

Q9HOW Propensity table help us

For a primary sequence, and a tentative 2’ structure, propensity table can help us compute the overall propensity. Product of propensity values is computed for overall propensity for each 2’ structure. An important point to note here is that 2’ structures are formed due to hydrogen bonding between amino acids.

So, we need to consider the neighboring amino acids as well.

You only need to compute propensities for a small number 2’ structures. The highest net propensity will be the most probably secondary structure that will be formed.

Q10 Factors That Participate In Folding Protein\

The external factors involved in protein denaturation or disruption of the native state include temperature, external fields (electric, magnetic), molecular crowding, and even the limitation of space, which can have a big influence on the folding of proteins. (Internet

Q11. Different Secondary Structures Of RNA?

The secondary structures of biological DNA's and RNA's tend to be different: biological DNA mostly exists as fully base paired double helices, while biological RNA is single stranded and often forms complex and intricate base-pairing interactions due to its increased ability to form hydrogen bonds stemming from the

Q12Role of amino acid

These structural properties are equally important in giving rise to protein structures Since some amino acids are hydrophobic, they may be employed in forming a stable core in a protein • Also, chemically inactive amino acids reduce chances of destabilizing reactions in core

Q13Briefly describe chou fasman algorithm? \

Ans: Chou – Fasman method: It is a technique for the prediction of secondary structures in proteins i.e. Alpha Helices, Beta Sheets and Turns is Chou – Fasman technique. The method is based on analyses of the relative frequencies of each amino acid in alpha helices, beta sheets, and turns based on known protein structures solved with X-ray crystallography. From these frequencies a set of probability parameters (in our handouts, it is propensity table) were derived For the appearance of each amino acid in each secondary structure type. To predict the probability that a given sequence of amino acids would form a helix, a beta strand, or a turn in a protein

Q14 Chou-fasman algorithm (alpha helix):

- For Alpha Helices, 4 contiguous amino acids are required.\
- Their Alpha-Helix propensity should be more than 1.0

- Once this propensity falls below 1.0, Alpha-Helix stops

Q15Use of mass spectrometer

Mass spectrometer is used to measure mass/charge ratio of ionized proteins and peptides. Data output from the MS comprises of m/z ratios and intensities of each molecule that is measured.

Q16Hydrophobicity of amino acid

mino acids have characteristics like polarity, hydrophobicity, and charge states. These characteristics are governed by the elemental composition of an amino acid's side chain (R group).

HYDROPHILIC AMINO ACIDS Since H and C introduce very little dipole moments in hydrophobic amino acids, these amino acids are non-polar. Hydrophobic amino acids are mostly found at the inside of folded proteins. Hydrophilic group contain the chain of C and H group in their R grouP

Q17Role of Amino Acids

We know that amino acids can be polar, charged and hydrophobic. Role of polar and charged amino acids in folding. Role of hydrophobic amino acids in folding

Q18Overall Goal of Folding

Anfinsen's thermodynamic hypothesis: Proteins fold for a unique, stable and minimum free kinetic energy structure. What other factors may come into play for satisfying Anfinsen hypothesis

Q19Need For Chou Fasman Algorithms?

The Chou-Fasman algorithm is simple in principle.The conformational parameters for each amino acid were calculated by considering the relative frequency of a given amino acid within a protein, its occurrence in a given type of secondary structure, and the fraction of residues occurring in that type of structure.

Q20 Role of Global Protein?

Global protein function prediction from protein-protein interaction networks. ... The availability of entire genome sequences and of high-throughput capabilities to determine gene coexpression patterns has shifted the research focus from the study of single proteins or small complexes to that of the entire proteome. (Internet)

Q21Chou – Fasman method?

t is a technique for the prediction of secondary structures in proteins i.e. Alpha Helices, Beta Sheets and Turns is Chou – Fasman technique. The method is based on analyses of the relative frequencies of each amino acid in alpha helices, beta sheets, and turns based on known protein structures solved with X-ray crystallography. From these frequencies a set of probability parameters (in our handouts, it is propensity table) were derived

 For the appearance of each amino acid in each secondary structure type. To predict the probability that a given sequence of amino acids would form a helix, a beta strand, or a turn in a protein.

Chou-fasman algorithm (alpha helix): For Alpha Helices, 4 contiguous amino acids are required. Their Alpha-Helix propensity should be more than 1.0 Once this propensity falls below 1.0, Alpha-Helix stops

Q22. What Is Ab Initio Modelling? Describe The Limitations Of This Model?\\

Ab initio methods have Anfinsen'sthermodynamic hypothesis at the center • These methods attempt to identify the structure with minimum free energy. • Applicable to any sequence • Not

very accurate biologically • Accuracy & applicability are limited by our understanding of the protein folding problem. Limitation • Computationally expensive • Suitable for proteins with less than 100 residues (PPT)

Q23 What is meant by “PK” value of an amino acid?

Ans: PK is the value for an amino acid is that PH at which exactly half of amino acids are charged and half are not charged

Limitation: Models can be conceptual, graphical or mathematical as they are used in science. It allows one to break down a concept into simpler terms with a visual component. A limitation of models in science is that they are usually simplified versions of the real situation or concept. (Internet)

Q24 Why folding is important?

Ans: Proteins fold spontaneously. Proteins fold

to achieve thermodynamic stability. Proteins fold to organize themselves for performing functions in cells

1. **Seven Steps to Homology Modelling?**

Template recognition and initial alignment • Alignment correction • Backbone generation • Loop modeling
• Side-chain modeling • Model optimization • Model validation

2. **Factors That Participate In Folding Protein?**

The external factors involved in protein denaturation or disruption of the native state include temperature, external fields (electric, magnetic), molecular crowding, and even the limitation of space, which can have a big influence on the folding of proteins.

3. **Role of Amino Acid?**

Protein plays a crucial role in almost all biological processes and amino acids are the building blocks of it. A large proportion of our cells, muscles and tissue is made up of amino acids, meaning they carry out many important bodily functions, such as giving cells their structure.

4. **Role of Global Protein?**

Global protein function prediction from protein-protein interaction networks. ... The availability of entire genome sequences and of high-throughput capabilities to determine gene coexpression patterns has shifted the research focus from the study of single proteins or small complexes to that of the entire proteome.

5. **Acidic and Basic Amino Acid?**

There are three amino acids that have basic side chains at neutral pH. These are arginine (Arg), lysine (Lys), and histidine (His). Their side chains contain nitrogen and resemble ammonia, which is a base. Their pKa's are high enough that they tend to bind protons, gaining a positive charge in the process.

6. **Points of Bottom up Proteomic?**

The major alternative workflow used in high-throughput proteomics is called top-down proteomics and does not use proteolytic digestion. Essentially, bottom-up proteomics is a means of determining the protein make- up of a given sample of cells, tissues, etc.

7. **Varies RNA Secondary Structure?**

Types of RNA Secondary Structures - I • RNA Complementary bases in primary structure bond together to fold RNA into a secondary structure

Types of RNA Secondary Structures - II • RNA 1' sequence (5' – 3') can fold onto itself to make 2' structures.
• Two previous examples of 2' structures include Helix and Hairpin structures.

Types of RNA Secondary Structures - III • RNA can be fold to form helices, bulge loops, and interior loops. An other 2' RNA structure is the Junction or Intersection

8. **Objective Base Method and Clustering Approach?**

Clustering Approach UPGMA WPGMA Neighbor Joining Single Linkage Complete Linkage
Objective based Methods Least Square Distances Maximum Likelihood Maximum Parsimony

9. Shot Gun and Peptide Mass Finger Printing?

Shotgun proteomics digests the entire protein mix 1st followed by peptide analysis & protein database search

Peptide mass fingerprinting involves protein separation followed by a single protein's peptide analysis.

10. Need For Chou Fasman Algorithms?

The Chou-Fasman algorithm is simple in principle.The conformational parameters for each amino acid were calculated by considering the relative frequency of a given amino acid within a protein, its occurrence in a given type of secondary structure, and the fraction of residues occurring in that type of structure.

11. Substitution and Indels?

Substitution at adjacent nucleotides (primarily substitutions at two adjacent nucleotides, but substitutions at three adjacent nucleotides have been observed.

Indels, being either insertions, or deletions, can be used as genetic markers in natural populations, especially in phylogenetic studies.

12. Advantages and Disadvantages Of Ab Initio Modeling?

Advantages • Ab Initio methods can fold any target sequence using only physical atomic properties • Predictions are mostly accurate and correctly describe the natural folding process.

Disadvantages • Ab initio methods are the very difficult to design (energy function) • These methods are slow due to the huge possibilities • An order of 1012 steps are needed to simulate protein folding for medium sized protein structures.

13. PK Value is Less than PH Why?

The lower the pKa, the stronger the acid and the greater its ability to donate protons. pH depends on the concentration of the solution. This is important because it means a weak acid could actually have a lower pH than a diluted strong acid. On the other hand, the pKa value is a constant for each type of molecule.

14. What Is Hydrophobicity?

In chemistry, hydrophobicity is the physical property of a molecule that is seemingly repelled from a mass of water. In contrast, hydrophiles are attracted to water. Hydrophobic molecules tend to be nonpolar and, thus, prefer other neutral molecules and nonpolar solvents.

15. Difference Between Folding Protein and Denature Protein, Role Of Folding Protein, and Forces Act on Protein / What Factors Are Used For Folding Proceeds?

Protein folding is the physical process by which a protein chain acquires its native 3-dimensional structure, a conformation that is usually biologically functional, in an expeditious and reproducible manner.

Denaturation of proteins involves the disruption and possible destruction of both the secondary and tertiary structures. Since denaturation reactions are not strong enough to break the peptide bonds, the primary structure (sequence of amino acids) remains the same after a denaturation process.

Protein folding is a process by which a polypeptide chain folds to become a biologically active protein in its native 3D structure. Protein structure is crucial to its function. Folded proteins are held together by various molecular interactions The amino acid sequence of a protein determines its 3D structure.

The forces involved in protein folding include:

• Electrostatic interactions, • van der Waals interactions, • Hydrogen bonds, • Hydrophobic interactions

16. Prediction of Protein Structure?

Protein structure prediction is the inference of the three-dimensional structure of a protein from its amino acid sequence—that is, the prediction of its folding and its secondary and tertiary structure from its primary structure.

17. How Can We Predict Amino Acid In Alpha Helix And Beta Sheets?

Proteins that convert from alpha helix to beta sheet: implications for folding and disease. The sequence of a protein normally determines which amino acid residues will form alpha helices, and which one beta sheets, to an extent that allows secondary structure prediction to be made with a reasonable reliability.

20. Chou Fasman (Loop Formation after Alpha and Beta Sheet)? / How Loops Are Found After The Finding Of Alpha And Beta Sheets?

Alpha helices and beta strands are connected by these turns and loops Loops and turns generally lie on
the surfaces of proteins so they often participate in interactions between proteins and other molecules. In a loop, there are no regular structures as can be found in helices or beta strands.

21. The Term Expansy Stands For?

Expert Protein Analysis System

Expasy • Developed by Swiss Bioinformatics Institute (SIB) • Website provides access to databases and tools

• Proteomics, Genomics, Phylogeny, Systems biology, Population genetics, transcriptomics etc. **(PPT)**

22. Differentiate Between Bottom Up Proteomics And Top Down Proteomics?

Bottom Up Proteomics: Measures peptide masses produced after precursor protein's enzymatic digestion.

Top Down Proteomics: Measures intact proteins followed by their peptides after fragmentation.

23. How Proteins Sequences Identified?

Predicting from DNA/RNA sequences. In biology, proteins are produced by translation of messenger RNA (mRNA) with the protein sequence deriving from the sequence of codons in the mRNA. .....Predicted protein
sequences are an important resource for protein identification by mass spectrometry.

24. Why Folding Is Important?

Proteins have several layers of structure each of which is important in the process of protein folding. The first most basic level of this structure is the sequence of amino acids themselves. The sequencing is important because it will determine the types of interactions seen in the protein as it is folding.

25. What Is Ab Initio Modelling? Describe The Limitations Of This Model?

Ab initio methods have Anfinsen'sthermodynamic hypothesis at the center • These methods attempt to identify the structure with minimum free energy. • Applicable to any sequence • Not very accurate biologically • Accuracy & applicability are limited by our understanding of the protein folding problem.

Limitation • Computationally expensive • Suitable for proteins with less than 100 residues (PPT)

Limitation: Models can be conceptual, graphical or mathematical as they are used in science. It allows one to break down a concept into simpler terms with a visual component. A limitation of models in science is that they are usually simplified versions of the real situation or concept.

26. Write Down The Following Terms Of PDB (3)

Title (Title of the structure)
Source (Identify which organism structure from) COMPND (Brief detail of the structure)
REVEDAT (The data of last revision)

27. What Happend When Ph Is Less Than The Pk Value?

If the pH is higher than the pKa, then the compound will be deprotonated. A further consideration is the charge on the compound. Acids are neutral when protonated and negatively charged (ionized) when deprotonated. Bases are neutral when deprotonated and positively charged (ionized) when protonated.

28. Define Ionization In Terms Of Proteins?

Any process by which a neutral atom gains or loses electrons, thus acquiring a net charge, as the dissociation of a substance in solution into ions or ion production by the passage of radioactive particles.

29. Mathematical Relation For Representing Phylogenetic Tree?

A phylogenetic tree is a diagram that represents evolutionary relationships among organisms In trees, two
species are more related if they have a more recent common ancestor and less related if they have a less recent common ancestor. Phylogenetic trees can be drawn in various equivalent styles.

30. Define MGF?

MGF – Mascot Generic Format • MGF is a simple human-readable format for MS/MS data developed by Matrix Science • Mascot Search Engine available at http://www.matrixscience.com/search_form_select.html.

31. What Is Difference Stack Energy And ---------- Energy?

The stacking-fault energy (SFE) is a materials property on a very small scale. ... A stacking fault is an interruption of the normal stacking sequence of atomic planes in a close-packed crystal structure.

32. Summary Of Alpha Helix By Chou's Algorithm?

?????????????

33. How Homology Modelling Is Used To Sequence Unknown. Protein Sequence?

Homology modeling is a procedure that generates a previously unknown protein structure by "fitting" its sequence (target) into a known structure (template), given a certain level of sequence homology (at least 30%) between target and template.

34. Chou Fasman Algorithm?

The Chou–Fasman method is an empirical technique for the prediction of secondary structures in proteins, originally developed in the 1970s by Peter Y. Chou and Gerald D. Fasman.

35. Improvements Of Chou Fasman Algorithm?

The Chou-Fasman algorithm, one of the earliest methods, has been successfully applied to the prediction. ...

We improved Chou-Fasman method in three aspects. (a) Replace the nucleation regions with extreme values of coefficients calculated by the continuous wavelet transform.

36. Limitations Of Chou Fasman?

The Chou–Fasman method has been widely used for predicting protein secondary structure. It is based on knowledge of the potential of amino acid residues to form α-helical or β-sheet regions in proteins. Our main interest in this study was to examine the reliability of these Chou–Fasman parameters.

37. Beta Sheet By Chou Fasman and Alpha Helix By Chou fasman?

The Chou–Fasman method predicts helices and strands in a similar fashion, first searching linearly through the sequence for a "nucleation" region of high helix or strand probability and then extending the region until a subsequent four-residue window carries a probability of less than 1.

The Chou–Fasman method is an empirical technique for the prediction of secondary structures in proteins, originally developed in the 1970s by Peter Y. Chou and Gerald D. Fasman. The method is based on analyses of the relative frequencies of each amino acid in alpha helices, beta sheets, ...

38. Propensity?

In the statistical analysis of observational data, propensity score matching (PSM) is a statistical matching technique that attempts to estimate the effect of a treatment, policy, or other intervention by accounting for the covariates that predict receiving the treatment.

39. Fragmentation Of Protein?

Protein fragmentation is a common practice in the study of. protein structure and function. 1'2 Specific cleavage of certain. peptide bonds in the primary structure of a protein is essential in.

40. Hydrophobic Amino Acids?

Hydrophobic Amino Acids. Amino acids are grouped according to what their side chains are like. The nine amino acids that have hydrophobic side chains are glycine (Gly), alanine (Ala), valine (Val), leucine (Leu), isoleucine (Ile), proline (Pro), phenylalanine (Phe), methionine (Met), and tryptophan (Trp).

41. Optimal Energy Function In Protein?

Results: For a single protein sequence, the probability of success (i.e. the probability that the folded state is the lowest energy state) is derived. We then maximize the average probability of success for a set of proteins to obtain the optimal potential energy function.

42. Blast And Fasta Tools In Sequence Alignmnedt And Its Differences?

BLAST is the most widely used tool for the local alignment of nucleotide and amino acid sequences. FASTA is a fine similarity searching tool which uses sequence patterns or words. ... The main difference between BLAST and FASTA is in the similarity searching strategies used in each tool.

43. Protein Scoring?

Protein digestibility-corrected amino acid score (PDCAAS) is a method of evaluating the quality of a protein based on both the amino acid requirements of humans and their ability to digest it.

44. Why Is It Difficult To Predict The Structure Of Protein?

It is much more difficult to compare two models that predict a protein whose structure is unknown and determine which one is more accurate." ... Many protein structures are unknown. Groups around the world are all attempting to determine the structures of proteins that are important for current research.

45. Silico Fragments Comparison?

Matching experimental fragments with in silico fragments is the final resort in protein search & identification

46. How Edman Degradation Work?

Edman degradation. Edman degradation, developed by Pehr Edman, is a method of sequencing amino acids in a peptide. In this method, the amino-terminal residue is labeled and cleaved from the peptide without disrupting the peptide bonds between other amino acid residues.

47. Integrative Scoring Scheme?

Three individual scores can be obtained including 1. MW Match Score 2. PST Match Score 3. In vitro & In silico Match Score.

48. Distinguish In Multiple Sequence Alignment And Pairwise Sequence Alignment? Multiple Sequence Alignment:

A multiple sequence alignment is a sequence alignment of three or more biological sequences, generally protein, DNA, or RNA. In many cases, the input set of query sequences are assumed to have an evolutionary relationship by which they share a linkage and are descended from a common ancestor.

Pairwise Sequence Alignment:

Pairwise Sequence Alignment is used to identify regions of similarity that may indicate functional, structural and/or evolutionary relationships between two biological sequences (protein or nucleic acid).

49. What Are Uses Of Mass Spectrometry?

Mass spectrometry (MS) is an analytical technique that ionizes chemical species and sorts the ions based on their mass-to-charge ratio. In simpler terms, a mass spectrum measures the masses within a sample. Mass spectrometry is used in many different fields and is applied to pure samples as well as complex mixtures.

50. Define Tblastx blastn blast?

tblastx (translated
BLAST) blastn
(translated BLAST)
BLAST performs searches by quick alignments on sequences.

51. Scoring In Homology Modelling?

The scoring functions (1) and (2) are sequence-based: score (1) compares the secondary structure prediction for the target sequence with the actual secondary structure assignment of the template protein, while score
(2) evaluates the sequence similarity between the target and template sequence.

52. Why Model Of Protein Structure Is Necessary?

Proteins are chains of amino acids joined together by peptide bonds. Many conformations of this chain are possible due to the rotation of the chain about each Cα atom. It is these conformational changes that are responsible for differences in the three dimensional structure of proteins.

53. How Positive And Negative Energy Relapsed In A Molecule?

The change in enthalpy is negative in exothermic processes, because energy is released from the system into its surroundings. Generally, a positive change in enthalpy is required to break a bond, while a negative change in enthalpy is accompanied by the formation of a bond.

54. Enlist Few Points In Scope Of Bioinformatics?

The scope of a bioinformatics project (or lab) varies widely depending on the balance of math/stats, computational/software/engineering, and biological/molecular training of the researchers involved and the data orsubject matter they deal with particularly DNA sequences, gene expression, protein, epigenetics.

55. Different Secondary Structures Of RNA?

The secondary structures of biological DNA's and RNA's tend to be different: biological DNA mostly exists as fully base paired double helices, while biological RNA is single stranded and often forms complex and intricate base-pairing interactions due to its increased ability to form hydrogen bonds stemming from the.

56. Silico And Vitro?

In silico is an expression meaning "performed on computer or via computer simulation" in reference to biological experiments.

biotechnologists

In vitro studies are conducted using components of an organism that have been isolated from their usual biological surroundings, such as microorganisms, cells, or biological molecules.

57. MS1 And MS2 Discuss?

MS1 reports the intact mass of molecules (proteins or peptides) in the sample • Intact mass can be compared with every protein's mass in database to identify the molecule in sample • Incase multiple candidate proteins are reported, **MS2** can be performed MS2 helps measure fragment peptide masses • MS2 data can used to extract peptide sequence tags

58. What Know About Proteomics?

Proteomics generally refers to the large-scale experimental analysis of proteins and proteomes, but often is used specifically to refer to protein purification and mass spectrometry.

59. Top Down Proteomics?

Top-down proteomics is a method of protein identification that uses an ion trapping mass spectrometer to store an isolated protein ion for mass measurement and tandem mass spectrometry analysis. Top-down proteomics is capable of identifying and quantitating unique proteoforms through the analysis of intact proteins.

60. Summary Of Structural Modelling?

Structural equation modeling is a multivariate statistical analysis technique that is used to analyze structural relationships. This technique is the combination of factor analysis and multiple regression analysis, and it is used to analyze the structural relationship between measured variables and latent constructs.

Protein structure prediction?

There are three different strategies for structure prediction

1. Homology Modelling
2. Threading/Fold Recognition
3. Ab Initio Modelling

Chou – Fasman method?

It is a technique for the prediction of secondary structures in proteins i.e.

Alpha Helices, Beta Sheets and Turns is Chou – Fasman technique. The method is based on analyses of the relative frequencies of each amino acid in alpha helices, beta sheets, and turns based on known protein structures solved with X-ray crystallography. From these frequencies a set of probability parameters (in our handouts, it is propensity table) were derived

- For the appearance of each amino acid in each secondary structure type.
- To predict the probability that a given sequence of amino acids would form a helix, a beta strand, or a turn in a protein.

Chou-fasman algorithm (alpha helix):

- For Alpha Helices, 4 contiguous amino acids are required.
- Their Alpha-Helix propensity should be more than 1.0
- Once this propensity falls below 1.0, Alpha-Helix stops.

Chou- Fasman (Loop form after alpha helix and beta sheet)?

After computing the propensity of alpha helices and beta sheets, we need to settle for loops. Let's see how we can find out the loops using Chou Fasman Algorithm.

For any jth residue in sequence, we calculate f (Total) = f(j) f(j+1) f(j+2) f(j+3) (tetra peptide).
If,
f (Total) > 0.000075.
The average value for P (turn) > 1.00 in the tetra peptide.
The averages for the tetra peptide are such P(a-helix) < P(turn) > P(b-sheet), It is a Turn!

Advantage and disadvantage of Ab initio? Advantages:

- Ab Initio methods can fold any target sequence using only physical atomic properties.
- Predictions are mostly accurate and correctly describe the natural folding process.

Disadvantages:

- Ab initio methods are the very difficult to design (energy function).
- These methods are slow due to the huge possibilities.

Bioinformatics role?
An interdisciplinary field
New but rapidly developing field
Low requirement on infrastructure and research equipment Vast opportunities for scientific discovery

PAM matrix?
PAM means "Point Accepted Mutations"
Point accepted mutation is a substitution of one amino acid by another such that the protein functions stays conserved.
PAM unit is a time in which about 1% of amino acids in a sequence undergo accepted mutations. PAM matrices are scoring matrices that are useful in computing sequence alignment scores.

CLUSTALW?
Performs alignment in slow/accurate or fast/approximate
Create multiple alignments, optimize existing alignments, profile analysis, and create phylogenetic trees. CLUSTALW can use multiple file formats including EMBL/SwissProt and Pearson (Fasta) etc

Function of Tandem MS?
Tandem MS helps in measurement of mass to the fragments as well. This process provides another step in further scoring and ranking and Protein identification thus becomes easier.

Steps present in flow chart of homology modeling?
1. Template recognition and initial alignment.
2. Alignment correction.
3. Backbone generation.
4. Loop modeling.
5. Side-chain modeling.
6. Model optimization.
7. Model validation

How homology modeling is used for knowing the sequence of unknown protein sequence?
If another protein which has a similar sequence also has its structure known, the structure of an unknown protein can be predicted based on that similar protein. So, it is then possible to identify unknown protein structures by just examining the homologous protein sequences. Good sequence

ensures that homology modelling will give accurate results. Thus, Homology modeling is used to predict structures of proteins having high sequence similarity with other proteins with known structures:

Silico fragment scoring?

- Count the matches between in silico and in vitro peaks.
- Give an equivalent score to the candidate protein.
- Weigh each of the aforementioned match by the mass error.
- Accumulate the score

How we can get scoring in BUP proteomic?

There are two approaches to perform BUP:

- Peptide Mass Fingerprinting.
- Shotgun Proteomics

ProSight pattern:

Kelleher et al have developed
an Engine i.e. Prosight PTM
ProSight PTM is the state of the art in top down proteomics search.
Using Prosight PTM, post-translational modifications can be accurately identified.

Top down proteomics:

Measures intact proteins followed by their peptides after fragmentation Score in Protein Fragmentation:- If we can:
Measure the mass of fragments using MS.
Calculate the Protein Fragmentation Techniques theoretical mass of the fragments. Then, we can award score on the basis of the similarity of experimental and theoretical mass.

Molecular biology?

It is the branch of biology that deals with the structure and function of the macromolecules (e.g. proteins and nucleic acids) essential to life. Molecular biology is the study of biology at a molecular level.

Protein Fragmentation techniques?

- Electron Capture Dissociation (ECD).
- Electron Transfer Dissociation (ETD).
- Collision Induced Dissociated (CID).

Types of secondary structures of RNA:
I. Single stranded:

technologist)
biotechnologists

- 3'end may fold on to the 5' end.

2. Helices:

- Double stranded RNA helix of stacked base pairs

3. Hairpin loop:

- The loop of the hairpin must at least four bases long to avoid steric hindrance with base-pairing in the stem part of the structure

4. Bulge Loops:

- Bulges, are formed when a double-stranded region cannot form base pairs perfectly.

5. Interior loop:

- Interior loops are formed by an asymmetric number of unpaired bases on each side of the loop.

6. Junctions or intersections:

- Junctions include two or more double-stranded regions converging to form a closed structure.
- The unpaired bases appear as a bulge.

Rooted Un Rooted Tree?

Rooted Tree:

- Each node with descendants represents the inferred most recent common ancestor of the descendants.
- The edge lengths in some trees may be interpreted as time estimates.
- Rooted trees can show temporal evolutionary direction.
- Expensive.

Unrooted:

- Only the relatedness of the leaf nodes.
- Do not require the ancestral root to be known or inferred.
- Less expensive.

Uses:

In bioinformatics, such as

- Rooted and unrooted trees can be used to show phylogenetic relationships between sequences.

1) Fasta Was Develop In(Ans1988)
2) In Corey Paulin Modal Of The Protein Each Atom Is Represented By(Solid Sphere)
3) How Many Storages Use For Structural Modling (3)
4) Modeler Software Use For Homology Modling Which Use(Phython File Format,Not Confrm)
5) Mass Of The Based Proteomic Begin With Measure Of(Intact Protein Amino Acid)
6) Protein Sequence Obtained From(Uni Port)
7) During The Formation Of The 5 Nucleotide Hydrogen Bond Energy Release(-12kcal/Mol)
8) Which Method Is Not Use For Making Polygenatic???Msa
9) Rna Dont Have Structer???Quatnaryy
10) Large Numbr Of Sequence Availble??Publically
11) Coding Region Of Dna Is???Exon
12) Blast Use For???Unknown And Known Protein
13) Expasy Develop By???Sib

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Variation In Sequence Develop By??? Insertion,Deletion And Substituin
Bond Form Between C And N Terminas??? Hydrogen Bond
X.Y.Z Position Of Protein Are Available By??? Pdb
Technique Use To Measure The Mass Of Protein???Mass Spactrometry
Large Amino Acid??? Tryptophan
Positive Charge Amino Acid??? Lysine And Arginine
Mrna Has??? Single Strand
C.Alpha Atom Are Traced To Recreat??? 3d Structer Of Protein
Bond Form Between Two Atom??? Energy Release
Provide Clue Of Precurser Protein Peptide Sequence??Pst
Amino Acid Requird For Betasheet??? 3-5
Are Formed By Bonding Betweenci And Ni+4n Protein Backbone??? Alpha Halix
Blasta Stand For??? Basic Local Alignment Search Tool
Amino Acid Have Uniwe Trait Which Can Help In??? Protein Folding
Align Comparison Betweendna/Dna Anddna/Protein??? Fasta.
Ratio Use To Calculate Mass Of Protein??? M/Q.
Treading Not Requied For??? Structer Of Entire Backbone.
Needle Man Wounch Man Or Smith Water Man Algorithm Use For??? Local And Global Alignmnt.

Q.1. In mass electrometer chamber protein is deflected according to its ……
a) charge b)velocity c)charge, mass d) volume

in MS chamber mass/charge ratio can be measured by …… of protein molecules.
a) volume b) mass c) charge d) speed

while scoring protein mass score is awarded on the base of ... b/w experimental and theoretical mass.
a) Distance b) closeness c) equality d) all of these

Tandem MS is used to..... of protein fragments.
a) Ionization b) fragmentation c) measure the mass d) sequence

Fragmentation techniques split the molecules of protein into……..
a) Two part b) three part c) four part d) multiple part

Tandem MS is typically called as,
a) MS b) MS1 c) MS2 d) ALL

Dianamic Programing Recombinant Nucleotide By??? Traseback.
Insertion And Deletion Cause??? Gap.
Information Required For Follding Of Amino Acid Is Present In??? Amino Acid.
Mgf Use For??? Proteomic.
Nj Algorith Use To Pridict The??? 2nd Structer Of Rna..

Each fragmentation technique differ in their work on bases of ,
a) Having different splitting sites
b) Having different time duration
c) Having different mass detection
d) all of these

Mass spectrometer use o measure the mass of protein and ……
A) Amino acids b) lipids c) peptides d) DNA

Protien ionization is the.....step in MS based proteomics.
a) first b) second c) third d) fourth

Ionization process involve gaining and removal of …….
a) electron b) proton c) ion d) atom

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molecular weight increases by..... Of proton

a) removing b) adding c) transfer d) All of these

We can score each protein hit in away that the better match gets a ……

a) higher score b) lower score c) equal score d) opposite score

ECD gives out "C" and.....ions,

a) B b) Y c) Z d) A

CID gives out "B" and..... ions,

a) Y b) Z c) B d) C

PST is abbreviation of…..

a) Protein sequence tags b) protein sequence tools c) Peptide sequence tags d) peptide sequence tools

PST is sequence of peptidesproduced during…..

a) MS b) MS1 c) MS2 d) ALL

Fragmentation at consecutive sits lead to a mass difference equal to that of a single

…..

a) peptide b) amino acid c) protein d) DNA

The data base protein which have matching PSTs to invitro protein are known as

……

a) Sample protein b) candidate protein c) data base protein d) PST protein

Experimental peptides also known as ……

a) data base peptide b) observational peptide c) invitro peptide d) sample peptide

Errors in peak matching also considered by…….

a) Sequence b) fragmentation c) ionization d) scoring scheme

Which one is the right sequence of steps to create a composite protein search engine.

a) PST – MW – In silico comparison b) MW – PST – In silico comparison

c) In silico comparison – MW – PST c) MW – in silico comparison – PST

Q. 22 We can´t add these three scoring comparison until we divide them by their……

a) length b) error c) numbers d) distance

Typtic peptides may reach upto an order of …….

a) 200,000 – 30,000 b) 30,00 – 40,00 c) 300,000 – 400,000 d) 400,00 – 500,00

large scale proteomics is a developed form of

a) mass spectrometry b) BUP technique c) TDP technique d) uniprot technique

Mass spectrometry use to measure.....Ratio of ionized protein or peptide.

a) mass/distanceb) charge/distance c) mass/charge d) distance/charge

Raw file in a format are often found in.....That we have to change in an open

format so as to make it readable.

a) Theoretical form b) decimal form c) percentile form d) binary form

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mzData was developed by ……

a) mzML b) HUPO-PSI c) RAW format d) MGF

MASCOT is an online ..... Search engine,

a) Bottom up proteomic b) top down proteomic c) both d) none

is a simple human readable format for MS/MS developed by Matrix Science.

a) MASCO b) ABI c) MGF d) RAW format

theoretical vs experimental fragments comparison helps us as the ..... Stage for short listing candidate protein list.

a) 1st b) 2nd c) d) 4rth

To ensure that all fragments of precursor molecules are also charged, we can use,

a) ESI b) RAW c) BOT d) NONE

Monoisotopic mass is the mass of ……..

a) single isotope b) average isotope c) least abundant isotope d) most abundant isotope

Q. 33 Proteins are assembled by ..... in cellular cytoplasm

a.golgi apparatus b.mitochndria **c.ribosomes** d.nucleaus

Q. 34.Each protein is synthesized by polymerization of .....

a.peptide bond **b.aminoacids** c.nucleotide sequance d.all

Q.35.The characteristics are governed by the elemental composition of an amino acid's .....

a.alpha carbon b.H group **c. R group** d.amino group

Q.36 Since ....and.....introduce very little dipole moments in hydrophobic amino acids, these amino acids are non polar

a.H &C b.H & Y c.Y & H d.A & Z

Q. 37 Mostly .....aminoacids found at the surface of folded proteins

a.polar b.nonpolar c.acidic d.basic

Q.38. At pH=7, five amino acids are charged, ... negatively and..... positively

a.3 & 2 b.3 & 3 **c.2 & 3** d.2 & 2

Q. 39.The charge state depends on the acceptance or donation of.....from the R group side chains

a.electron **b.proton** c.neutron d.both a & b

Q. 40. Amino acids have property such as .....

a.charge state b.polarity c.hydrophobicity **d.all**

Q. 41 Tendem MS helps to measure the moleculer weight of ……

a) isotops b) precursor protein c) ionized fragments d) invitro protien

Q. 42.If pH >pK for aminoacids carboxyle side losses the proton and bocome negative which is .....

a.basic b.neutral c.less acidic **d.acidic**

Q. 43. ..... are key factors participating in protein folding

a.solublity b.reactivity **c.both** d.none

Q. 44.The charge state depends on the acceptance or donation of.....from the R group side chains

a.electron **b.proton** c.neutron d.both a & b

Q. 45. Amno acid with the side chain containing organic acids are soluble in ……

a) ethanol b) water c) lipids d) a and c

Q. 46 .....residue has no R group.

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a) cystine b) proline c) glycine d) adinine
Q. 47 proline residue..... the flexibility of polypeptide chain.
a) increase b) decrease c) stabilize d) no effect
Q. 48 A peptide unit has a backbone of polar NH and C=O which is highly…….
a) hydrophilic b) hydrophobic c) neutral d) acidic
Q.49 Functional group of protein always attach to its hydrophobic …….
a) surface b) coat region c) core region d) backbone
Q. 50 After polymerization.....are formed.
a) 3D structure b) alpha helix c) beta sheets d) All of these
. 51 The force involve in protein folding ……
a) electrostatic interactions b) van der waals interactions c) hydrogen bonding d)all
Q. 52 .....folded form of protein is thermodynamically more stable than others.
a) Combined b) native c) partially d) a and b
Q. 53 3´ structural can combine with other polypeptide units to form ……
a) secondary structure b) tertiary structure c) quaternary structure d) all
Q. 54 There is a..... variety of protein structure exist.
a) rare b) restrict c) limited d) huge
Q. 55 Folded sub structures are called ……
a) primary structure b) secondary structure c) tertiary structure d) All
Q. 56 The protein super structure are called ……
a) primary structure b) secondary structure c) tertiary structure d) All
Q. 57 Lysine and tryptophan have respectively..... 1-letter codes.
a) L – T b) S – F c) K – W d) N - A
Q. 58 C- terminus is ..... charge.
a) negatively b) positively c) neutral d) none
Q. 59 ..... bonds are the reason of 2' structure formation.
a) covalent b) hydrogen c) ionic d) All
Q. 60 Loops and Coils are very important 2' structures in that they form ..... of proteins.
a) active site b) inactive site c) acidic site d) basic site
Q. 61 ..... Enforce the 3' structure.
a) Covalent bonds b) Hydrogen bonds c) hydrophobic interaction d) all
Q. 62 Proteins with..... chain can form 4' structures.
a) single b) double c) linear d) multiple
Dihedral Angles always form between ……
a) two points b) three points c) four points d) five points
angles are taken up as a result of folding of protein.

a) phi b) psi c) both d) none

.64 ..... can be used to construct the backbone of a protein towards its visualization.
a) N b) C c) C Beta d) C Alpha
Q. 65 Atoms of phosphorus,sulfur, and chlorine are .....in covalent radius.
a) ~1 Å b) ~2 Å c) ~3 Å d) ~4 Å
Q. 66 The angstrom is used to express ……
a) small biological structures b) lengths of chemical bonds,
c) arrangement of atoms in crystals. d) All

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Q. 67 Distances between C-Alphas are measured in the unit "Angstrom" by the process of …….
a) adman degradation b) X Ray crystallography c) Blossom matrix d) PAM Matrix

Q. 68 1 A resolution is better than ……..
a) 9 A b) 8 A c) 7A d) 10 A

Q. 67 PDB contains protein……information
a) structure b) size c) distance d) function

Q. 68 PyMOL and Cn3D is a python-script based tool for visualizing the ……..
a) protein size b) protein structure c) amino acids size d) a and b

Q.69 BS Models have atoms as colored balls and intermediate bonds as ……..
a) straw b) spindle c) stick d) small line

Q. 70 In CPK diagrams, each atom is represented by a …….
a) dot sign b) Solid Ball c) doted circle d) solid sphere

According to "Anfinsen's thermodynamic hypothesis" protein folds for making a
……..
a) stable structure b) structure having minimum free energy c) structure having maximum free energy d) a and b

Hence, the basic idea of thermodynamic stability is to .....bonding in order to …… the free energy.
a) minimizing, maximizing b) maximizing, minimizing c) maximizing, stable
d) b and c

To determine the number of each type of interaction within a protein, we need to find its ……
a) type of amino acid b) types of bonds c) inter-atomic distances d) atomic mass

by looking up at the bond/energy table, we can compute the …….
a) overall energy b) molecular mass c) atomic mass d) bond sequence

If two atoms are participating in a covalent bond, their distance is ……
a) ~0.95 A b) ~0.96A c) ~0.97 A d) ~0.98 A

"Helix Formers" are generally.....amino acids (M, A, L…)
a) acidic b) basic c) hydrophilic d) hydrophobic

Alpha Helices are formed by hydrogen bonding (O-H)between …… and ……. atoms in the protein backbone
a) C – H b) H – N c) C – N d) P - N

resides are needed to make a Beta Strand,typically
a) 3 – 4 b) 5 – 10 c) 6 – 7 d) 5 – 6

Loops are the..... of secondary structure
a) 1st b) 2nd c) d) 4rth

what is made by a single beta sheet when it twists and coils upon itself.
a) Beta Barrel b) beta sandwich c) loops d) coils

are flexible & can adopt multiple conformations.

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a) coils b) helices c) loops d) All

are apparently disordered regions with random structure and random length.

a) coils b) helices c) loops d) beta sheets

are Non-functional Combinations of 2' structures)

a) loops b) coils c) beta sheets d) motifs

Domains are semi-independent functional structures in a protein having over ……..

a) ~30 residues b) ~40 residues c) ~50 residues d) ~60 residues

is one such system in which proteins are organized into classes, architecture, topology & homology.

a) CATH b) FSSP c) LOCK d) DOLI

By comparing .....can help us evaluate if the proteins do similar function.

a) sequence b) free energy c) structure d) All

For comparing the complete or partial protein structures, the position of ……. can be used.

a) N terminal b) C terminal c) Alpha Carbon d) functional group

Whole protein structures can be compared by calculating the.....difference b/w their Alpha Carbons positions.

a) distance b) root mean squared c) energy d) All

The.....the RMSD, the similar are the proteins.

a) Equal b) higher c) lower d) random

RMSD tells us about the quality of the matches

a) matches b) mismatches c) sequence d) bonding

There are over a million protein sequences but only about .....protein structures.

a) 10,000 b) 100,000 c) 10,0000 d) 1,000

if we have several..... in the sequence, then we can anticipate that a helix may be

formed by them.

a) Proline b) Thiamin c) Alanines d) Adenine

Product of propensity values is computed for .....for each 2' structure.

a) individual propensity b) overall propensity c) greatest propensity d) All

For Alpha Helices.....contiguous amino acids are required.

a) 6 b) 5 c) 8 d) 4

Q. 98 Alpha-Helix propensity should be more than ..... Once this propensity falls, Alpha-Helix stops.

a) 1.0 b) 1.5 c) 2.0 d) 2.5

Q. 99 …… amino acids are needed to start an Alpha Helix and ..... amino acids for Beta Sheet.

a) 3 , 4 b) 4 , 5 c) 5 , 6 d) 4 , 6

Q. 100 Regions where overlapping alpha-helices and betasheets occur are declared helices.

a) loops b) coil c) helix d) helices

In chou fasman algorithm Alpha Helices can be finalized if their propensity is …….

to the propensity for Beta Sheets.

a) equal b) less c) higher d) different

Chou Fasman Algorithm helps us to predict,

a) Alpha Helices b) Beta Sheets c) Turns d) All

Beta sheets can be predicted from …… amino acid sequences

a) primary b) secondary c) tertiary d) quaternary

Q.104Chou Fasman Algorithm is based on statistical occurrence of.....in known structures.

a) protein b) Amino Acids c) nucleotide d) a and b

Q. 105 Functional evaluation of proteins can be performed by……

a) Structure visualization b) , classification c) prediction d) All

Functional evaluation of proteins is important for understanding …….

a) disease b) designing drugs c) treatment d) ALL

X-Ray Crystallography and NMR Spectroscopy are used to find the structures of ……..

a) DNA b) Amino acid c) proteins d) nucleotides

Q.108it is then possible to identify unknown protein structures by just examining the …… protein sequences

a) heterogeneous b) homologous c) multiple d) binary

Q.109In homology modelling, proteins with similar.....are considered.

a) 1' sequences b) 2' sequences c) 3' sequences d) 4' sequences

Q.110Homology modelling is used to predict structures of proteins having high sequence similarity with other proteins with.....!

a) known structures b) unknown sequence c) similar sequence d) a and c

Q111 Protein structure can be predicted by…..... methods:

a). Homology Modelling b). Threading/Fold Recognition c). Ab Initio Modelling d). All

in homo logy modeling,Definition of Template (……..) & Target (.....)

a) known – unknown b) unknown - known c) known – similar d) unknown - similar

One way to handle loops is to take some residues before and after the insertion as ……. residues.

a) sample b) predicted c) anchor d) similar

The backbone of tyrosine strongly prefers .....rotamers & the real side-chain may fit one of them.

a) 1 b)2 c)3 d)4

in homo logy modeling when protein backbone is ready,then next, ……. were modelled and used to bridge gaps.

a) α helices b) β sheets c) coils d) loops

Loop modeling is ……step of Homology modeling.

a) 2nd b) 3rd c) 4th d) 5th

Homology Modelling ..... study conformational changes.

a) can b) cannot c) may be d) some times can

Q.188 We can overcome the limitations of homology modeling by ……

a) Threading b) Bowie Algorithm c) Ab Initio modeling d) a and c

Q. is a software for homology modelling.

a) Homolog b) Modeller c) iTASSER d) a and b

There are ..... stable folds of protein in nature.

a) 30,000 b) 4,000 c) 40,000 d) 5,000

Several tools are available to perform homology modelling in a programmatic or automated way such as…..

a) Swiss Model b) Robetta c) 3D Jigsaw d)all

Fold recognition is also called …….

a) scoring sequencing c) Threading d) modeling

Homology modelling fails to predict quality structures .....is a technique for predicting these structures.

a) Fold recognition b) Swiss Model c) Threading d) a and c

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Threading Scoring typically involves using a.....function based on energy of a molecule

a) Z-Score b) C-score c) A-score d) Y-score

cannot be predicted using threading.

a) invtro protein b) primary protein c) 2ndary protein d) Novel proteins

3D-1D Bowie Algorithm Proposed by Bowie et al in ……

A) 1997 B) 1991 C) 1995 D) 1998

3D-1D Identify amino acids based on ……..

a) protein core b) side chain positioning c) solubility d) All

3D-1D methods convert structure and environment information into ……..

a) software b) packets c) profiles d) subunits

Ab initio modeling is Computationally..... Suitable for proteins with less than …… residues

a) cheap – 500 b) expensive – 100 c) cheap – 100 d) expensive - 500

Ab Initio Modelling methods attempt to identify the structure with minimum ……

a) charge ratio b) distance c) molecular weight d) free energy

Energy released during the.....process is computed for predicting structure.

a) Sequencing b) folding c) breakdown d) bonding

Ab initio methods base their predictions on.....models of folding mechanisms

a) Chemical b) statistical c) physical d) all

Global optimization finds the most …….

a) matching sequence b) stable structure c) similar pattern d) a and c

structure not always at the global minimum.

a) Native b) invitro c) insilico d) stable

Ab initio methods are the very ..... to design.

a) easy b) difficult c) suitable d) accessible

In homology modeling first we perform ……. and then ……..

a) Loop modeling - Backbone generation b) Backbone generation - Loop modeling

There are .....Steps of Homology Modelling.

A) 5 b) 6 c) 7 d) 8

Ab initio modeling is Computationally ……. And..... process.

a) easy and expensive b) Expensive and complex c) cheap and easy

d) difficult and cheap

Ab initio methods predict protein structures by folding proteins based on each constituent atom's ……..

a) volume b) charge c) mass d) all

UPGMA is a ..... method.

a) clustering b) non clustering c) both

Homology modelling is performed in cases of .....identity and alignment score • For the "Twilight zone", other strategies are employed

a) high b) low c) equal d) overall

maximum Parsimony etc are ..... methods (not included in this course).

a) clustring b) nonclustering c) mapping d) visualizing

Online tools: ……..

a) mascot b) sequest c) prosight PC d) all

Number of known protein sequences is..... as compared to known proteins structures

a) larger b) much larger c) smaller d) mech smaller